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Oil 2025

Analysis and forecast to 2030

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Abstract

Global oil markets have so far had a turbulent 2025. Heightened trade tensions and uncertainty have weighed on the world economy and, by extension, oil demand growth. Combined with the recent OPEC+ decision to accelerate the unwinding of oil production curbs that have been in place for several years, these factors have recently pushed international oil prices to four-year low in April and early May. At the same time, shifts in energy policies are affecting oil producers and consumers alike, with oil supply security remaining high on the international energy policy agenda.

Oil 2025 looks beyond the short-term horizon covered in the IEA's monthly *Oil Market Report (OMR)* to provide a comprehensive overview of evolving oil supply, demand, refining and trade dynamics through 2030. This *Report* presents detailed forecasts and analysis of oil demand fundamentals across fuels, sectors and regions as well as the supply outlook from planned upstream and downstream projects around the world. The results provide valuable insights on the prospects for spare oil production capacity, product supply and trade flows through 2030. *Oil 2025* also explores the implications of surging output of natural gas liquids (NGLs) in an era of petrochemical-driven demand growth.

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Executive summary

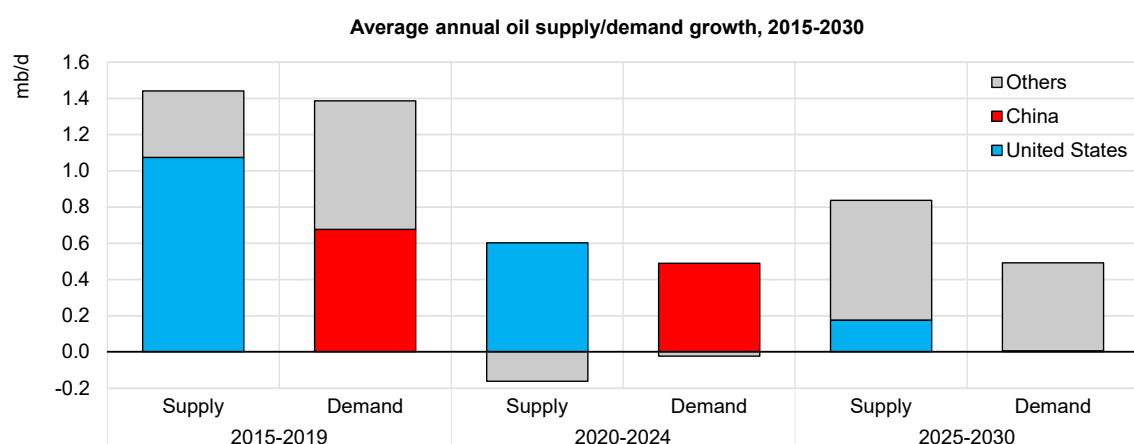
Turbulent times in oil markets

Heightened geopolitical risks, unresolved trade tensions, and policy shifts have added myriad uncertainties to the oil market outlook. Since the start of the year, major economic forecasters have cut their outlooks for world GDP growth in 2025 by roughly half a percentage point to around 2.8% and see a below-trend pace of about 3% annually for the remainder of the decade, with knock-on implications for oil demand. With conflicts in the Middle East region at risk of intensifying and trade negotiations ongoing, uncertainties surrounding our forecasts are substantial.

At the same time, a decision by the OPEC+ producer group, led by Saudi Arabia, to start unwinding oil production curbs in May 2025 is resetting oil supply trajectories over our 2024-30 forecast period. The anticipated output increase from OPEC+ and the impact of higher tariffs on trade pushed oil prices to four-year lows in April and early May. As a result, oil executives are recalibrating investment plans. However, oil prices have since rebounded after an exchange of air strikes between Israel and Iran started on 13 June 2025. With geopolitical and economic uncertainties affecting oil producers and consumers alike, oil supply security remains high on the international energy policy agenda.

Oil markets are going through a fundamental transformation as the drivers of global oil supply and demand patterns shift. Over the past decade, oil market dynamics have been defined by the parallel growth in US oil supply and Chinese oil demand.

US and China's shares in global oil supply and demand growth fade



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From 2015 to 2024, the United States accounted for 90% of the increase in global supply, with the shale boom lifting US oil production by more than 8 mb/d to over 20 mb/d. At the same time, Chinese oil demand rose by nearly 6 mb/d, accounting for 60% of the global increase in oil use. The picture to 2030 looks very different. Following an extraordinary surge in EV sales, the continued deployment of trucks running on liquified natural gas (LNG), as well as strong growth in the country's high-speed rail network, along with structural shifts in its economy, Chinese oil demand is on track to peak this decade. For supply, the pace of expansion in US oil production is slowing as oil companies scale back investments but it nevertheless remains the largest contributor to non-OPEC+ growth in the forecast.

A peak in global oil demand is still on the horizon

Global oil demand is forecast to rise by 2.5 mb/d from 2024 to 2030, reaching a plateau around 105.5 mb/d by the end of the decade. However, annual growth slows from roughly 700 kb/d in 2025 and 2026 to just a trickle over the next several years, with a small decline expected in 2030, based on today's policy settings and market trends. This is driven by below-trend economic growth, weighed down by global trade tensions and fiscal imbalances, and the accelerating substitution away from oil in the transport and power generation sectors.

Despite some recent headwinds, global electric car sales have continued their remarkable growth trajectory. They exceeded 17 million in 2024 and are expected to surpass 20 million in 2025, representing around one-quarter of all cars sold, according to the IEA's [Global Electric Vehicle Outlook 2025](#). This analysis shows that EVs are set to displace 5.4 mb/d of global oil demand by the end of the decade. Substitution away from oil will also feature prominently in power generation during the forecast period – particularly in Saudi Arabia, where displacement of oil burning by natural gas and renewables drives the single largest decline in oil demand for any country through 2030.

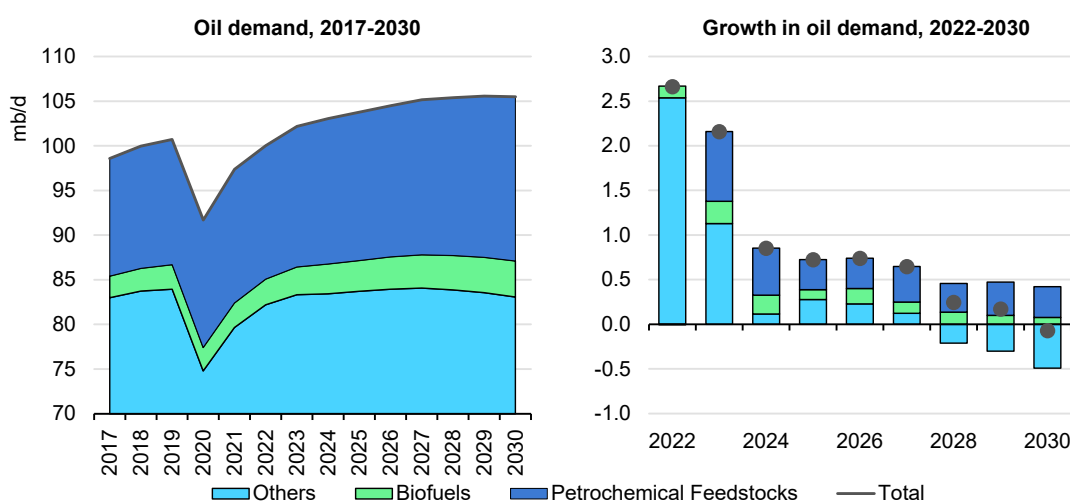
As the transport and power generation sectors continue to diversify towards other fuels, the petrochemical industry is set to become the dominant source of global oil demand growth from 2026 onwards. Increased petrochemical consumption is closely linked to the burgeoning supply of natural gas liquids (NGLs). China and the United States lead the build out of petrochemical capacity, at the expense of Europe and other economies in Asia. Globally, the production of polymers and synthetic fibres will require 18.4 mb/d of oil by 2030, or more than one in every six barrels. Demand for oil from combustible fossil fuels – which excludes petrochemical feedstocks and biofuels – may peak as early as 2027.

Robust oil demand growth of 4.2 mb/d in emerging and developing economies over the 2024-30 period contrasts with a continued contraction in advanced economies. Asian markets dominate growth, with India's expected 1 mb/d

increase the largest of any single country by far, though rising oil use in Southeast Asian economies is also significant. By contrast, oil consumption among OECD countries is forecast to decline by 1.7 mb/d through 2030.

While the overall picture for global oil demand is broadly unchanged from [last year's forecast](#), this masks notable changes among the world's two largest consumers. China's total oil consumption in 2030 is now set to be only marginally higher than in 2024, compared with growth of around 1 mb/d forecast previously. By contrast, lower gasoline prices and a loss of momentum in EV adoption in the United States, the world's largest oil consumer, have led to an increase in forecast oil demand of 1.1 mb/d by 2030 compared with last year's report.

World oil demand forecast to plateau this decade



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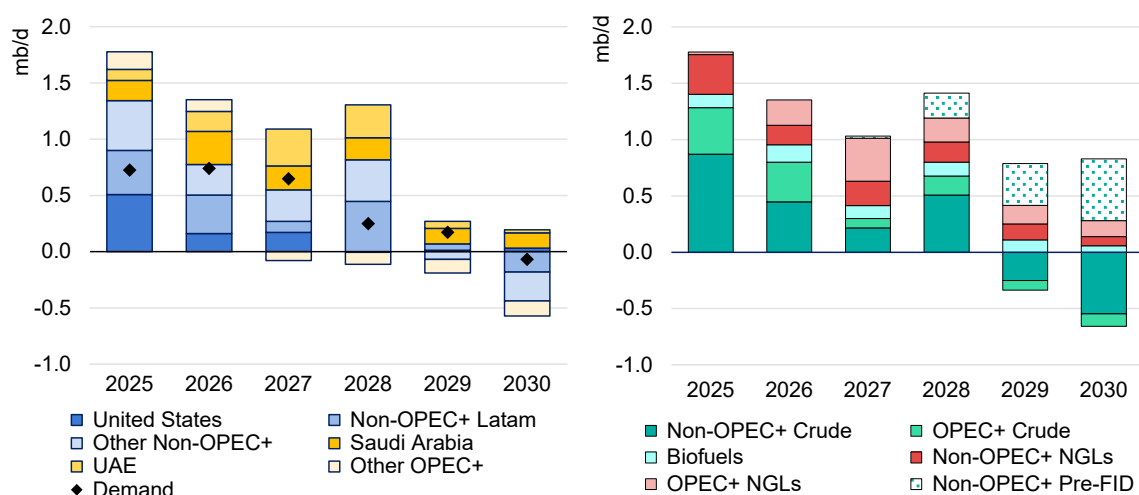
Lower oil prices set to slow upstream expansions

The precipitous drop in oil prices in the early months of 2025 and an uncertain investment climate have prompted oil executives around the world to reevaluate their upstream priorities. In addition, after nearly five years of production restraint, the OPEC+ alliance has begun to unwind voluntary supply cuts of more than 2 mb/d that have been in place since 2023. This has put additional pressure on producers outside the bloc to rebalance the market.

Upstream oil investment is set to fall by 6% to around USD 420 billion in 2025, with some of the largest declines in light tight oil in the United States. Investment in conventional projects – both existing and new – is expected to prove more resilient in 2025. Nonetheless, lower oil prices and higher production costs, due to tariffs and inflated costs for essential materials, could mean larger cuts to investment are still to come, while a return to durably higher prices could boost spending.

World oil production capacity is forecast to rise by 5.1 mb/d to 114.7 mb/d by 2030, led by Saudi Arabia and the United States – significantly outpacing the projected 2.5 mb/d increase in global oil demand. Mirroring demand trends, supply capacity growth is heavily frontloaded, slipping from 1.8 mb/d in 2025 to contraction after 2029 as the pipeline of non-OPEC+ projects wanes.

Global oil supply capacity forecast, year-on-year change, 2025-2030



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Notes: OPEC+ NGLs include condensates. Crude includes processing gains and non-conventional volumes. The right-hand chart includes pre-sanctioned projects, listed in the Annex tables.

Global NGL production is expected to increase by 2.3 mb/d to 20.1 mb/d by 2030, accounting for nearly half the total increase in world oil production capacity. A strategic focus on natural gas developments by producers in the Middle East is projected to boost regional NGL supply by 1.4 mb/d to 2030. While independent US producers are set to slow spending, increasing associated gas from the shale patch is expected to buoy US NGL output by 860 kb/d. Biofuel supply gains led by Brazil, India and Indonesia are forecast to add another 680 kb/d by 2030. Crude oil capacity is set to rise by 1.8 mb/d globally, led by the United Arab Emirates (+720 kb/d) and Iraq (+560 kb/d), while the biggest decline comes from Mexico.

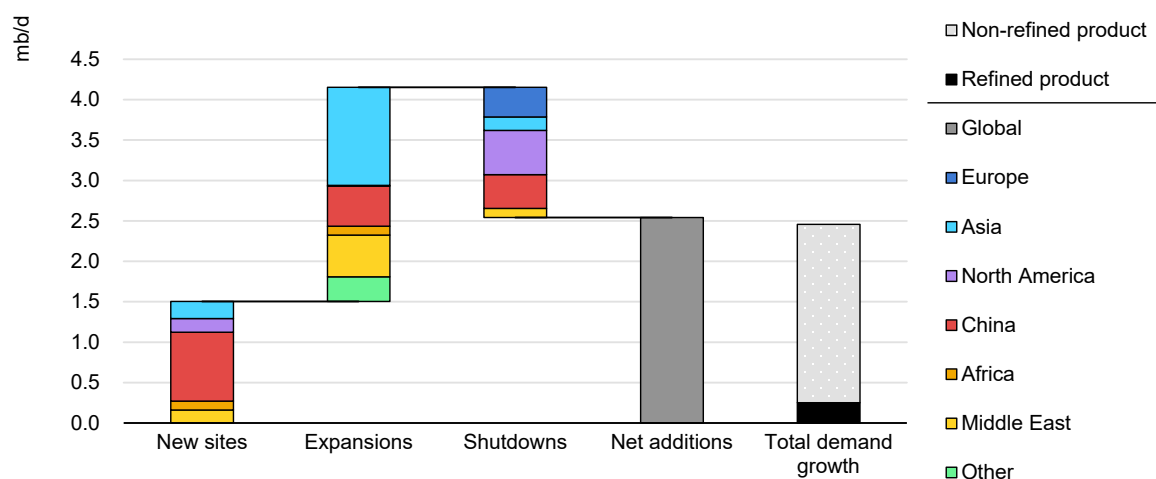
Total non-OPEC+ oil supply is forecast to climb by 3.1 mb/d by 2030, despite the number of approved projects tailing off after 2027. The Americas dominate the expansion even though the outlook for US and Brazilian production has dimmed. Combined with a forecast slowdown in oil demand, the call on OPEC+ crude in 2030 is 1.2 mb/d below the average 2025 call. If OPEC+ crude oil supply is sustained at current rates, all else being equal, global oil supply would rise to 107.2 mb/d by 2030, 1.7 mb/d higher than projected demand, suggesting prices would have to drop to prevent an untenable stock build.

The refining industry faces tepid growth for products

The refining sector is set to be increasingly challenged by demand growth that is underpinned almost exclusively by petrochemicals produced from non-refined products such as NGLs. Global refined products demand is projected to peak in 2027 at 86.3 mb/d, only 710 kb/d above 2024 levels. Thereafter, accelerating declines in gasoline and diesel use outweigh growth in naphtha and jet fuel.

Despite tepid demand growth projections, 4.2 mb/d of new refining capacity is expected globally by 2030, partly offset by 1.6 mb/d of closures. Even so, net capacity growth is set to far exceed refined product demand, with increases in Asia, especially China and India, outpacing shutdowns in Europe and the United States. This indicates that more capacity will have to shut, with high-cost plants in Europe and on the US West Coast expected to be hardest hit.

Refinery expansion and closures, and demand growth, 2024-2030



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Note: Refined product demand net of CTL/GTL, additives, biofuels, NGLs and direct use of crude.

Rising oil demand in Asia boosts long-haul oil imports

With European and US refining activity set to slow over the remainder of the decade, and with oil supply in the Americas continuing to expand, the Atlantic Basin crude surplus is expected to widen by approximately 870 kb/d to 7.1 mb/d by 2030. Atlantic Basin product exports are poised to rise by 320 kb/d to 3.5 mb/d, as regional demand for refined products contracts more rapidly. Meanwhile, East of Suez refinery activity will not keep up with the rise in refined product demand, marginally widening the product supply gap. Even so, the Middle East is projected to contribute an additional 860 kb/d to global product supply through 2030, further consolidating its role as a key export-oriented refining hub.

Demand

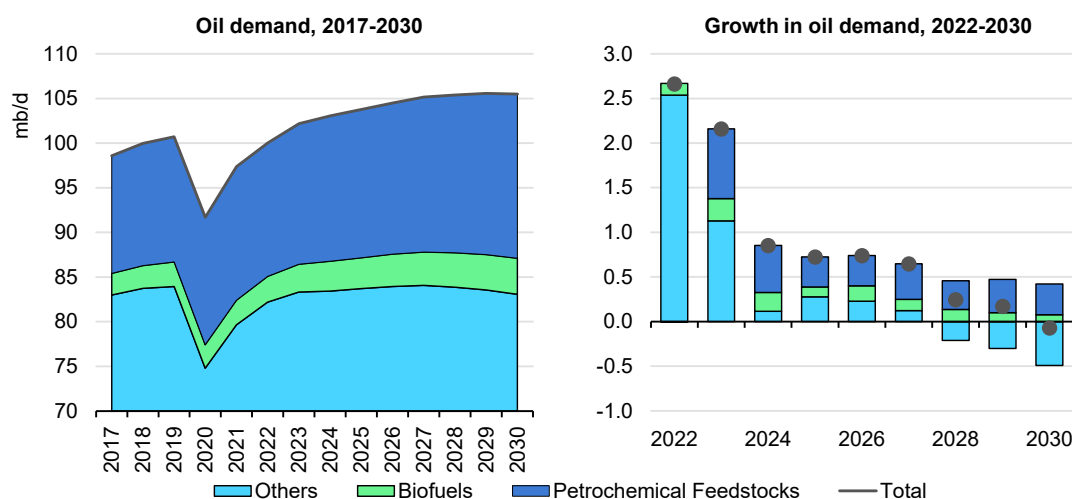
Global summary

World oil demand plateauing amid structural changes

Global oil demand is forecast to increase by an aggregate 2.5 mb/d over our 2024-30 outlook period, to reach 105.5 mb/d by the end of the decade. However, these gains are strongly front-loaded, with growth slowing from around 720 kb/d in 2025-26 to a mid-period plateau, before entering a narrow contraction in the final year. Subpar economic growth and the accelerating substitution away from oil in the transport and power generation sectors are the main structural drivers of this slowdown.

Petrochemicals, led by the steady increase in global natural gas liquids (NGLs) supply, and a continued shift towards biofuels account for the overwhelming majority of growth from 2026 onwards. As a result, demand for oil from combustible fossil fuels – which excludes petrochemical feedstocks and biofuels – looks set to plateau at around 84 mb/d. As a consequence, CO₂ emissions from oil use are now forecast to hit an apex by 2027.

World oil demand forecast to plateau this decade



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Emerging economies will account for the entirety of additional demand growth. Amid broad expansion across the product spectrum, non-OECD oil demand is set

to rise throughout the forecast period, by an average 1.2% per year, for total gains of 4.2 mb/d. Asian countries will dominate growth, with India's 1 mb/d increase the largest of any country by far.

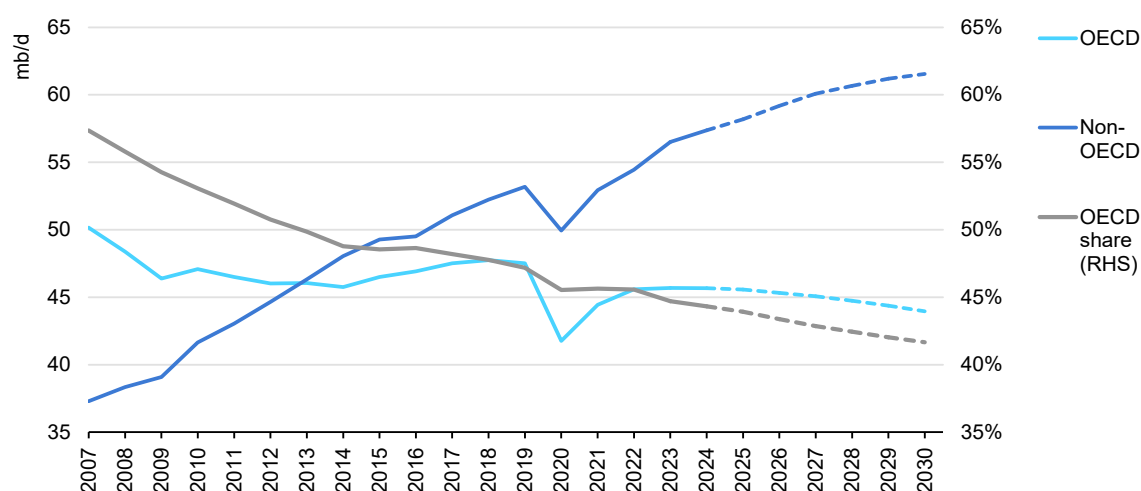
World oil demand by region, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
North America	24.9	21.9	23.7	24.3	24.6	24.5	24.6	24.5	24.4	24.3	24.1	24.0	-0.4%	-0.6
S&C America	6.7	5.7	6.3	6.6	6.7	6.8	6.9	7.0	7.1	7.2	7.4	7.4	1.4%	0.6
Europe	15.8	13.7	14.4	14.9	14.8	14.9	14.9	14.8	14.6	14.5	14.3	14.1	-0.8%	-0.7
Africa	4.2	3.9	4.4	4.5	4.6	4.6	4.8	4.9	5.0	5.2	5.3	5.4	2.9%	0.9
Middle East	8.9	8.3	8.6	9.1	9.2	9.2	9.4	9.5	9.6	9.6	9.4	9.2	-0.1%	0.0
Eurasia	4.3	4.0	4.3	4.4	4.4	4.3	4.4	4.4	4.5	4.6	4.6	4.6	1.1%	0.3
Asia Pacific	36.0	34.2	35.6	36.3	38.0	38.6	38.9	39.4	39.8	40.1	40.4	40.7	0.9%	2.0
World	100.7	91.7	97.4	100.0	102.2	103.0	103.8	104.5	105.1	105.4	105.6	105.5	0.4%	2.5
Annual change	0.7	-9.0	5.7	2.7	2.2	0.9	0.7	0.7	0.6	0.2	0.2	-0.1		

Conversely, OECD oil consumption, having completed its post-pandemic rebound in 2023, will decline by a net 1.7 mb/d from 2024 to 2030. Progress in the region's vehicle fleet transition and structurally weak economic growth weigh on transport and industrial fuels alike, with gasoil and gasoline consumption falling by around 1 mb/d each. Only LPG/ethane and jet/kerosene will be able to escape the general bearishness, with annual growth rates of around 1% each. As a consequence, the group's share of global oil use will fall to 42% in 2030, from 44% in 2024, having dropped below 50% in 2013.

Underscoring the ever-growing importance of the petrochemical sector, naphtha and LPG/ethane will account for 24% of total oil demand in 2030 compared to 22% in 2024 and 20% in 2019, LPG/ethane's growth will be the highest among products in both regions.

OECD and non-OECD oil demand, 2007-2030

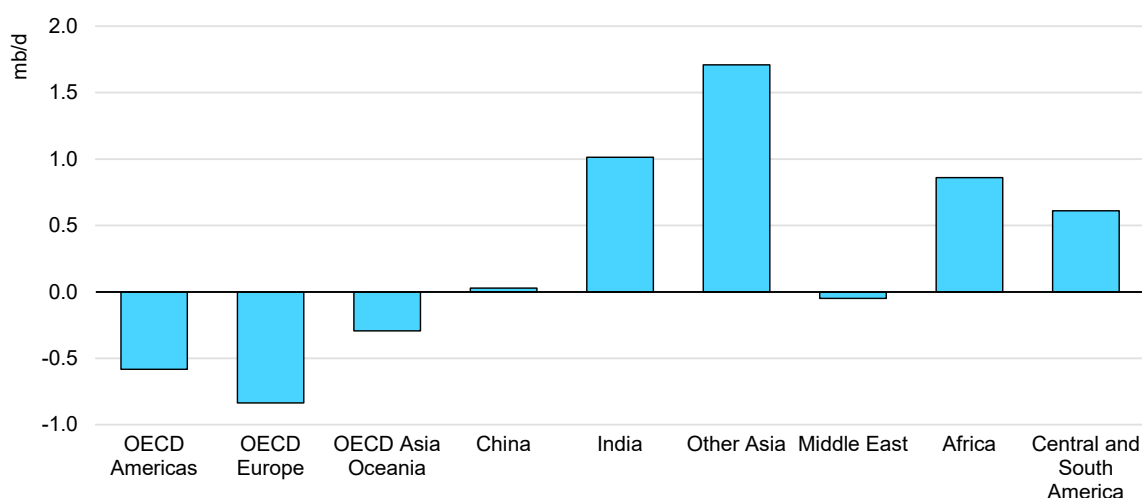


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While the overall global demand growth trajectory is broadly similar to last year's [Oil 2024](#), this masks offsetting shifts in individual countries, especially the United States, the People's Republic of China (hereafter, "China") and Saudi Arabia. In our current forecast, Chinese oil demand will rise by a very modest 30 kb/d, concluding the decade at 16.7 mb/d, compared to the 18.1 mb/d expected last year. This downward adjustment mainly reflects EV sales charging ahead at breakneck speed, eclipsing last year's assumed pace, driven by new policy initiatives introduced to hasten the uptake of EVs as part of the country's wider economic stimulus measures announced in 2024. By contrast, US oil demand in 2030 has been upgraded by 1.1 mb/d to 20 mb/d, as a faster GDP growth outlook this year (2.1% vs. 1.7%) and lower pump prices combine with a significant loss of projected momentum in EV adoption to boost gasoline demand. In addition, the substitution away from oil in power generation is accelerating, led by Saudi Arabia. The country's aggressive plans to phase out oil use in its electricity sector in favour of natural gas and renewables will slash oil burn in power generation by 1 mb/d over the forecast, offsetting growth in other products and resulting in Saudi Arabia's consumption falling by 620 kb/d – the largest decline in oil demand in absolute terms for any country in the 2024-30 outlook.

Foremost among the assumptions underlying our demand forecasts are economic growth, oil prices and EV adoption. For each of these factors, the recent escalation in trade tensions augurs a protracted period of heightened uncertainty. This pertains especially to the unsettled global regulatory and policy environment, with domestic and international governing directives, such as EV subsidies and tariff measures, affecting all three concurrently. Considering projected global oil consumption's relatively limited gains, even minor changes in these variables would materially shift the timing and gradient of this trajectory.

Growth in world oil demand by selected countries and regions, 2024-2030



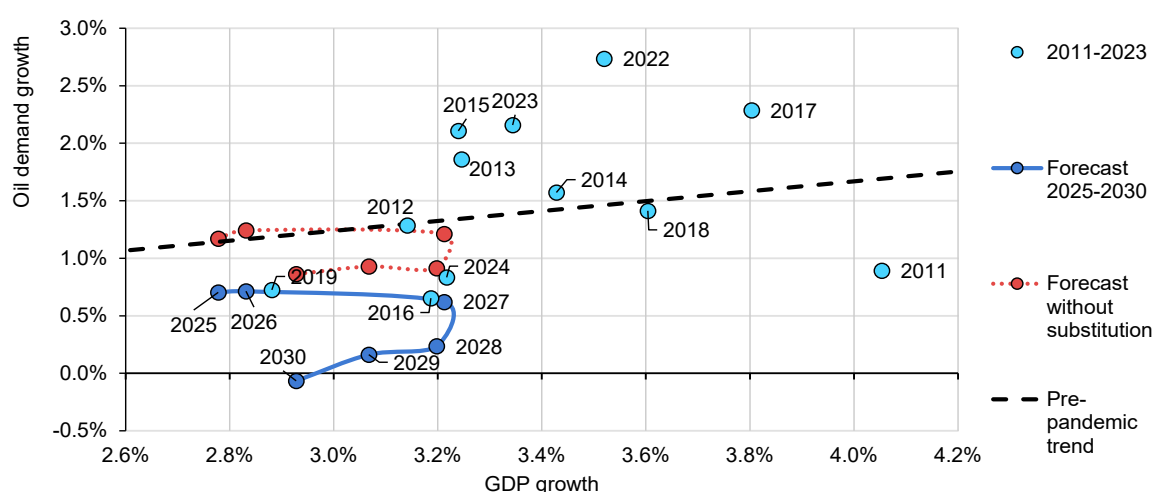
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GDP still drives oil demand but decoupling already underway

Our GDP outlook, with average global growth of about 3.0% over the forecast period, incorporates a significant regional disparity, with the OECD's annual rate of 1.8% only half the non-OECD's 3.9% pace. The global rate is almost half a point below the 2010s trend, as adverse demographics and deglobalisation undermine the outlook for economic growth and trade. This deceleration is especially sharp in China – its annual GDP expansion is almost four points lower than in the 2010s – amid myriad structural demographic and economic challenges.

With the pandemic period (when public health policy upended oil demand) now concluded, GDP has reasserted its traditional role as the key determinant of oil use – albeit only briefly. Oil consumption increases in 2025 and 2026 are roughly in line with the level implied by (subpar) GDP growth. However, this long-standing relationship subsequently diminishes, with oil demand forecast to flatline and then move towards a small contraction by the end of the decade as substitution away from oil in transport and power generation increasingly offsets GDP's contribution.

Growth in global oil demand and GDP, 2011-2030



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Note: Excludes 2020 and 2021 due to Covid-19 distortions.

Source: IEA analysis based on data from Oxford Economics.

Emerging Asia and petrochemicals power global gains

The substitution of oil by electrification is particularly prominent in the transport sector, where liquid fuels consumption is already plateauing. Despite slowing penetration in some advanced economies, global electric car sales will continue to increase sharply as China's extraordinary EV sales momentum consolidates. Rising EV use is forecast to displace more than 5 mb/d of gasoline and diesel consumption globally by 2030. Post-Covid behavioural shifts further depress the outlook for oil demand in transport. Teleworking, now entrenched in advanced

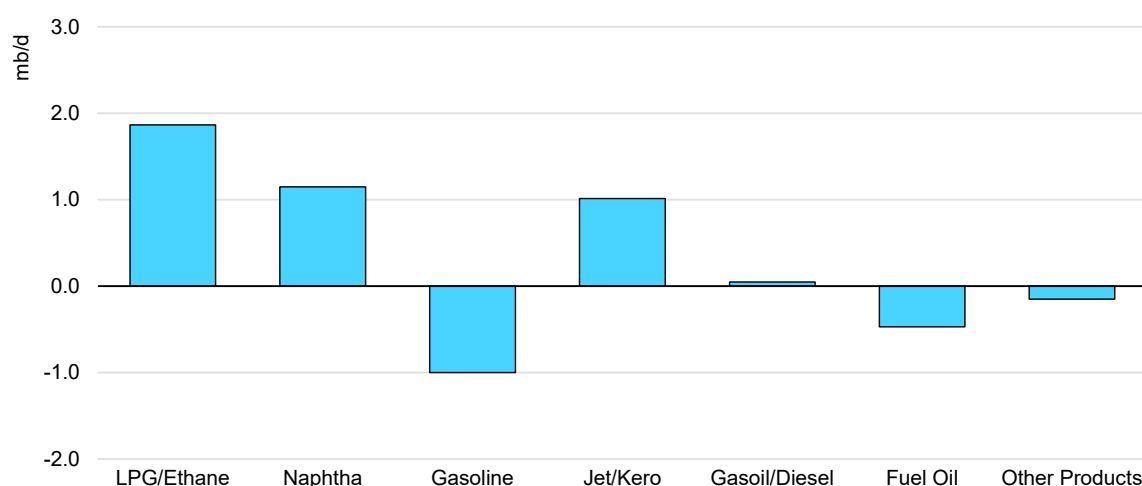
economies, saved around 800 kb/d in transport fuel during 2024, compared with 2019. Public transport use has been similarly transformed by the pandemic, with mass transit ridership remaining well below 2019 levels in many major western cities. Here too, China's post-reopening mobility boom eclipses the recovery in other countries, with its public transport rebounding at a much faster pace.

World oil demand by product, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	13.2	13.3	13.6	14.2	14.6	15.1	15.4	15.6	16.1	16.4	16.7	17.0	2.0%	1.9
Naphtha	6.7	6.6	7.0	6.8	7.1	7.3	7.5	7.8	7.9	8.1	8.2	8.4	2.5%	1.2
Gasoline	26.9	23.9	26.0	26.6	27.1	27.2	27.4	27.3	27.2	27.0	26.7	26.2	-0.6%	-1.0
Jet/Kerosene	7.9	4.6	5.2	6.1	7.2	7.6	7.7	7.8	8.0	8.2	8.4	8.6	2.1%	1.0
Gasoil/Diesel	28.7	26.7	28.2	28.7	28.4	28.3	28.3	28.4	28.5	28.5	28.4	28.3	0.0%	0.0
Residual fuel oil	6.2	5.8	6.3	6.5	6.5	6.5	6.5	6.5	6.4	6.3	6.2	6.1	-1.2%	-0.5
Other products	11.1	10.8	10.9	11.2	11.2	11.0	11.0	11.0	11.0	11.0	11.0	10.9	-0.2%	-0.2
Total products	100.7	91.7	97.4	100.0	102.2	103.0	103.8	104.5	105.1	105.4	105.6	105.5	0.4%	2.5
Annual change	0.7	-9.0	5.7	2.7	2.2	0.9	0.7	0.7	0.6	0.2	0.2	-0.1		

Oil use in aviation and marine shipping has so far been relatively insulated from the direct substitution of electrification and other non-oil alternatives impacting the road fuels. However, a reasonably constructive underlying demand outlook for both sectors will be substantially eroded by fuel efficiency gains. As a result, jet/kerosene use may not regain its pre-Covid levels until 2027 – four years after flight activity passed the same mark. At the same time, emission standards mandated by the United Nations' (UN) International Maritime Organization (IMO) will cause bunkering growth to flatline.

Growth in world oil demand by product, 2024-2030



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On a sectoral level, petrochemical feedstocks are set to continue their domination of global demand growth. Subject to fewer structural headwinds than the major

transport fuels, we estimate that the production of products like polymers and synthetic fibres will require 18.4 mb/d of oil in 2030, or more than one in every six barrels used globally. Almost half of the uplift will come from ethane or LPG and the petrochemical sector remains the major end user of burgeoning NGLs supply.

Feedstocks will also be the major growth catalyst of China's oil demand, with their 1.1 mb/d increase offsetting a roughly equivalent contraction in gasoline. Accordingly, Chinese oil use will barely grow over the forecast period – a far cry from the 2010s, when the country's oil demand grew by 6% annually. Instead, India will lead global gains, rising by 1 mb/d over the forecast. However, India will fall well short of attaining China's spectacular pace of expansion, with its per capita GDP and oil consumption remaining at a fraction of its northern neighbour.

Fundamentals

A new era of protectionism tempers GDP growth

Amid structural headwinds such as ageing populations, increased protectionism and stretched government finances, global GDP growth is projected to average near 3% over 2025-30 – almost half a point lower than during the 2010s. Emerging economies will account for the bulk of this expansion, with average non-OECD growth of 3.9%, compared to 1.8% for OECD countries. By 2030, the non-OECD share of global GDP will climb to 59%, from 56% in 2024.

Well before the United States shifted to strongly protectionist trade policies in early 2025, the past decade had already witnessed a sharp increase in measures hampering trade and investment flows amid a broader process of deglobalisation. The ratio of cross-border trade and investments to GDP has essentially stalled over the past decade amid a global retreat from free trade towards mercantilist policies, tariffs and subsidies. Besides the direct stagflationary impact of higher tariffs and other restrictions to exports and imports, the accompanying uncertainty acts as a deterrent for trade and investment over the forecasting period. A case in point is the [Baker, Bloom and Davis Global Economic Policy Uncertainty Index](#) surging to the highest level ever in April 2025.

Before the latest uncertainty surrounding tariffs took the spotlight this year, the global economy had displayed considerable resilience in the wake of the unprecedented central bank interest rate hikes of 2022-23. Post-pandemic demand and supply shocks had propelled inflation in developed economies to 9-10% by mid-2022 – a 40-year high. Inflation has since made a retreat, and most major central banks have now pivoted to an accommodative monetary stance. The main exception in this regard is the United States, where easing has, for now, come to a halt due to inflation proving stickier than initially expected. However,

with breakeven inflation rates derived from bond markets pricing annual US inflation in at around 2.4% over a five-year horizon, it is not yet at risk of becoming psychologically entrenched.

Still, a range of structural factors may pressure inflation upward and weigh on growth in the longer term, making for a stagflationary cocktail and conclusively putting an end to the low-interest rate environment that prevailed for most of the past two decades. At the heart of these issues are adverse demographics in developed countries, where an ageing population is set to depress labour productivity, strain fiscal balances and boost public debt – the latter aspect already stretched by ballooning spending for defence, healthcare and social welfare. Spiralling budget deficits and unsustainable fiscal policies are increasingly burdening economic growth, as a greater share of government revenues is allocated to debt refinancing and interest rate payments. In this regard, the International Monetary Fund (IMF) has pointed to the urgent priority of tackling an “unforgiving combination of low growth and high debt” as global public debt reached a record USD 100 trillion in 2024.

In parallel, geopolitical tensions are reshaping trade patterns. Besides conflicts in Europe and the Middle East, China is increasingly the focal point amid accusations of unfair trade practices as the country essentially exports its domestic economic imbalances. Additionally, China’s territorial claims have led to progressively fraught relationships with its neighbours. Trade tensions with the United States have risen in parallel, concentrated in technology and intellectual property.

Real GDP growth assumptions

	2011-2019	2023	2024	2025	2026-30	Vs 2010s
United States	2.4%	2.9%	2.8%	1.2%	2.3%	-0.1%
Europe	1.9%	1.4%	1.7%	1.4%	1.5%	-0.4%
Japan	0.9%	1.5%	0.1%	0.8%	0.2%	-0.6%
China	7.3%	5.4%	5.0%	4.1%	3.8%	-3.5%
India	6.8%	8.9%	6.6%	6.5%	6.4%	-0.4%
Africa	3.6%	2.7%	3.3%	3.7%	3.9%	0.2%
OECD	2.1%	1.8%	1.7%	1.2%	1.9%	-0.2%
Non-OECD	4.7%	4.6%	4.4%	4.0%	3.9%	-0.7%
World	3.4%	3.3%	3.2%	2.8%	3.0%	-0.3%

Source: Oxford Economics.

Assumptions about GDP growth and its impact on oil demand are a key component of our estimates. The average GDP elasticity of global oil demand used for the modelling (before substitution effects) is about 0.3 for OECD and 0.6 for non-OECD, reflecting the greater oil-intensity of emerging markets.

Additionally, expectations of future oil prices are paramount for demand estimates, with forecasts sensitive to both the absolute price level and intertemporal price changes. Estimates are based on the ICE Brent forward price curve (rising from

USD 64/bbl in 2026 to USD 67/bbl in 2030). For 2025, prices are prorated between realised Brent prices and the forward curve. These are then discounted to real terms.

GDP in the **United States** is assumed to grow by an average 2.1% over the forecast period, building on its strong post-Covid recovery. The country continues to outperform other developed economies, with growth and productivity buoyed by a healthy demographic profile, flexible labour market, consumer-led economy, supply-side dynamism and technological innovation. Conversely, unsustainable fiscal spending, surging national debt and uncertainty regarding trade policies act as headwinds. The 2024 budget deficit topped USD 1.8 trillion, or 6.4% of GDP, while publicly held federal debt of USD 28 trillion approaches 100% of GDP. This is expected to keep inflation elevated and monetary policy relatively restrictive, causing GDP growth to decelerate from 2.6% in 2028 towards 2% at the end of the forecast period.

The long-ailing **euro zone** is set to gradually emerge from the stasis that characterised 2023-24 when members' economies barely expanded. Growth will average 1.5% over the forecast period, led by Germany's recovery to its trend rate of around 2%. In this regard, the country's recent commitment to revive its moribund economy through fiscal spending looks to be a game changer. With inflation having recently decelerated towards the European Central Bank's 2% target, accommodative monetary policy will be a key tailwind, boosting real incomes and domestic demand. However, this makes for a still-subdued pace of expansion, about half a point below the 2010s average, as the bloc faces the spectre of so-called 'Japanification'. Amid low productivity and grim demographics, the region's working-age population is about to enter contraction. Growth will be led by the service sector, especially in Southern European economies, where surging tourism is a major source of support. By contrast, industrial production, which has been in decline for more than three years, remains muted – well below pre-Covid levels – amid soft external demand. Protectionist US policies and increased competition from China may crowd out EU countries in global trade as its competitiveness erodes.

Japan's expansion will gradually subside from around 1% mid-decade towards zero at the end of the forecast period, weighed down by chronically low productivity and dire demographics as an ageing population translates into a shrinking labour force.

China's growth will decline steadily over the outlook period as the country faces myriad structural headwinds. GDP expansion will slide to 3.7% in 2030 – half the 2010s pre-pandemic trend. Besides a shrinking population (set to decline by 1.3% between 2025 and 2030), a protracted property slump and massive overcapacity, there is the ever-present risk of a deflationary spiral. China's

export-led industrial model in solar panels, automobiles and batteries is increasingly coming under threat, with important trading partners reluctant to absorb the cheap exports with which the country is flooding global markets. As consumer spending remains cautious, authorities have stepped up fiscal and monetary stimulus. However, this will not be sufficient to structurally revive sluggish domestic activity.

India's stellar expansionary trajectory will continue essentially unabated, with GDP growth averaging 6.4% over the forecast period. This is the highest by far of any major economy, propelled by structural advantages such as an expanding middle class with growing spending power and young population dynamics. Additionally, improved infrastructure will help to boost mobility and car ownership. Oil demand will grow at a relatively fast rate as changing spending patterns, urbanisation and industrialisation make India's economy more energy intensive.

Economic and technological shifts slow the rise in per capita fuel use

Perhaps the single most important driver of oil demand is economic growth. While higher GDP typically leads to increased oil consumption, this connection can be uneven and changes over time. The relationship between demand for the three major transport fuels (gasoline, gasoil and jet fuel) and GDP is illustrative in this respect.

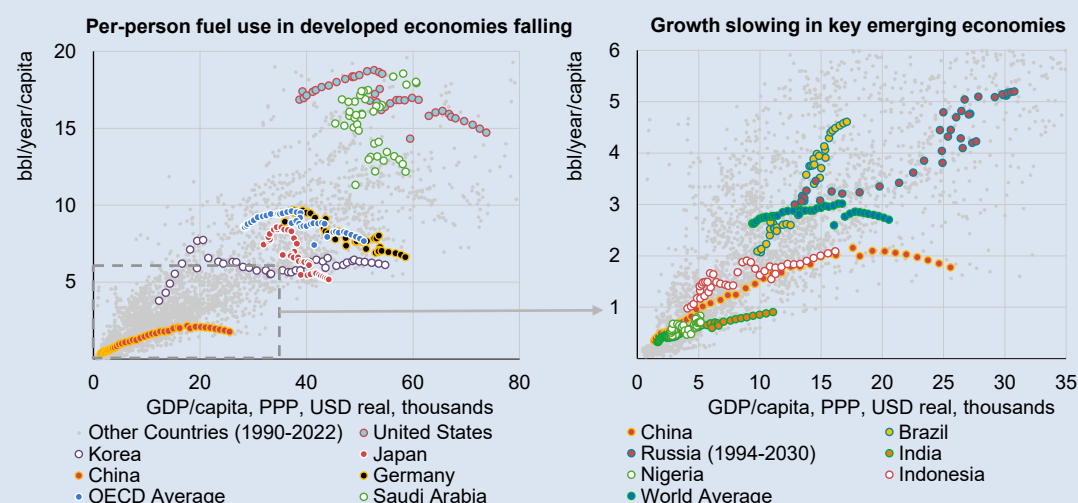
Rising utilisation of fuels produced from oil have historically played a key role as GDP per capita increases. Gasoil use is crucial in expanding the industrial and extractive sectors and the establishment of associated logistic networks involved in an economy moving to middle income status. The fuel is also essential in construction and related fixed capital formation. Gasoline and jet fuel demand goes up significantly as middle class disposable incomes, car ownership and opportunities for leisure travel rise.

Over time, as the service sector begins to comprise a larger piece of the economy, the share of growth in fuel-intensive sectors declines. The amount of personal travel is also fundamentally limited by time and distances so growth in distance driven per person slows at higher incomes.

The United States has comparatively steep fuel use per capita, amongst the highest of all major economies and more than double that of Germany, Japan or the United Kingdom. This results from a combination of several factors. The United States has very high average incomes combined with a low population density and sprawling cities. It also has a comparative lack of well-developed public transport networks. As well, the country's status as a very large oil producer helps to keep consumer prices relatively low while other countries apply substantial fuel taxes.

In these terms, nations like Saudi Arabia, Canada and Australia are more like the United States than other highly industrialised nations in Europe and Asia. Amongst large middle-income economies, it better reflects the direction of travel for Brazil and the Russian Federation (hereafter, “Russia”) than densely populated China and India.

GDP vs. total gasoline, jet/kerosene and gasoil demand per capita, 1990-2030



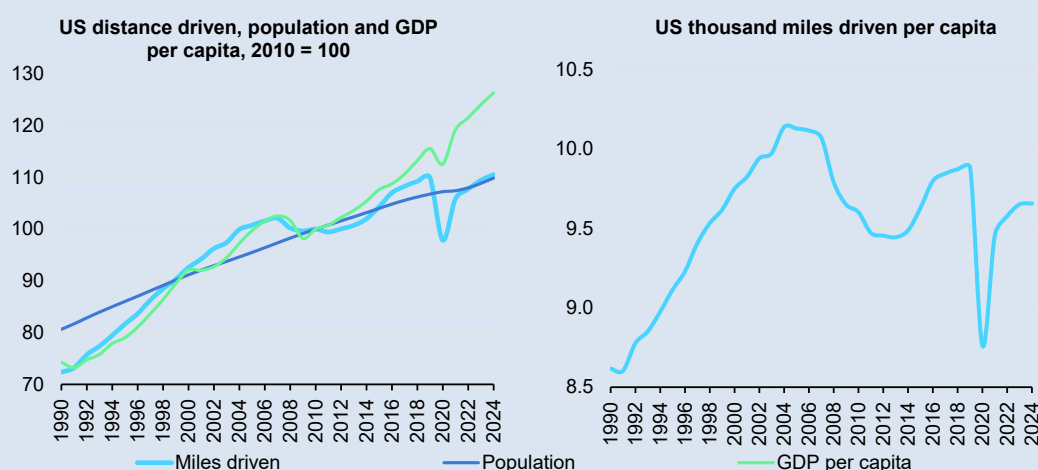
IEA. CC BY 4.0.

Note: PPP = Purchasing Power Parity.

Source: IEA analysis based on data from Oxford Economics.

US fuel demand per capita has been steadily declining since the early years of this century. The same is true for the OECD as a group and comes despite widespread GDP growth per head. Some wealthy non-OECD states, notably Saudi Arabia, also display similar trajectories as other sources of energy are increasingly used in power generation and industry, and the role of less oil-intensive sectors rises.

US vehicle miles travelled (VMT), reported by the Federal Highway Administration, are a key indicator for global oil demand, with US drivers accounting for roughly one-third of all gasoline consumed worldwide. Before the Global Financial Crisis (GFC) in 2008, VMT rose steadily in line with rising incomes. Since then, VMT/capita has gone up, albeit unevenly, slightly lagging overall population growth and well below increases in income. Notably, neither the post-GFC nor the post-Covid rebound returned average driving rates to earlier levels. The 2024 VMT/capita was nearly 5% below pre-GFC levels despite an almost 25% real terms increase in GDP per capita. Higher fuel prices, especially in 2007-08 and 2011-14, likely played a role, as has an ageing population and the significantly increased prevalence of teleworking. With US distances driven reaching what appears to be a saturation point, improving fuel economy and EV substitution are gradually eroding demand.



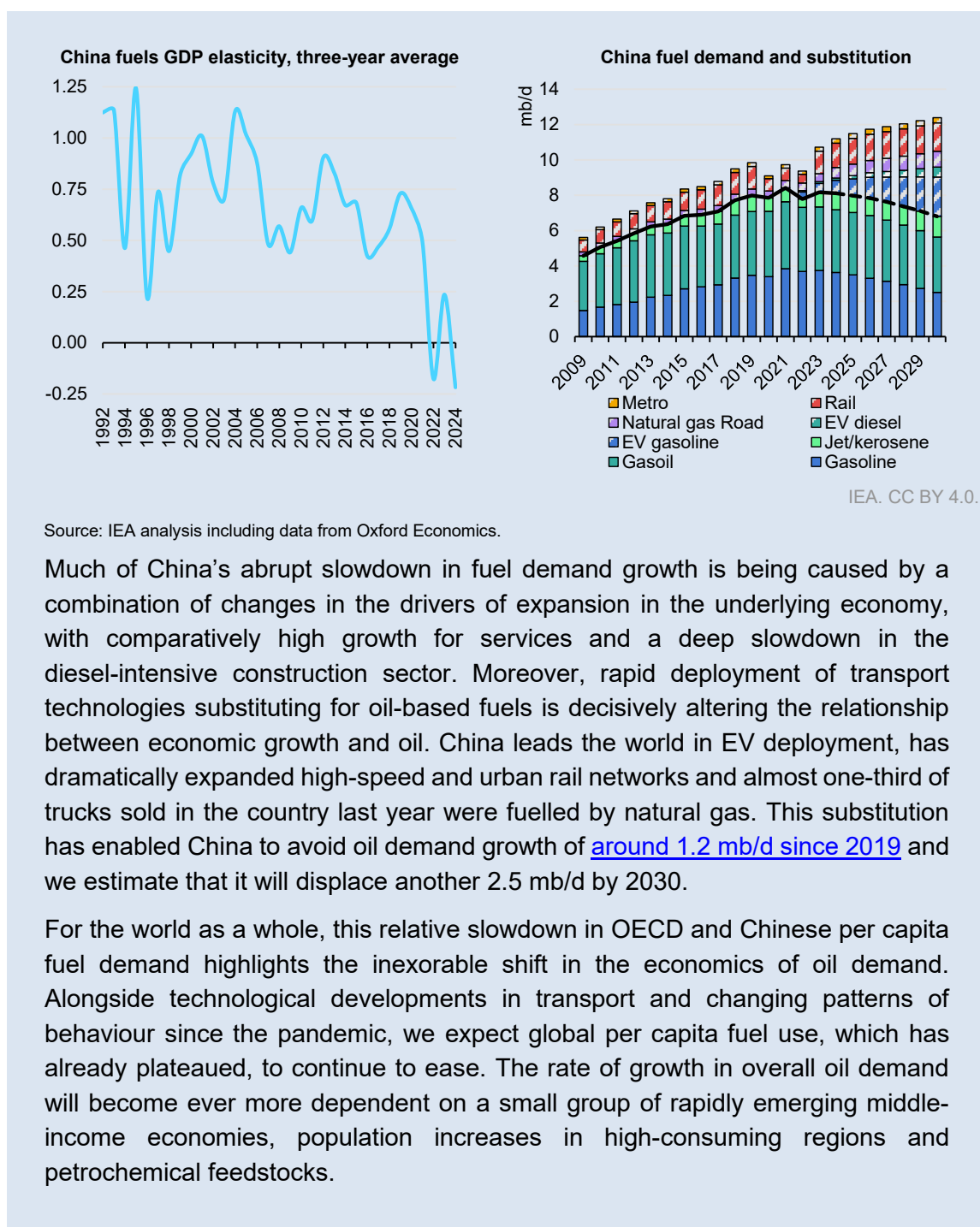
IEA. CC BY 4.0.

Sources: US Federal Highway Administration and Oxford Economics.

In Japan, a combination of a rapidly ageing population, the end of a construction boom and excellent public transport infrastructure (including a world-class high-speed rail network) means that fuel use per person has fallen by more than 25% since 1990, despite a 30% increase in GDP per head. Very strong car fleet efficiency gains (see *Hybrid-driven vehicle fuel efficiency gains cutting Japanese gasoline use*) have added to this brisk downward trajectory. Many of these factors also apply to Korea, which is even more densely populated than its eastern neighbour. Korean per capita GDP has grown spectacularly since 1990, from a level equivalent to present-day Indonesia, to surpass countries like Japan, the United Kingdom and France, centred on an enormous rise in industrial output. Nevertheless, fuel demand per head has fallen by 20% since peaking in 1996. The experiences of Japan and Korea are invaluable reference points for understanding unfolding developments in China and other emerging and developing Asian economies.

Over recent decades, Chinese economic output per capita has increased with remarkable speed, vaulting the country out of a comparatively low average-income range and firmly into middle-income status. Between 1990 and 2024 China's real GDP per capita (PPP) increased by more than 14 times. No major economy comes close to growth on this scale. For example, Indian GDP per head rose by just under five times and Korea's less than four. Perhaps unsurprisingly, China has been the lynchpin of world fuel demand growth through most of the period, with 5.7 mb/d of demand added this century alone, more than one-third of the global total.

While impressive, this fuel demand growth is on a slightly smaller scale than might be implied by the GDP gains. For most economies, the ratio of fuel to GDP, or GDP elasticity, is around 50% and can be much higher in middle-income countries. In China, this was just below 37% for 1990-2024 on average, falling to zero in recent years amid growing substitution in transport.



Road fuel demand

EVs continue to erode gasoline and diesel consumption

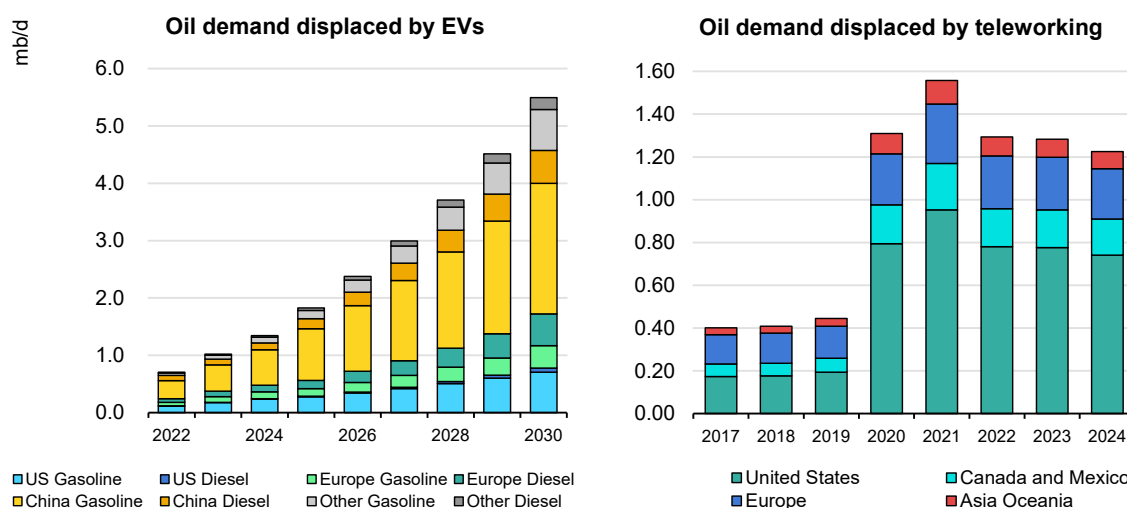
Global electric car sales will continue their remarkable growth trajectory, with EVs set to displace a total of 5.4 mb/d of global oil demand by the end of the decade, up from 1.3 mb/d saved in 2024. EV sales exceeded 17 million in 2024, and are

expected to surpass 20 million in 2025, representing around one-quarter of all cars sold globally, according to the IEA's [Global EV Outlook 2025](#).

However, these headline numbers disguise significant disparities across regions, with China's buoyancy contrasting markedly with slowing momentum elsewhere. Chinese EV sales were up nearly 40% last year, accounting for almost two-thirds of global sales. Almost half of Chinese cars sold were electric, underpinned by strong government policy support and a wide range of competitively priced models.

Sales elsewhere were less stellar last year, with stagnant growth in Europe and only a modest 10% y-o-y increase in the United States – down from 40% in 2023. In many advanced economies EVs have had trouble extending their appeal beyond urban, environmentally-conscious motorists. High prices in both absolute terms and compared to conventional cars, concerns about the lack of charging infrastructure and driving range, as well as falling second-hand values have kept some buyers on the sidelines. The phasing out or reduction of subsidies in some countries has also added to buyer reluctance. Accordingly, the global EV narrative remains predominantly a Chinese story, with its mass market adoption contrasting with EVs comparative niche status elsewhere.

Oil demand displacement by EVs and by teleworking



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By and large, these regional disparities, although becoming slightly less pronounced, will persist over the forecast period, with China's dominance set to continue. In 2030, the global share of electric car sales is set to exceed 40%. EVs will represent 80% of Chinese sales, with the corresponding shares for Europe (60%) and, especially, the United States (20%) more modest. China will account

for about half of the projected 5.4 mb/d oil demand savings by 2030, with Europe and the United States each capturing about 14%. About three-quarters of the reductions in demand will be in gasoline, with the remainder in diesel.

Considerable uncertainty surrounds our outlook, not least due to the volatile regulatory landscape. This applies to both domestic markets (e.g. US tax credits could disappear under recently proposed legislation) and internationally. China's dominance of global clean technologies has created frictions with its trading partners, amid accusations that, backed by disproportionate government aid, the country is exporting its structural overcapacity. Additional EU and US tariffs that came into effect last year are a case in point.

Working-from-home jobs transform gasoline demand outlook

Five years after the Covid-19 pandemic created an acute need for teleworking, working-from-home (WFH) employment has become firmly established in western economies – particularly in the United States. We see the aggregate impact of post-pandemic work-related behavioural changes having reduced global road fuels demand by around 800 kb/d compared to 2019 levels, for a total displacement of around 1.2 mb/d, with this level of savings holding roughly steady through our forecast period to 2030. This reduction is dominated by the United States, where almost 800 kb/d of fuel use is being avoided.

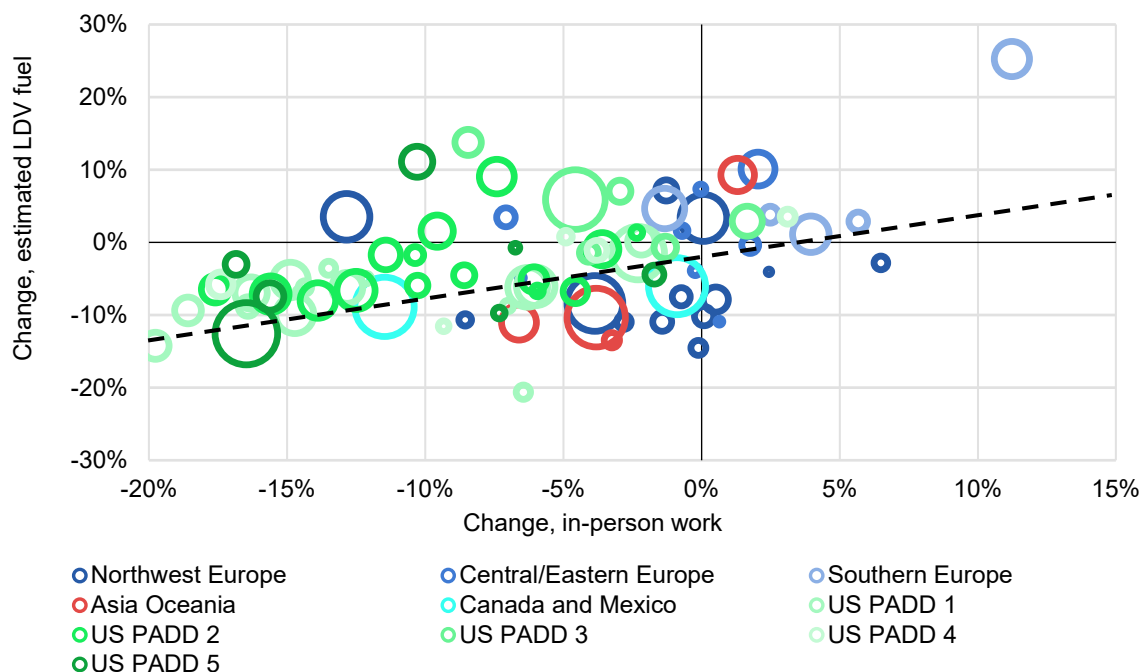
After soaring to 60% at the height of 2020 lockdown restrictions (from below 10% pre-Covid), WFH's share of paid workdays in the United States has remained fairly stable from 2022 onwards, at around 25-30%. US office vacancy rates – currently at a record high of 13.9% and up from 9.3% in 2019 – also reflect this transformation. WFH has proved popular with employees keen to improve their work-life balance by avoiding long commutes. Financial savings for costly travel as well as living in lower-cost areas are other incentives. Amid large differences across age groups, incomes and industries, WFH is more widespread with younger and better-educated urban employees in information technology, finance and the media. This contrasts with sectors such as hospitality and manufacturing where work cannot easily be performed remotely.

Still, hybrid work's ascension to a new normal may be somewhat premature. In 2024, there were numerous reports of large companies reversing their stance on remote and hybrid work, calling corporate staff back to the office five days a week. These employers stepped up their efforts to reclaim control amid worries that remote work weighs on productivity and innovation, with employees disconnected at home seen as less reactive and approachable. This issue remains to be settled, with studies of changes in productivity and teleworking over 2020-23 not showing

a significant correlation. Still, the recent cooling of the US labour market, especially in the tech industry, has buttressed employers' bargaining power to roll back remote work.

However, this return to office reversal among large companies does not yet show up in teleworking statistics. The number of US firms requiring five days in the office has actually fallen by six points over 2024, from 38% to 32%, according to data from the research platform [Flex Index](#). Even if 2025 is likely to see tighter policies come into effect, increased teleworking is likely to remain a permanent legacy of the pandemic, with the right to flexible working arrangements now enshrined in law in several European countries. Moreover, ongoing technological advances in videoconferencing may boost WFH further in the longer term.

OECD in-person work and road fuel demand, 2019 vs. 2023



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Notes: LDV = light-duty vehicle. PADD = US Petroleum Administration for Defense District. US PADD 1 = East Coast, US PADD 2 = Midwest, US PADD 3 = US Gulf Coast, US PADD 4 = Rocky Mountain and US PADD 5 = West Coast.

Sources: IEA analysis including data from [US Census Bureau](#), [US Federal Highway Administration](#), [Eurostat](#) and [WFH Research Survey of Working Arrangements and Attitudes](#).

The impact of WFH on gasoline demand has proved most pronounced in the Anglosphere, where it has gained broad cultural acceptance. In fact, cultural individualism emerges as perhaps the main driver for variance in [WFH rates across countries](#), with other factors such as lockdown stringency, population density and GDP per capita also correlating positively to WFH uptake. In early 2025, full-time employees worked from home in excess of 1.5 days per week in the United States, Canada and the United Kingdom, leading other developed

economies, according to the [Global Survey of Working Arrangements](#). In Europe, teleworking is most prevalent in northern cities, with London, Stockholm and Amsterdam reporting the highest share of employees working from home in 2023, with more than 50% of residents teleworking either “usually” or “sometimes”, according to the EU’s Eurostat.

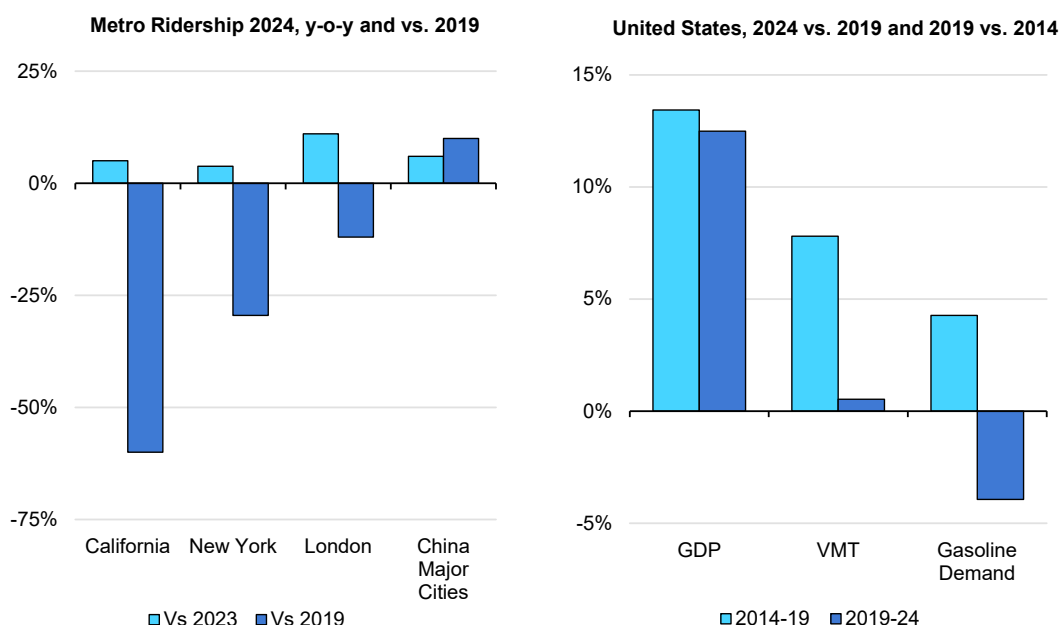
Nonetheless, WFH’s link to driving and road fuels demand is not clear-cut. Research shows that, while teleworking reduces both overall distance and vehicle miles travelled in aggregate, teleworkers make more non-commute trips on teleworking days in their freed-up time that offset a significant portion of the reduction in commuter travel. However, these additional trips are typically shorter than commuting and likely more geared towards non-motorised transport modes such as walking and cycling. Daily US travel metric data from the Bureau of Transportation Statistics shows a growing share of trips of less than one mile. These constituted 23.9% of all trips in 2019 and 27.6% in 2023. Along the same line, the average working population staying at home (i.e. not moving more than one mile away from home) has increased by 11% over this period, from 63.4 million to 70.1 million. Additionally, state-level workforce and VMT data reveal substantial variation within the United States, with states in the Northeast and on the West Coast exhibiting higher levels of teleworking and larger declines in driving and implied gasoline consumption.

In the United States, WFH has also accelerated the disconnect between mobility and macroeconomic variables such as GDP and employment trends, with fewer miles being driven per dollar of output. VMT in the United States have lagged GDP for decades, as driving gradually approached saturation – a development subsequently accelerated by teleworking. While cumulative US GDP growth of 13% between 2019 and 2024 is roughly the same as in the five years before the pandemic, mobility growth, weighed down by WFH, has flatlined post-Covid compared to healthy growth in the preceding five years. Gasoline use has fallen even faster than implied by the reduction in VMT, and moved into contraction in the 2020-24 period, with demand 4% below 2019 levels as fuel efficiencies and EVs reduced gasoline consumption per mile travelled. The estimated fuel economy of new US vehicles produced in 2024 increased to an average of 28 miles per gallon (mpg), from 24.9 mpg in 2019, according to the 2024 [EPA Automotive Trends Report](#) – despite increasing consumer preferences towards larger, heavier and more powerful vehicles.

Even more than reduced vehicle driving, the increased prevalence of remote working is very apparent in public transport use, with 2024 ridership in major European cities still 5-10% below pre-Covid levels. Declines are even larger in the United States, with subway ridership 30% lower in New York and down 60% on California’s Bay Area Rapid Transit system (BART). In cities where public transport competes with commuting by automobile, public transport lost ground to

car journeys during the pandemic as health risk-averse travellers switched to driving. This has resulted in a semi-permanent recalibration between modes of transport.

Mobility indicators, pre- and post-pandemic periods



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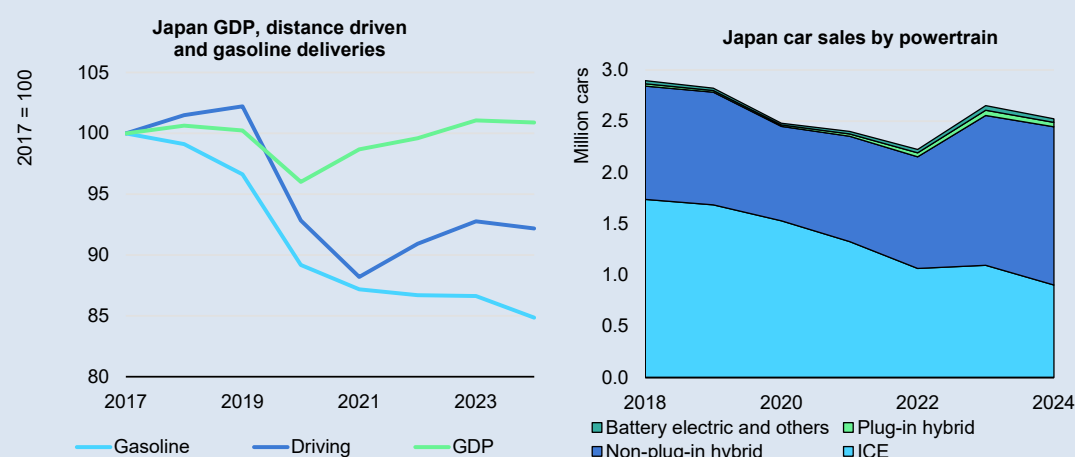
Notes: California = Bay Area Rapid Transit system (BART). VMT = vehicle miles travelled.

Sources: IEA analysis based on data from California's [BART](#) system, [New York City Transit Authority](#), [Transport for London](#), China's [metroDB.org](#), Oxford Economics and [US Federal Highway Administration](#).

China, where mobility displayed an opposite evolution to western patterns, is the main exception in this regard. This is largely due to WFH being far less prevalent in the country, partly resulting from a very different pattern of lockdowns compared with western economies. Short initial lockdowns were followed by severe restrictions in 2022, culminating in an abrupt release of pent-up mobility after the country's reopening in early 2023. Urban transit use was quick to recover, returning to pre-pandemic levels in 2023 and rising 10% above pre-pandemic levels in 2024. Along the same line, railway passenger volumes were 7.4% above 2019 levels in 2024, according to data from China's National Bureau of Statistics (NBS), while domestic flights have exceeded pre-pandemic levels since early 2023, and in 2025 are up by one-third over 2019 levels. International flights were slower to rebound after the 2023 reopening, rising by around 20% compared to 2019. Holiday travel during China's main festivals averaged 15-20% above pre-pandemic levels between 2023 and early 2025. Conversely, highway passenger numbers in 2024 remained around 10% below 2019, according to the NBS.

Hybrid-driven vehicle fuel efficiency gains cutting Japanese gasoline use

Gasoline demand in Japan has consistently fallen for several years and was almost 15% below pre-pandemic levels in 2024. This decline rate represents a sizeable underperformance compared with GDP, which increased narrowly (+0.6%). While reported distances driven have also significantly decreased, down 9% compared to 2017-19, the faster drop in gasoline deliveries reflects the substantial improvements in the car fleet's fuel economy resulting from extensive hybrid vehicle sales.



IEA. CC BY 4.0.

Note: ICE = internal combustion engine.

Sources: [E-Stat.go.jp](https://e-stat.go.jp), Oxford Economics and [Japan Automobile Dealers Association](https://www.jada.or.jp).

Japan has long led the world in the development and deployment of hybrid electric vehicles (HEV). At 61% of total car purchases in 2024, dominated by non-plug-in hybrids, sales were higher than in any other major economy. The result has been efficiency gains of around 1% per year, close to the rate achieved in other OECD countries, but from a more efficient starting point. Conversely, EV sales have been lower than in other developed markets, leaving hybrid engines as the main technological driver of reductions in fuel demand.

The substantial fall in overall driving levels reflects a steady decline in the country's working-age population. This demographic limit on underlying mobility is likely to become a more widespread trend as increasing numbers of developed and middle-income countries grapple with similar problems. Teleworking is a much less important drag on mobility in Japan compared with other wealthy countries. Japan's highly urbanised population and excellent public transport networks also help to mitigate the pass-through from economic growth to kilometres driven.

This combination of reduced driving and 1% per year fleet efficiency improvements has led to an average annual decline in gasoline use of around 2% in recent years. Japan's comparatively stable GDP draws these trends into sharper focus, but similar patterns can be seen elsewhere. In the United States, the largest gasoline

market in the world, deliveries were well below their 2019 level in 2024, despite increased driving, GDP and employment. Fleet efficiency gains across the OECD are also close to 1% per year, but these owe more to tightening standards for ICE vehicles and more substantial and growing substitution by EVs, while more prevalent teleworking has seen mobility lag other macroeconomic indicators.

Petrochemical demand rising strongly

Feedstocks remain the backbone of global oil demand growth

Rising use of oil products as raw materials for the production of plastics and synthetic fibres cemented petrochemical's dominance of overall oil demand growth in 2024, accounting for about three-quarters of net growth. We expect this increase to continue over the remainder of the decade, and beyond, with petrochemical feedstocks' share of total oil consumption rising from 15.8% last year, to 17.4% by 2030. Reaching 18.4 mb/d by the end of the forecast period, feedstock demand will be 2.1 mb/d above 2024 levels – split across naphtha (10.2 mb/d, +1.1 mb/d) and LPG/ethane (8.2 mb/d, +990 kb/d). The rapid increase in LPG and ethane processing, especially in the United States, has been underpinned by burgeoning NGLs supply.

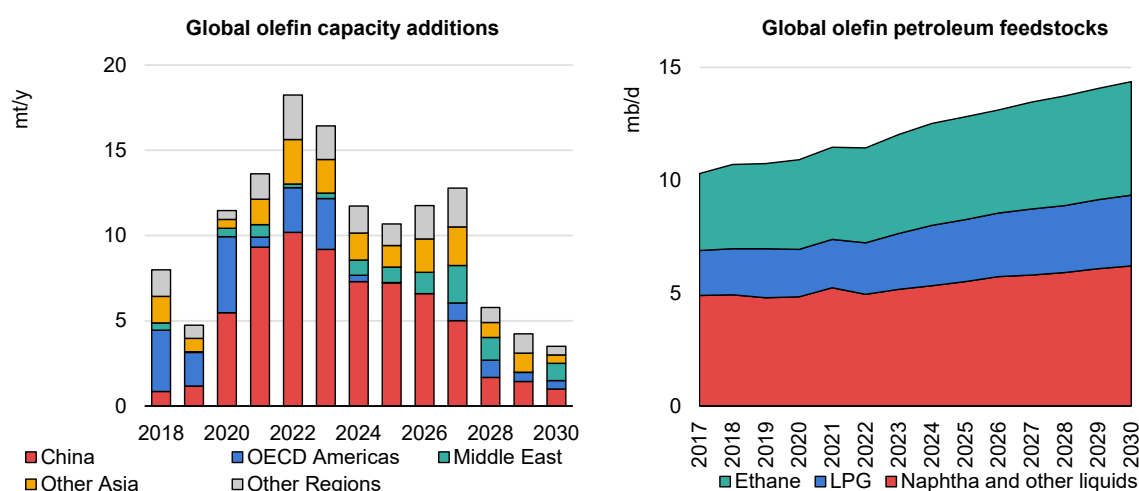
The petrochemical sector was already the overwhelming driver of demand growth during the last five years. Between 2019 and 2024, consumption of oil-based feedstocks rose by an estimated 2.3 mb/d. This is equivalent to more than 95% of the net increase in overall demand and petrochemicals' share of total oil use increased markedly during the pandemic.

Underlying use of plastics and fibres – for sectors like packing, manufactured goods and construction – will continue to rise steadily, especially in emerging market countries. This is driven by trends like urbanisation, increasing average incomes and higher consumer spending. Growth in online delivery services and new manufacturing industries (including clean energy technologies such as EVs and solar photovoltaic panels) also boost feedstock requirements. We estimate an average annual 2024-30 increase of about 2.1% in oil consumed in the petrochemical sector.

The preponderance of petrochemical inputs in total oil demand growth does not result from a profound acceleration in the consumption of polymers. Rather, it reflects steady growth, in contrast to the structural slowdown underway in key transportation sectors. Plastics use is not subject to factors comparable to vehicle efficiency gains or widespread substitution effects on a scale similar to the electrification of transport. A more rapid rise in the amount of recycled material

substituting for virgin polymers could begin to dampen the connection between growing demand from key applications – which include packaging, manufactured goods or construction – and final oil demand. However, while plastic waste collection is improving in many countries, we do not expect this to significantly limit oil use during the forecast period.

Olefin production, capacity and feedstocks, 2017-2030



IEA. CC BY 4.0.

Sources: IEA analysis of data from S&P Global and ICIS.

Chinese and US investments bearing fruit

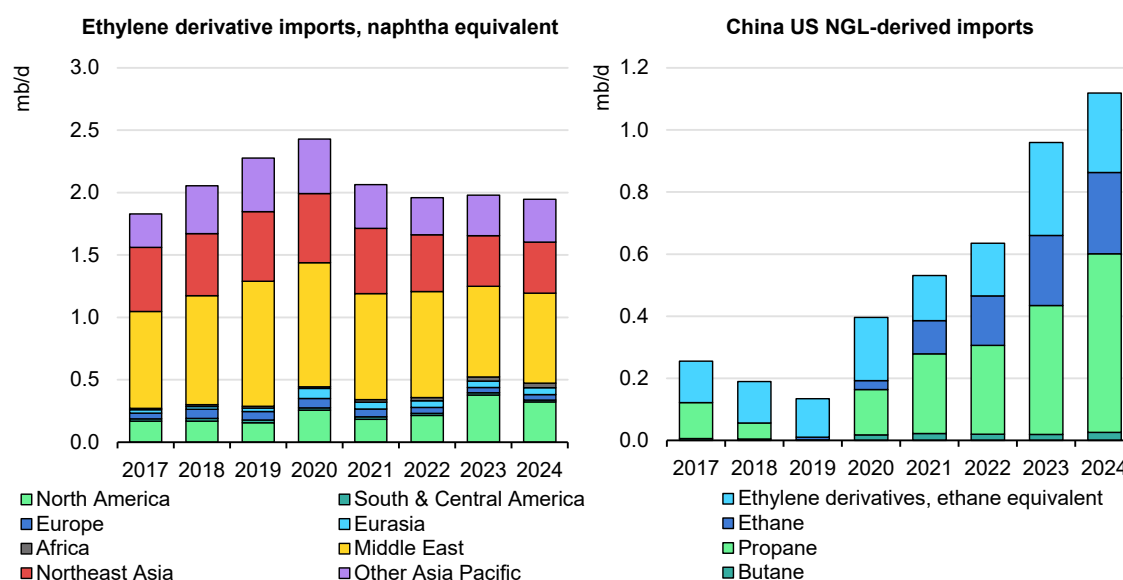
The central role of petrochemicals in oil demand growth is mirrored by the role of China in the sector itself. A major wave of capacity additions, still ongoing, has seen a 2.2 mb/d rise in oil-derived feedstock intake for Chinese plants from 2019 to 2024. This increase, close to the global rise over the same period, saw a 15% (330 kb/d) drop in feedstock-equivalent petrochemical imports, according to data from ICIS.

China's new capacity has come in the form of a combination of refinery-integrated naphtha crackers, alongside coastal propane dehydrogenation (PDH) plants and steam crackers based on imported NGL feedstocks. Therefore, the 2019-24 increase in feedstock demand has been split between naphtha (+1.3 mb/d) and LPG/ethane (+890 kb/d). Since naphtha crackers produce about one tonne of propylene for every two tonnes of ethylene and PDH plants produce only propylene, this also means that national petrochemical output of both ethylene (roughly +19 million tonnes per year (Mt/yr), or +110%) and propylene (about +17 Mt/yr, +130%) went up dramatically over the five years to 2024. Other recent waves of capacity additions, for example in the United States and Middle East, have focussed on monetising local ethane supply. Ethane cracking produces very little propylene, offering naphtha crackers some respite when competing in

international markets. However, dramatically increased Chinese propylene output has now become a key contributor to reduced output in regions like Europe and the rest of East Asia and enabled the higher run rates at export-focussed US ethane crackers.

In addition to new capacity, patterns of petrochemical demand growth have been shaped by changing feedstock availability. US output of NGLs has risen by more than any other source of supply in recent years and this material has enabled the major increases in US and Chinese petrochemical intake of LPG and ethane. US ethane demand has gone up by 790 kb/d from 2019 to 2024. Much of this production of polymers goes for export, but also meets rising domestic demand and displaces other feedstocks. Polymer and intermediate export shipments rose by 38%, or the equivalent of 350 kb/d of ethane, between 2019 and 2024. Significant volumes flow to Latin America, Europe and various non-OECD Asian markets (including China). Over the same period, US exports of ethane and LPG have gone up by 1.1 mb/d to 2.6 mb/d, with almost three-quarters of this rise directed to China.

Olefin production, capacity and feedstocks, 2017-2030



IEA. CC BY 4.0.

Sources: IEA analysis of data from ICIS and Kpler.

In total, including polymer imports, the equivalent of 1.1 mb/d of US NGLs were shipped to feed Chinese demand for plastics and fibre last year. As such, the primary drivers for global supply and demand, respectively, are woven together. While increased trade tensions and tariffs could put this at risk due to the significant China-US petrochemical synergies, our base case is that these trade flows continue mostly intact. European and Indian NGL import volumes have risen

in recent years and are set to continue growing through 2030, but only the Chinese petrochemical system has the capacity to process LPG/ethane imports on the scale implied by US exports.

Demand falling elsewhere amid intense competition

The way that Chinese and US producers have expanded market share, via import substitution and exports, respectively, means that the pressure of increasing global overcapacity has largely fallen on other regions. Higher-cost regions, like Europe and East Asia, have been targets for additional flows from traditional exporters. Although some lower-cost producers, including those in the Middle East, have suffered amid more competitive global trade, this trend has disproportionately impacted naphtha crackers. This results in LPG and ethane substituting for naphtha consumption on a global level.

The European petrochemical industry has been perhaps the hardest hit by increased competition, with an estimated 19% fall in total feedstock consumption from 2019 to 2024. Regional industrial demand for plastics softened with import volumes rising simultaneously. While naphtha crackers had relatively limited direct exposure to sharply higher natural gas prices in 2022, production of energy-intensive ethylene derivatives like polyvinyl chloride (PVC) and styrene appears to have been displaced on a larger scale, undermining local olefin demand.

Along with a shift in the average cracker feedstock mix, with operators (especially in coastal locations) using more LPG because of favourable pricing, the fall in operating rates substantially weakened European naphtha demand. Deliveries of the product averaged less than 900 kb/d in 2023, a multi-decade low, before a partial rebound in 2024. We expect further declines (totalling about -10%) over the rest of this decade driven by capacity closures – some already in effect and others announced for 2025 and 2026 – with additional cutbacks likely.

Run cuts or outright closures will be heavily concentrated in high-cost zones like Europe. Smaller declines have been recorded in OECD Asia, despite the latter's more direct exposure to competition from new plants in China. While Europe and Northeast Asian steam crackers have a similar cost base, many European plants are run by major companies that operate in several parts of the world and aim to optimise output across their global portfolio. Conversely, Japanese and Korean firms tend to be more domestically concentrated.

Korean plants also benefit from being highly integrated with local refineries and downstream demand as well as being larger and more modern, while falls in feedstock use for the generally older, smaller Japanese fleet have been more comparable to their European peers. The resilience of Korean plants, and the start-up of the new Shaheen petrochemical complex will see deliveries edge

higher over the remainder of the decade. Korea's Shaheen project, which we assume will begin operations in 2027, is a so-called 'crude oil to chemicals' (CTC) refinery, including a 1.8 Mt/yr mixed-feed steam cracker. The project is being developed by S-Oil, which is 63% owned by Saudi Aramco. As such, it is reflective of an intensification of the long-standing trend for greater downstream integration by major oil producers.

Production in the Middle East appears to have been impacted by global overcapacity, although limited data availability makes the region's petrochemical operations especially opaque. Taking advantage of very low feedstock and energy costs, Middle Eastern steam crackers are generally estimated to be the most cost competitive in the world. Local demand is quite low compared to the feedstock resources available, so plants in the region are overwhelmingly geared to produce commodity polymers for export, especially polyethylene (PE) and polypropylene (PP). China, the world's largest plastics import market, has long been the most important outlet, especially for Saudi Arabian and Iranian production. Both these countries have seen substantial falls in their exports since 2019 (as reported by importing countries). China has become a virtual monopsonist for Iranian petrochemical exports and recent years have seen a fall in the flow between the two countries, although a rise in domestic use may have mitigated some of this decline.

Saudi ethylene derivative exports to the world fell by about 30% between 2019 and 2024 in ethylene equivalent terms, according to *ICIS* data. In part, this may reflect increased domestic consumption, with solid non-oil GDP growth and substantial construction, infrastructure and clean energy investment programmes underway. Nevertheless, we estimate that Saudi feedstock use dropped by about 80 kb/d over the period, concentrated in LPG/ethane, and it is possible that the fall was even larger. It may be that national priorities around the provision of natural gas, especially for power generation (see box, *Saudi plans lead push to slash oil-fired electricity generation*), have seen less ethane extracted from gas streams. Reported Saudi Aramco NGLs data suggest that available ethane volumes may have declined by 70 kb/d, or 18%, between 2019 and 2024. There have been sporadic reports of feedstock shortages at Saudi ethane plants, and it is difficult to reconcile strong estimated cracker margins with a decline in activity.

Efficiency gains slow underlying growth in aviation and marine demand

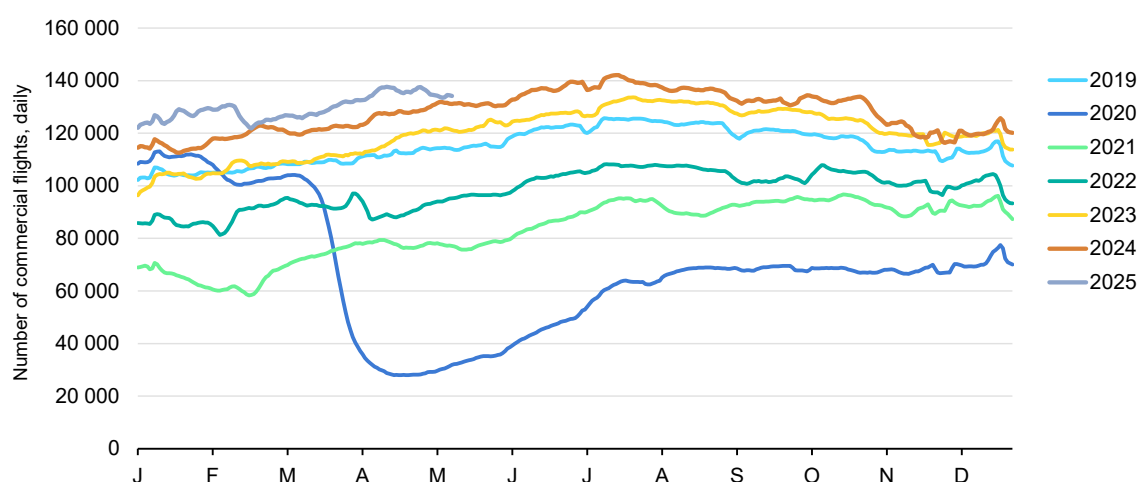
Jet fuel demand on course to regain pre-Covid levels

Amid record-high airline passenger numbers and revenues, the commercial aviation industry has completed its recovery from the pandemic crisis, the worst in

its history. On a global level, air travel had already regained pre-pandemic levels in 2023 by metrics such as the number of flights, passengers and miles flown. Domestic travel was the first to complete this rebound, with international travel following in early 2024. On a regional basis, the main exception is Asia Pacific, where international revenue passenger kilometres (RPKs) in 2024 remained 8.7% below 2019, according to data from the International Air Transport Association (IATA). Connectivity for the region took longer to recover as travel restrictions were lifted later than elsewhere. However, Asia Pacific is also set to regain pre-pandemic levels in 2025.

Overall jet/kerosene consumption gains of 130 kb/d in 2025 are only slightly behind the average pace of the 2010s, further suggesting that the post-Covid boom has concluded. Still, jet/kerosene uptake will remain 2% below pre-pandemic levels in 2025 as improved efficiencies counterbalance robust underlying growth in air travel.

Global number of commercial flights (daily), 2019-2025



IEA. CC BY 4.0.

Source: IEA analysis based on data from *FlightRadar* and *Bloomberg*.

Besides operational efficiencies such as increases in passenger load factors, airplane utilisation and seat density, this is chiefly due to large strides in fuel savings weighing on demand growth. Advanced technologies such as improved aerodynamics, the use of lightweight materials and enhanced engine propulsion all combine to reduce drag and eke out fuel economies. This technological transformation accelerated during the pandemic period after airlines retired a large share of their fleets and replaced them with new narrow-body flagship models such as the Airbus320neo (from 2016) and the Boeing 737 MAX (2017). These are about 20% more fuel-efficient than those they are replacing.

As ongoing fleet renewal combines with technological advances, improvements in fuel burn will slow, but not subvert, jet/kerosene's upward demand trajectory. Besides a decelerating pace in efficiency gains (our models see a gradual easing in the annual rate of efficiency savings over the forecast period, to around 1% by 2030), strong demand for air travel will support moderate growth, with jet/kerosene uptake only reaching pre-pandemic demand levels by 2027. In our outlook, demand is forecast to rise by a cumulative 1 mb/d in 2024-30, and by an average annual growth rate of 2.1%. By growing throughout the forecast period, jet/kerosene's outlook is relatively upbeat, at least when compared to road and marine transport fuels. Assumptions about the global economy are central to our forecast, given jet/kerosene is the most dependent on the macro backdrop among the refined products. The current subpar global economic growth outlook amid uncertainties on trade policies could yet weigh further on demand.

The Asia Pacific region is projected to dominate travel growth. IATA sees an annual expansion of 5.1% in passenger numbers – roughly double the rate of mature markets such as North America's 3.1% and Europe's 2.5%. Besides a fragmented geography suited for air travel, healthy population growth and a growing urban middle class with higher disposable income and wealth are boosting appetite for air travel, perceived as a luxury good. In turn, this reliable customer base supports the case for capacity investment in airports and airplanes.

Subdued trade and IMO efficiencies erode marine fuel demand growth

Demand for marine fuels, which is comprised of maritime bunkers and domestic navigation, is expected to flatline at around 5 mb/d over the forecast period, as subpar underlying growth in shipping combines with rising costs in the wake of the ongoing tightening of marine decarbonisation standards.

Recent years have seen significant disruptions to global maritime traffic, as attacks on vessels in the Red Sea forced some vessel operators to avoid the Suez Canal and seek alternative routes. Additionally, a severe drought at the Panama Canal reduced transits. While longer journeys and faster speeds initially supported bunker sales, international bunkering gains of 140 kb/d in 2024 were only slightly above trend, as soaring freight and insurance rates disincentivised maritime transport. Weak economic growth also acted as a headwind, with Germany's protracted slump weighing on European bunker sales.

The tariff turmoil unfolding in 2025 will make for an even harsher environment for global trade and shipping, with marine fuels disproportionately affected by lacklustre economic expansion. This merely accelerates the ongoing disconnect between GDP growth and maritime trade – the latter has lagged the former for more than a decade – amid a slower pace of globalisation and a GDP-shift from goods to services.

Our models assume an aggregate increase of around 10% in tonne-kilometres (tkm) over the forecast horizon, but this will be counterbalanced by efficiency improvements. Accordingly, we see a flattish demand profile for marine bunkering.

Emission standards mandated by the UN's IMO will continue to tighten. Under IMO 2020 regulations, limits already reduced the sulphur content of fuel oil to 0.5%. This prompted a switch from heavy fuel oil (HFO) to very low sulphur fuel oil (VLSFO) and to marine gasoil (MGO), as well as the installation of exhaust gas cleaning systems, known as 'scrubbers.'

Additionally, the Mediterranean became a Sulphur Emission Control Area (ECAMED) in May 2025, cutting fuel sulphur limits to 0.10% – matching the stricter standards seen in Northern Europe and North America. In parallel, under IMO regulations approved in April 2025, standards are set to become even more restrictive towards the end of the decade with the introduction of a global greenhouse gas (GHG) pricing system. Merchant ships will have to pay a penalty for above-target GHG emissions starting in 2028, encouraging use of lower emission fuels like biofuels, ammonia and hydrogen.

Demand developments by region

North American, European oil demand declines by 1.3 mb/d

North American oil demand will end the decade at 24 mb/d, after plateauing at around 24.6 mb/d in 2025. The region's consumption will fall by an average 0.4%, or 570 kb/d, over the forecast period. The **United States**, representing around 80% of North American oil consumption, will account for the bulk of the decline, with smaller contractions in **Canada** and **Mexico**. Although US oil demand has hovered around 2019's level in recent years, the region's overall consumption is not expected to regain its pre-pandemic high-water mark. Gasoline will be the main drag on use, declining by 790 kb/d between 2024 and 2030, with its outlook undermined by an expanding EV fleet, burgeoning vehicle efficiencies and the assumed absence of a decline in WFH employment. Still, this marks a much smaller contraction than *Oil 2024's* expected decline of 1.6 mb/d due to lower assumptions about EV penetration. Electric vehicles are expected to account for 20% of US total car sales in 2030, down from 55% assumed last year. Amid stagnation or declines elsewhere in the product spectrum, the only ones posting steady growth in our 2024-30 forecast are jet/kerosene and LPG/ethane, for aggregate increases of 180 kb/d and 370 kb/d, respectively. For LPG/ethane, the average growth rate of 1.7% stands in marked contrast to its 4% annual increase between 2019 and 2024. This slowdown is mainly due to the lack of further substantial petrochemical capacity expansions that characterised the early 2020s.

North America oil demand by product, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	3.5	3.6	3.8	3.9	4.1	4.2	4.4	4.4	4.5	4.5	4.6	4.6	1.4%	0.4
Naphtha	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.5%	0.0
Gasoline	11.0	9.4	10.2	10.4	10.5	10.5	10.4	10.4	10.2	10.0	9.8	9.5	-1.6%	-0.9
Jet/Kerosene	2.0	1.2	1.5	1.8	1.9	2.0	2.0	2.0	2.1	2.1	2.1	2.2	1.7%	0.2
Gasoil/Diesel	5.1	4.6	4.9	5.1	5.0	4.9	4.9	4.9	4.9	4.8	4.8	4.8	-0.6%	-0.2
Residual fuel oil	0.5	0.5	0.6	0.5	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	-0.2%	0.0
Other products	2.5	2.3	2.4	2.4	2.3	2.3	2.3	2.2	2.2	2.2	2.2	2.2	-0.3%	0.0
Total products	24.9	21.9	23.7	24.3	24.6	24.5	24.6	24.5	24.4	24.3	24.1	24.0	-0.4%	-0.6
Annual change	-0.2	-3.1	1.8	0.7	0.3	0.0	0.0	0.0	-0.1	-0.1	-0.2	-0.2		

European oil demand peaked in 2022 and will contract throughout the forecast by an aggregate 730 kb/d, or about an annual average decrease of 120 kb/d over the period. The industry-linked fuels naphtha and gasoil will show the fastest rate of declines, each down around 2% annually, in the wake of subpar GDP growth of 1.5% over the forecast period. The switching away from diesel to gasoline-engine cars acts as an additional headwind to gasoil use. In parallel, gasoline demand will post only a marginal decline (-50 kb/d by 2030) as a result of this substitution. Both powertrains have falling market shares, as reported by the [European Automobile Manufacturers' Association](#). The combined share of new gasoline (33.0%) and diesel (10.4%) cars constituted less than half of sales in the passenger market in 2024, while simple hybrid cars accounted for another 31.4%. The only products materially defying the overall slump are jet/kerosene (+1.2% annually) and LPG/ethane (+0.8%), with the latter supported by the start-up of a new ethane-based cracker in Belgium, expected by late 2026.

Europe oil demand by product, 2019-2030 (mb/d)

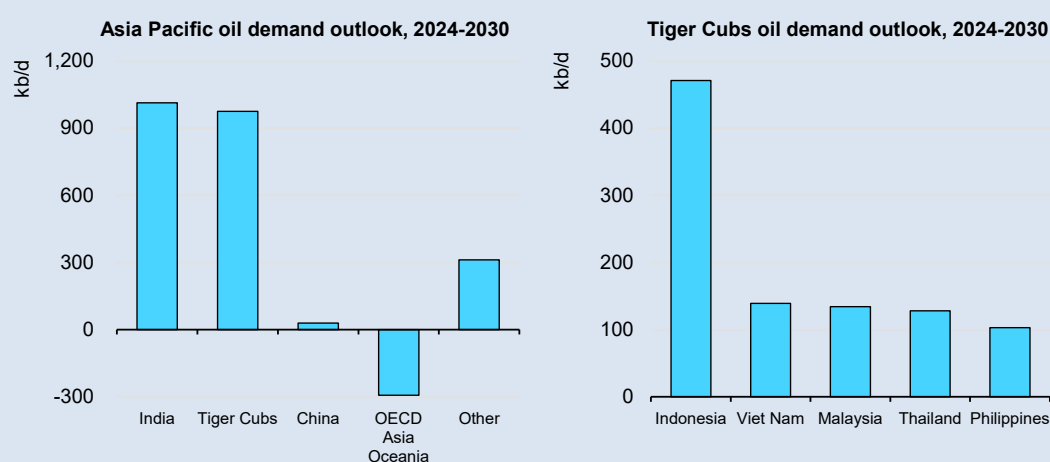
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	1.3	1.2	1.2	1.1	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.3	0.8%	0.1
Naphtha	1.1	1.1	1.1	1.0	0.9	1.0	1.0	0.9	0.9	0.9	0.9	0.9	-1.6%	-0.1
Gasoline	2.3	2.0	2.2	2.3	2.4	2.5	2.6	2.6	2.6	2.6	2.5	2.4	-0.3%	0.0
Jet/Kerosene	1.6	0.8	0.9	1.4	1.5	1.6	1.6	1.6	1.6	1.7	1.7	1.7	1.2%	0.1
Gasoil/Diesel	7.1	6.5	6.9	6.9	6.6	6.5	6.4	6.3	6.1	6.0	5.9	5.8	-2.0%	-0.7
Residual fuel oil	1.0	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.8	0.8	0.8	-0.8%	0.0
Other products	1.4	1.4	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3	-0.1%	0.0
Total products	15.8	13.7	14.4	14.9	14.8	14.9	14.9	14.8	14.6	14.5	14.3	14.1	-0.8%	-0.7
Annual change	0.0	-2.0	0.7	0.4	-0.1	0.1	0.0	-0.1	-0.2	-0.2	-0.1	-0.2		

Small group of Asian countries propels global demand growth

Oil demand in the **Asia Pacific** will increase by 2 mb/d, to 40.7 mb/d, over the forecast period, against a backdrop of robust GDP growth of 4% on average for the region. With Chinese consumption flatlining and OECD Asia Oceania contracting by 290 kb/d (weighed down by Japan's drop of 260 kb/d), India will account for around half of this rise at 1 mb/d. A group of five Southeast Asian economies will contribute a comparable amount.

Tiger Cubs roar, driving Southeast Asian oil demand higher

A small cluster of countries – Indonesia, Viet Nam, Malaysia, Thailand and the Philippines – will account for an increasing share of global oil demand growth, rising by a combined 980 kb/d to 6.3 mb/d by 2030. These gains, both in absolute and in relative terms, are led by Indonesia. The country's oil consumption is set to grow by 3.8% annually, for a cumulative 470 kb/d in 2030 – the second highest globally after India. The other four countries will see aggregate increases of around 100-150 kb/d each.



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This group of five is sometimes referred to as the Tiger Cubs, after the original four Asian Tigers (Hong Kong, Singapore, Korea and Chinese Taipei) that posted exceptional economic growth in the second half of the 20th century. Their export-driven economic models, characterised by technological investments and industrial development, are currently being emulated by the Cubs. Youthful demographics and large domestic consumer markets (of around 600 million people in total, of which 280 million are in Indonesia, the world's fourth most populous country) are other advantages, making for an above-par economic outlook. Our models assume a 5% average annual GDP expansion for the group over the forecast period, by and large continuing the 2010s pre-pandemic trend.

Moreover, the group's oil demand will grow at a relatively fast rate as the energy intensity of their economies accelerates from the low base that is typical for emerging countries. This pertains especially to the Philippines and Viet Nam, categorised as lower middle-income countries by the World Bank, with Indonesia, Thailand and Malaysia classified as higher middle-income.

Collectively, jet/kerosene (+3.2% annually, +80 kb/d in total) and gasoil (+2.5%, +310 kb/d) will propel the group's overall fuel consumption growth – the former buoyed by an emerging middle class eager to travel, the latter by a process of rapid industrialisation. This is in part due to the Cubs' emergence as Asia's key nearshoring hubs. Attracted by low labour costs and relative political stability,

companies have increasingly been migrating their manufacturing and supply chains to the region, often by diversifying away from China. To supply this expanding industrial base and rising demand for packaging and construction, the outlook period will see a handful of steam crackers coming online. Naphtha use will rise by 11.4% annually for a total increase of 290 kb/d, led by Indonesia (+170 kb/d) and Viet Nam (+70 kb/d).

Asia Pacific oil demand by product, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	4.4	4.4	4.6	4.9	5.2	5.5	5.6	5.7	5.9	6.0	6.1	6.2	2.0%	0.7
Naphtha	4.6	4.5	4.8	4.8	5.2	5.2	5.4	5.7	5.8	6.0	6.1	6.3	3.0%	1.0
Gasoline	7.6	7.3	7.8	7.9	8.1	8.0	7.9	7.8	7.7	7.6	7.4	7.2	-1.7%	-0.8
Jet/Kerosene	2.9	1.9	1.9	1.8	2.5	2.7	2.8	2.8	2.9	3.0	3.1	3.2	2.6%	0.5
Gasoil/Diesel	9.6	9.3	9.6	9.6	9.7	9.7	9.8	9.9	10.0	10.1	10.1	10.1	0.7%	0.4
Residual fuel oil	2.4	2.4	2.6	2.6	2.7	2.7	2.6	2.7	2.6	2.6	2.6	2.6	-0.4%	-0.1
Other products	4.5	4.5	4.5	4.7	4.7	4.7	4.7	4.8	4.8	4.9	5.0	5.0	1.0%	0.3
Total products	36.0	34.2	35.6	36.3	38.0	38.6	38.9	39.4	39.8	40.1	40.4	40.7	0.9%	2.0
Annual change	0.6	-1.8	1.4	0.7	1.6	0.7	0.3	0.5	0.4	0.3	0.3	0.2		

Chinese oil demand plateauing

China's oil use will hover around 16.8 mb/d throughout the forecast period, increasing by an almost negligible 30 kb/d between 2024 and 2030 – a sea change compared to the preceding decade when the country accounted for 60% of the global rise in oil demand. The barely positive growth rate pales in comparison to the 6% annual pace during 2010-19. However, the flattish headline numbers mask a sharp divergence among the different product segments, as a contraction in road fuels contrasts with robust growth in petrochemical feedstocks.

The slowdown in oil consumption is particularly unusual considering China's status as a middle-income country, with considerable growth in income and prosperity (our models assume average GDP expansion of 3.9% compared with 7.3% in the 2010s). This decoupling between oil use and GDP occurs due to substitution away from oil in transport amid mass market electrification, which is far more advanced in China than elsewhere. Rapid EV penetration has already caused gasoline demand to peak in 2021, at 3.8 mb/d. An annual consumption decline of a massive 6% will leave the fuel ending the decade at 2.5 mb/d – compared to 3 mb/d assumed under Oil 2024. Gasoil, which peaked in 2021, will see a similar, if less pronounced pattern, with an average annual decline of 2.1% as expanding sales of trucks powered by liquified natural gas (LNG) combine with anaemic construction activity and sluggish consumer spending. Here too, 2030 demand of 3.1 mb/d is well below last year's expected of 3.8 mb/d. Conversely, jet/kerosene will rise by a robust 3.9% on average. This growth is largely due to international air traffic, with increases in domestic flights undermined by the rapid

expansion of China's high-speed rail network. Already the largest in the world, it will expand further by 20% to 60 000 km between 2025 and 2030.

The deepening slump in road fuels will leave petrochemical feedstocks as the chief catalyst of China's oil demand growth, with LPG/ethane and naphtha rising by an annual 2.5% and 4.6%, respectively. This will result in a total increase in feedstock demand of 1.1 mb/d between 2024 and 2030 – equal to the decline in gasoline use. China's unprecedented wave of petrochemical additions is set to continue, albeit at a gradually slower pace as the decade progresses.

China oil demand by product, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	1.8	1.8	1.9	2.2	2.5	2.6	2.7	2.7	2.9	2.9	3.0	3.1	2.5%	0.4
Naphtha	1.4	1.5	1.6	1.8	2.3	2.3	2.5	2.7	2.8	2.9	3.0	3.0	4.6%	0.7
Gasoline	3.5	3.4	3.8	3.7	3.7	3.6	3.5	3.3	3.1	2.9	2.7	2.5	-6.0%	-1.1
Jet/Kerosene	0.9	0.8	0.8	0.5	0.8	0.9	1.0	1.0	1.0	1.1	1.1	1.2	3.9%	0.2
Gasoil/Diesel	3.6	3.7	3.8	3.6	3.6	3.6	3.5	3.5	3.5	3.4	3.3	3.1	-2.1%	-0.4
Residual fuel oil	0.4	0.5	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.2%	0.0
Other products	2.6	2.7	2.7	2.8	2.9	3.0	3.0	3.0	3.1	3.1	3.1	3.2	1.0%	0.2
Total products	14.2	14.3	15.2	15.2	16.5	16.6	16.7	16.9	16.9	16.9	16.8	16.7	0.0%	0.0
Annual change	0.7	0.2	0.8	0.0	1.3	0.2	0.1	0.2	0.1	-0.1	-0.1	-0.2		

India leads global demand growth by a large margin

India's oil demand will increase by a steep 1 mb/d over the forecast period – more than any other country – in the wake of stellar GDP expansion, at an average annual rate of 2.8%. India will remain the world's fastest growing major economy in 2025 for a fourth year running and is projected to overtake Japan as the world's fourth largest economy this year.

Moreover, while all key products will contribute to the expansion, transport fuels will lead the gains – a global anomaly. In relative terms, jet/kerosene will rise the fastest, at almost 6% annually. The fuel, starting from a low base, stands to benefit the most from population growth of 5% between 2025 and 2030 and its rapidly expanding middle class keen to spend on luxury goods and services, including foreign travel. A similar dynamic propels annual gasoline demand growth of 4%, which has significant scope for expansion given low levels of car ownership. Our models assume a 40% increase in the size of the car fleet by 2030 – a rate of expansion that comfortably outstrips the impact of efficiencies and EVs, with growth in the latter category mainly in two- and three-wheelers. A risk to our outlook is that the consumption boom driving demand for India's transport fuels is to a large extent debt fuelled. Borrowing has become increasingly commonplace among India's middle class. Household debt rose to 43% of GDP in 2024, up from 35% in 2022, and a possible credit spiral could derail expansion.

Demand for industrial-linked products will only marginally lag the growth rate for retail-oriented transport fuels, with annual increases in gasoil, naphtha and

LPG/ethane of 3.3%, 2.0% and 2.5%, respectively. For gasoil, making up around one-third of India's total oil use, this amounts to an aggregate gain of 380 kb/d over the forecast period, buoyed by secular trends such as urbanisation, industrialisation and the building out of the country's infrastructure. Expansion in the petrochemical feedstocks tracks the launch of new projects coming on stream, with the widespread adoption of clean cooking acting as an additional boost to LPG demand. Government subsidy programmes have been instrumental in this respect, in particular the Pradhan Mantri Ujjwala Yojana scheme launched in 2016 that aims to provide LPG connections across the country, replacing traditional cooking methods.

India oil demand by product, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	0.8	0.9	0.9	0.9	0.9	1.0	1.1	1.1	1.1	1.1	1.1	1.2	2.5%	0.2
Naphtha	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	2.0%	0.0
Gasoline	0.7	0.7	0.8	0.9	1.0	1.0	1.1	1.1	1.2	1.2	1.3	1.3	4.0%	0.3
Jet/Kerosene	0.2	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	5.6%	0.1
Gasoil/Diesel	1.6	1.5	1.5	1.7	1.7	1.8	1.8	1.9	2.0	2.0	2.1	2.2	3.3%	0.4
Residual fuel oil	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.7%	0.0
Other products	1.1	1.0	1.0	1.2	1.2	1.2	1.1	1.1	1.2	1.2	1.2	1.2	1.0%	0.1
Total products	5.0	4.6	4.8	5.3	5.4	5.6	5.8	5.9	6.1	6.2	6.5	6.7	2.8%	1.0
Annual change	0.0	-0.3	0.2	0.5	0.1	0.2	0.1	0.2	0.2	0.2	0.2	0.2		

African oil demand to post robust growth

African demand will increase by a solid 2.9% annually over the forecast period – a comparable pace to the continent's population growth (by 15% in total, to reach 1.32 billion in 2030). GDP expansion is equally firm, set to average a higher 3.8% annually over 2024-30, from 3.6% during the 2010s. This consumption increase will be relatively balanced across the product spectrum, with gasoline, gasoil and jet/kerosene all growing by around 2.8% annually. Still, in absolute terms, the role of the latter fuel will remain relatively marginal, at 270 kb/d on average, as air travel remains beyond reach for the great majority of the population. The jet/kerosene pace of expansion pales in comparison to LPG/ethane's 5.1% growth rate, with consumption approaching 900 kb/d in 2030, propelled by soaring use in clean cooking.

Among countries, Egypt will end the decade at 1.2 mb/d in 2030, accounting for more than 20% of the continent's overall consumption. Other leading countries are Algeria, Nigeria and South Africa, each representing around 600 kb/d in 2030. However, the relative fragility of the continent's demand forecast bears emphasising, with Egypt and Nigeria experiencing severe sovereign debt and currency crises as recently as last year, leading to IMF/World Bank bailouts. Still, with its finances now on surer footing, Nigeria's cumulative consumption growth of 26% over the forecast period will be the highest of any major African country, benefitting from a low per capita base in energy consumption, with industrialisation

and young population dynamics further propelling growth. This contrasts with South Africa, where oil demand will grow only marginally over the forecast period. Besides a higher baseline, the country's underperformance is set to continue, its protracted power grid crisis reflective of its chronic economic malaise.

Africa oil demand by product, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	0.5	0.6	0.6	0.6	0.7	0.7	0.7	0.7	0.8	0.8	0.8	0.9	5.1%	0.2
Naphtha	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6%	0.0
Gasoline	1.2	1.2	1.3	1.3	1.2	1.2	1.2	1.3	1.3	1.3	1.4	1.4	2.6%	0.2
Jet/Kerosene	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	2.8%	0.0
Gasoil/Diesel	1.7	1.5	1.7	1.7	1.8	1.8	1.8	1.9	1.9	2.0	2.0	2.1	2.8%	0.3
Residual fuel oil	0.3	0.2	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	1.1%	0.0
Other products	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	2.4%	0.0
Total products	4.2	3.9	4.4	4.5	4.6	4.6	4.8	4.9	5.0	5.2	5.3	5.4	2.9%	0.9
Annual change	0.0	-0.3	0.5	0.1	0.1	0.0	0.2	0.1	0.2	0.1	0.1	0.1		

Central and South American oil demand will grow by an average 1.4% over the forecast period – a relatively subdued rate by emerging market standards. This is in line with equally muted average GDP growth of 2.4%, compared to 3.9% for the non-OECD as a whole. To a large extent, this reflects the continent's structural problems: institutional and legal framework weaknesses, poor labour productivity and an underdeveloped manufacturing base amid heavy reliance on agriculture and mining.

Central and South America oil demand by product, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	0.6	0.6	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.8%	0.0
Naphtha	0.2	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	1.8%	0.0
Gasoline	2.0	1.7	1.9	2.0	2.1	2.1	2.2	2.2	2.2	2.3	2.3	2.3	1.6%	0.2
Jet/Kerosene	0.3	0.2	0.2	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4	2.7%	0.1
Gasoil/Diesel	2.3	2.1	2.3	2.4	2.4	2.4	2.5	2.5	2.6	2.6	2.6	2.7	1.5%	0.2
Residual fuel oil	0.5	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.6%	0.0
Other products	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.7	1.2%	0.0
Total products	6.7	5.7	6.3	6.6	6.7	6.8	6.9	7.0	7.1	7.2	7.4	7.4	1.4%	0.6
Annual change	0.0	-1.0	0.6	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1		

Brazil, responsible for about half of the region's total oil demand, is a case in point in this respect. Overall oil consumption will increase by around 1.3% annually between 2024 and 2030, with gasoil (responsible for more than a third of Brazil's total demand) expanding at a similarly pace. Farming, dependent on diesel-intensive machinery during harvest and planting seasons, accounts for around one-seventh of this growth. An agribusiness powerhouse, this also represents a vulnerability of Brazil's economy, with agricultural exports complemented by a lack of competitiveness elsewhere.

Argentina, which saw oil demand contract each year between 2014-24, with the exception of the 2020-22 Covid years, will see consistently positive growth of around 10 kb/d per year over the 2025-30 forecasting period. This is in line with

solidly positive GDP growth of 3% on average, reflecting the country's resurgent economic outlook in the wake of the Milei Administration's economic reforms.

Eurasian oil demand is set to increase by 300 kb/d over the forecast period, concluding the decade at 4.6 mb/d. Russia, which represents about 80% of the region's consumption, will see growth of only 110 kb/d over the 2024-30 outlook, concentrated in petrochemical feedstocks. Conversely, aggregate demand for gasoil, gasoline and jet/kerosene will be essentially flat amid muted average GDP growth of 0.6% per year.

Middle Eastern oil consumption will fall marginally, by 50 kb/d to 9.2 mb/d over the 2024-30 outlook period. Total demand is forecast to peak at 9.6 mb/d in 2027, after which a rapid decline sets in, exclusively due to an accelerated drop in oil use for power generation. This exceptional decline masks relatively healthy oil demand gains of around 2-3% in the other main fuels, buttressed by GDP growth of about 3.5% and an increase of 1.5% in the region's population. LPG/ethane leads gains at 410 kb/d, followed by gasoline at 280 kb/d while jet fuel/kerosene and naphtha are each up 120 kb/d. This combined growth of 890 kb/d is nevertheless outweighed by plummeting oil use for power generation of a projected 1.1 mb/d, with aggregate declines of 400 kb/d for fuel oil and 540 kb/d for 'other products', dominated by direct crude burning. These massive contractions are largely concentrated in Saudi Arabia, in line with the country's stated policy to phase out oil use for electricity and desalination plants with the substitution of natural gas and renewables (see Box below, Saudi plans lead push to slash oil-fired electricity generation.) Saudi Arabia's oil consumption, which accounts for about 40% of the region's demand, is forecast to decline by 620 kb/d over the forecast period – the most of any country. Iraq is the only other Middle Eastern country posting a drop in oil demand, by 50 kb/d, also due to lower oil burn for power generation. This contrasts with other major economies in the region seeing steady growth, with Iran, Qatar and the UAE expanding by around 150 kb/d each.

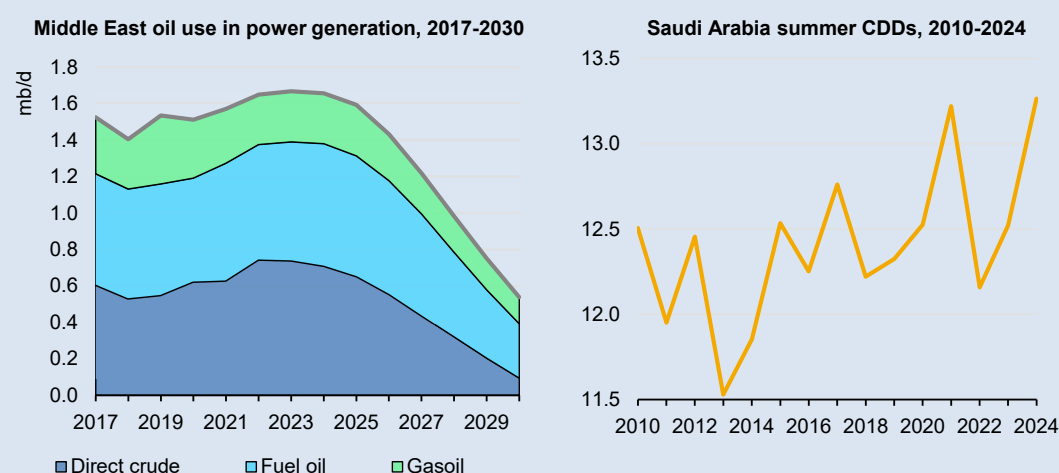
Middle East oil demand by product, 2019-2030 (mb/d)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2024-30 Growth Rate	2024-30 Growth
LPG/Ethane	2.1	2.0	2.0	2.0	1.9	1.9	2.0	2.0	2.1	2.2	2.3	2.3	3.3%	0.4
Naphtha	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	4.2%	0.1
Gasoline	1.8	1.5	1.7	1.7	1.8	1.8	1.9	1.9	2.0	2.0	2.1	2.1	2.4%	0.3
Jet/Kerosene	0.5	0.3	0.3	0.4	0.5	0.5	0.5	0.6	0.6	0.6	0.6	0.6	3.4%	0.1
Gasoil/Diesel	1.8	1.6	1.7	1.7	1.7	1.7	1.8	1.8	1.7	1.7	1.7	1.7	-0.4%	0.0
Residual fuel oil	1.4	1.3	1.4	1.5	1.5	1.6	1.6	1.6	1.5	1.4	1.3	1.2	-4.6%	-0.4
Other products	1.1	1.2	1.1	1.3	1.3	1.2	1.2	1.2	1.1	1.0	0.9	0.7	-9.3%	-0.5
Total products	8.9	8.3	8.6	9.1	9.2	9.2	9.4	9.5	9.6	9.6	9.4	9.2	-0.1%	0.0
Annual change	0.2	-0.7	0.4	0.4	0.1	0.1	0.1	0.1	0.1	-0.1	-0.2	-0.2		

Saudi plans lead push to slash oil-fired electricity generation

The Middle East is the primary region for oil-fired electricity generation globally but concerted efforts in recent years to ramp up new natural gas supplies will sharply lower oil burning in power plants during our forecast period. Crude, fuel oil and gasoil use in power generation is forecast to plunge by two-thirds before the end of the decade. Oil use for power in the Middle East was estimated at almost 1.7 mb/d in 2024, including 670 kb/d of fuel oil, 710 kb/d of crude and 280 kb/d of gasoil. Based on the delivery of announced plans, we estimate that this will fall by a steep 1.1 mb/d to just 540 kb/d by 2030 as more gas developments, as well as renewable projects, come online. Saudi Arabia, the world's largest consumer of oil for power generation, will account for around 1 mb/d of this decline.

Liquid fuels play an important role in a number of Middle Eastern countries, notably in Saudi Arabia, Iraq, Kuwait and Iran, and are essential for managing peaks in power consumption, as well as meeting base load in areas with lower gas availability. A push to make better use of underutilised gas resources and develop renewable generation is set to reduce this dependence in the region's two largest users of oil in power generation, Saudi Arabia and Iraq. This would come with air-quality and economic benefits, and freeing additional oil for export.



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Source: IEA analysis including *Bloomberg* data.

The most ambitious plans come from Saudi Arabia which is significantly investing in gas and renewables to meet rising electricity demand and targeting the phase out oil burn in power plants. The Kingdom's Liquid Displacement Programme aims to replace roughly 1 mb/d of oil consumed in power generation, desalination plants and agriculture by 2030 with natural gas, renewables and electrification. Progress thus far has seen several new gas plants already operational or under development and the first signs of this push bearing fruit can be seen in latest data. Saudi use of direct crude and fuel oil was essentially flat from 2023 to 2024, despite a 6% increase in cooling degree days (CDDs) amid extremely hot summer

weather. Nevertheless, these strategic investments will likely have their biggest impact on oil demand in the final years of this decade.

The Iraqi power system has come under strain in recent years, with imports of electricity and gas from neighbouring countries insufficient to reduce the need to burn large volumes of crude and fuel oil. The country aims to alleviate these issues by better use of existing gas resources and investments in solar generation. The Gas Growth Integrated Project (GGIP), in partnership with TotalEnergies, will recover gas from the Ratawi oil field, which is currently being flared. Projects like this are forecast to cut Iraqi oil use by about 100 kb/d between 2024 and 2030.

Electricity demand and, by extension, oil-fired power generation was boosted by extremely hot summer temperatures during 2024. Air conditioning is a major contributor to spikes in power use and therefore periods of intense heat and increased CDDs are often associated with sharply higher oil consumption. These extreme weather patterns had [a major impact](#) on regional and global patterns of demand in 2024 and variation in temperatures will likely continue to result in uneven Middle Eastern demand over the medium term.

Supply

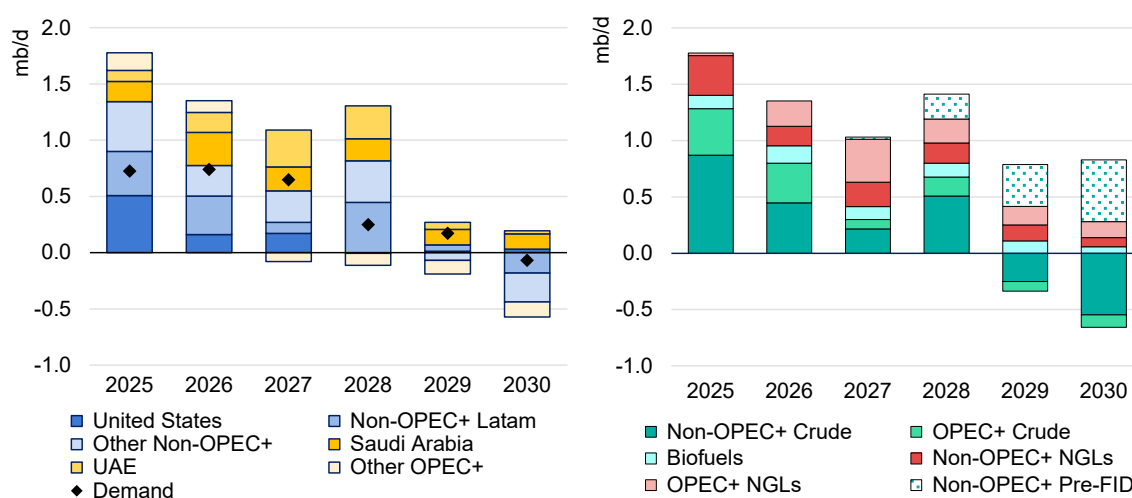
Global summary

Oil production capacity growth dominated by a surge in NGLs, led by the United States and Saudi Arabia

World oil production capacity is forecast to rise by 5.1 mb/d to 114.7 mb/d by 2030, with growth dominated by robust gains in NGLs and other non-crude liquids, and propelled higher by the world's two largest suppliers, Saudi Arabia and the United States. The strategic shift towards increased non-crude capacity is driven by strong global demand for petrochemical feedstocks and the development of liquid-rich gas resources.

Non-OPEC+ producers will provide roughly two-thirds of the global increase, boosting capacity by 3.1 mb/d, while OPEC+ adds nearly 2 mb/d of capacity over the 2024-30 outlook period. The United States will continue to lead the ramp up in non-OPEC+ production, led by strong growth in NGLs, even as its dominance in global supply expansions wanes with shale producers scaling back activity at lower oil prices. Saudi Arabia leads OPEC+ capacity increases, almost exclusively in NGLs. Combined, Saudi Arabia and the United States will contribute 40% to total global oil capacity growth in the forecast period.

Global oil supply capacity forecast, year-on-year change, 2025-2030



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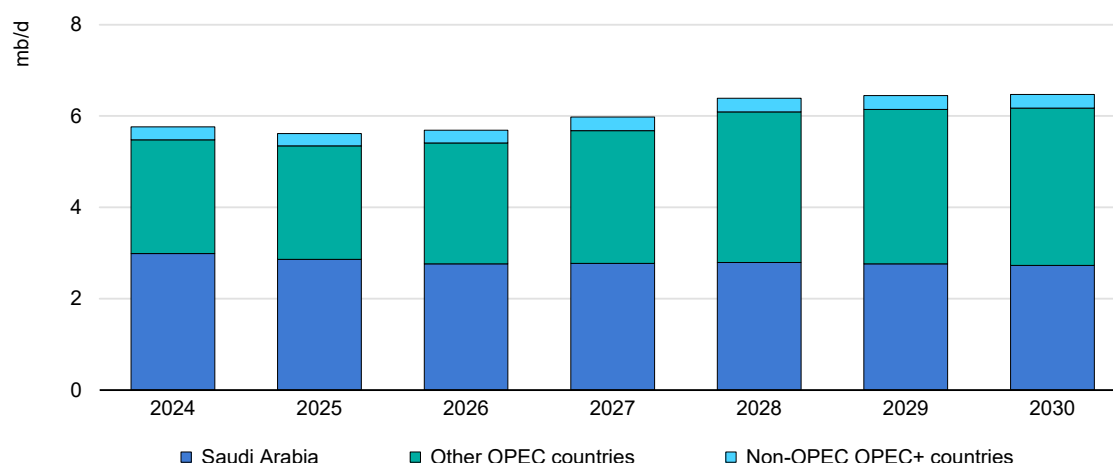
Notes: Assumes Iran, Russia and Venezuela remain under sanctions. OPEC+ NGLs include condensates. Crude includes processing gains and non-conventional volumes. The right-hand chart includes pre-sanctioned projects, listed in the Annex tables.

Total non-crude liquids supply capacity rises by 3.1 mb/d to 2030, of which NGLs account for nearly three-quarters of the increase at 2.3 mb/d. Growing natural gas production from Middle Eastern producers will boost regional NGL supply by 1.4 mb/d to 2030. Despite US light tight oil (LTO) producers cutting spending amid a lower price outlook, increasingly gassy reservoirs will buoy US NGL growth by 860 kb/d. Biofuels are forecast to increase by around 680 kb/d – led by Brazil, India and Indonesia, while non-conventional output and processing gains will add a further 200 kb/d over the forecast period.

By contrast, global crude and condensate capacity expands sharply early in the forecast, underpinned by gains in non-OPEC+ Americas. Overall non-OPEC+ crude and condensate capacity builds by 1.2 mb/d to 2030 despite the number of sanctioned projects tailing off after 2027. OPEC+ net crude capacity, excluding condensates, rises by 820 kb/d, with strong growth in the UAE and Iraq offset by field declines most notably in Mexico.

Net capacity growth through 2030 is roughly twice the projected 2.5 mb/d increase in oil demand over the forecast period. Based on the current OPEC+ supply trajectory laid out at its May 2025 meeting, we estimate OPEC+ spare crude capacity, excluding Iran, Venezuela and Russia, to reach 6.5 mb/d in 2030 and implied stock builds of nearly 2 mb/d at the end of the forecast period. Saudi Arabia holds around half of OPEC+ spare capacity now and it keeps its level roughly steady through the forecast.

OPEC+ spare crude capacity, 2024-2030



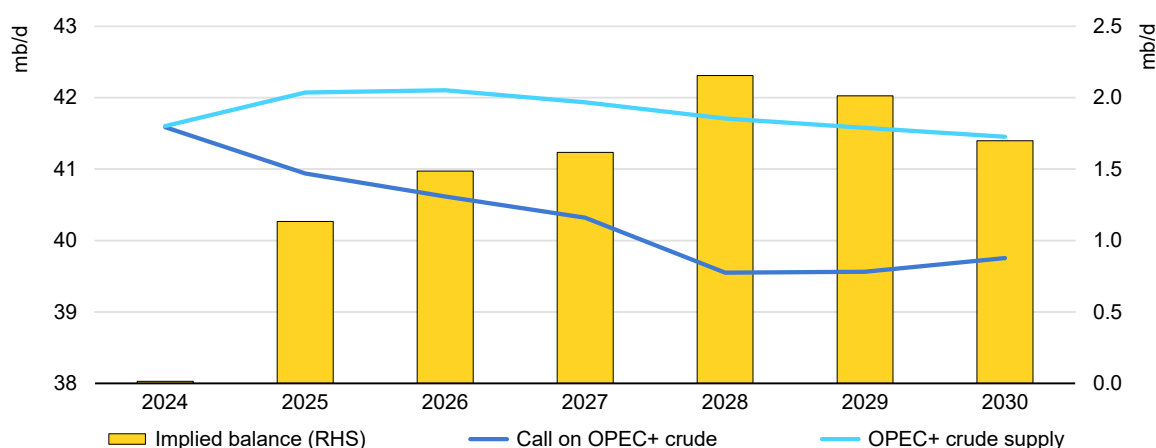
IEA. CC BY 4.0.

Notes: Based on OPEC+ crude supply targets effective July 2025, excluding Iran, Russia and Venezuela

Non-OPEC+ dominates supply growth

Global oil supply to the market, in contrast to capacity, is projected to rise by 4.1 mb/d to 107.2 mb/d by 2030. Producers in the Americas account for nearly all the 3.1 mb/d growth projected for non-OPEC+ over the forecast period. US LTO supply growth stalls, with output in 2030 only 360 kb/d above 2024 levels. By contrast, Argentinian LTO more than doubles – albeit from a lower base – from 400 kb/d in 2024 to 890 kb/d by the end of the decade. Production growth in Guyana and Brazil will lose momentum from 2028 onwards as the number of sanctioned projects tails off. Meanwhile, new Qatari LNG buildouts and Saudi gas developments bring on more associated NGLs. Taken altogether, these additions, combined with a forecast slowdown in oil demand, will reduce the call on OPEC+ crude oil by 1.8 mb/d from 41.6 mb/d in 2024 to 39.8 mb/d in 2030.

Call on OPEC+ crude oil, OPEC+ supply and implied oil market balance, 2024-2030



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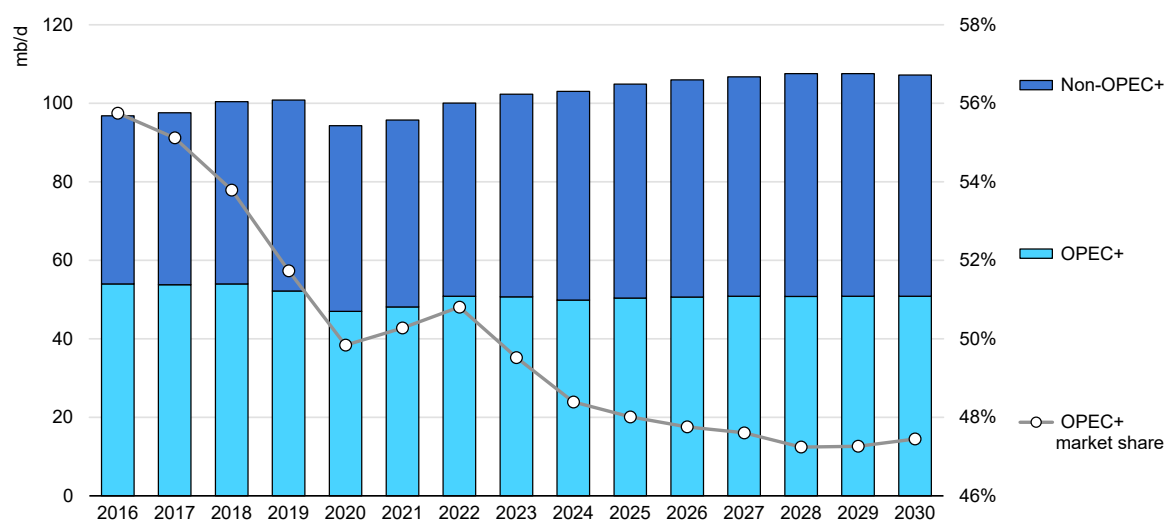
Note: Forecast OPEC+ crude supply based on quota targets effective July 2025.

OPEC+ oil output, including NGLs and condensates, grows by 1 mb/d from 2024 to 2030. For this *Report*, our OPEC+ oil supply outlook is based on the group's crude production level announced on 31 May. The group of eight OPEC+ producers – Saudi Arabia, Russia, Iraq, the UAE, Kuwait, Algeria, Oman and Kazakhstan – that had voluntarily agreed in November 2023 to scale back an additional 2.2 mb/d of production announced 1.4 mb/d of those cuts will be reversed by July. While not all of this supply may hit the market as some members enforce compensation plans, we estimate that only Saudi Arabia has room to substantially raise production, bringing about an additional 180 kb/d onto the market each month from May to July.

Even so, OPEC+ may struggle to regain substantial market share as non-OPEC+ oil supply continues to see robust growth in the early years of the forecast period.

Non-OPEC+ gains slip into contraction after 2028, however, as the pipeline of projects delivered wanes by the end of the decade. This *Report* estimates that an additional 1.1 mb/d of potential pre-FID non-OPEC+ supply, excluding LTO, could come online between 2028 and 2030 should projects move ahead swiftly.

OPEC+ struggles to reclaim market share through the forecast period



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Notes: Forecast OPEC+ crude supply based on quota targets effective July 2025. Russia, Iran and Venezuela remain under sanctions. Russian and Iranian production levels are held at current levels through the forecast period. Venezuelan output is assumed to decrease to 600 kb/d by 2026 based on current sanctions. Chart is based on current OPEC+ composition throughout the 2016-2030 period.

World Oil Production by Region (mb/d)

	2024	2025	2026	2027	2028	2029	2030	2024-30
Africa	7.2	7.4	7.4	7.5	7.5	7.4	7.3	0.1
Latin America	7.4	7.7	7.9	8.0	8.4	8.5	8.3	0.9
North America	28.3	28.9	29.0	29.3	29.3	29.2	29.2	0.9
China	4.3	4.4	4.4	4.4	4.3	4.3	4.2	-0.1
Other Asia	3.1	3.0	3.0	2.9	2.9	2.8	2.8	-0.3
Europe	3.3	3.4	3.4	3.2	3.1	3.0	2.8	-0.5
Eurasia	13.5	13.6	13.7	13.7	13.6	13.5	13.5	0.0
Middle East	30.2	30.6	31.1	31.6	32.1	32.4	32.7	2.5
Total Oil Production	97.3	99.0	99.9	100.5	101.2	101.1	100.6	3.4
Processing Gains	2.4	2.4	2.5	2.5	2.5	2.5	2.5	0.1
Global Biofuels	3.4	3.5	3.7	3.8	3.9	4.0	4.1	0.7
Total Supply	103.1	104.9	106.0	106.8	107.6	107.6	107.2	4.1
OPEC Crude	27.2	27.6	27.7	27.7	27.6	27.6	27.7	0.4
OPEC NGLs ¹	5.5	5.7	5.9	6.3	6.5	6.7	6.9	1.3
Non-OPEC OPEC+	17.1	17.1	17.1	16.9	16.7	16.5	16.3	-0.8
Total OPEC+	49.9	50.4	50.6	50.8	50.8	50.8	50.9	1.00
Memo: Call on OPEC+	41.6	40.9	40.6	40.3	39.6	39.6	39.8	-1.8

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Notes: Forecast OPEC+ crude supply based on quota targets effective July 2025. ¹ Includes condensates reported by OPEC countries, oil from non-conventional sources, e.g. GTL in Nigeria and non-oil inputs to Saudi Arabian MTBE.

Investment and exploration

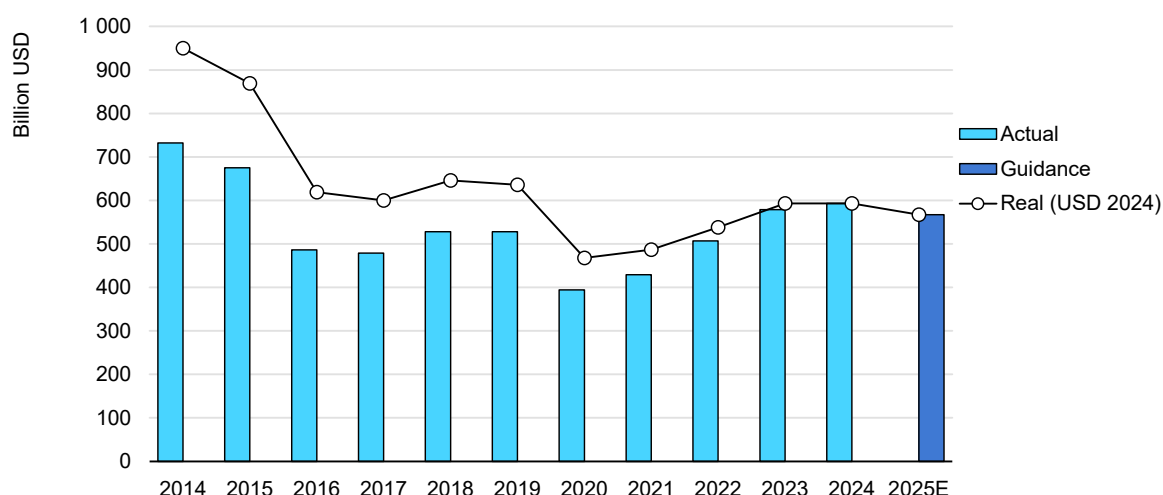
Capex to retreat in 2025 as price risks mount

Global upstream oil and gas investment is set to decline by 4% in 2025, to around USD 565 billion, with US independent shale producers accounting for the majority of the reductions. Upstream oil investments are on track to fall by nearly 6% to USD 420 billion, according to company guidance while spending on gas increases 3% to USD 145 billion. Recent oil price fluctuations, financial market volatility and trade uncertainties could further impact development costs and investment decisions.

In 2024, global upstream oil and gas capex was flat in nominal terms at USD 590 billion, and down by USD 10 billion in real terms, compared to 2023 levels. Nominally, 2023 to 2025 capex levels are at the highest level since 2015, but still below 2019 in real terms.

Global upstream investment continues to reflect more cautious capital deployment across the industry, as it has since 2015. Companies no longer pursue aggressive growth, and instead place greater emphasis on capital discipline, profitability and cost optimisation. This mantra has perhaps intensified in the past twelve months.

Global upstream oil and gas capital spending



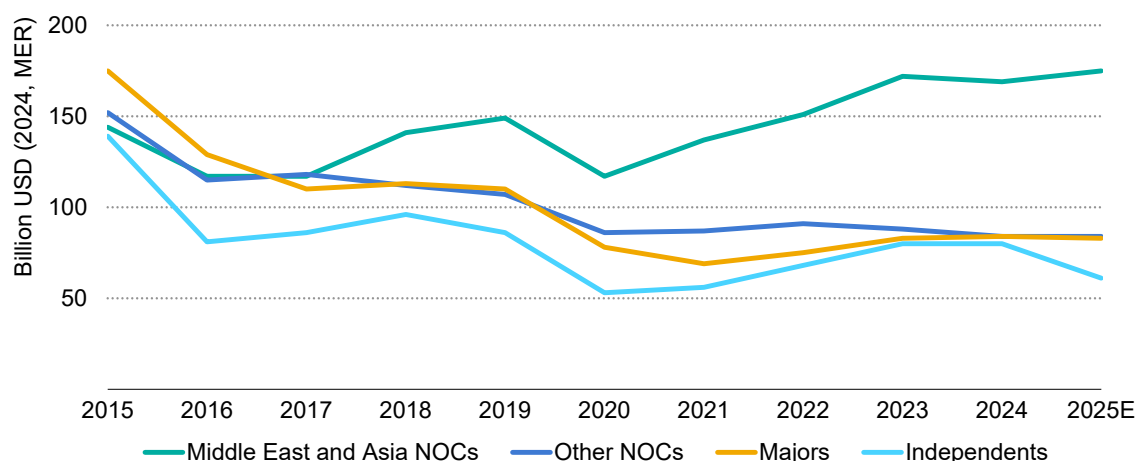
IEA. CC BY 4.0.

Source: IEA, [World Energy Investment 2025](#).

Investment priorities are given to high-potential, low-cost basins such as those in West Africa, deepwater Latin America and the US federal offshore. In developed offshore basins, exploration activity focuses near existing infrastructure to lower development expenditures. Despite the broader trend of investment restraint,

upstream spending continues to rise in the Middle East and among certain major oil companies, underscoring their strategic focus on long-term resource development.

Global upstream oil and gas capital spending by operator segment



IEA. CC BY 4.0.

Source: IEA, [World Energy Investment 2025](#).

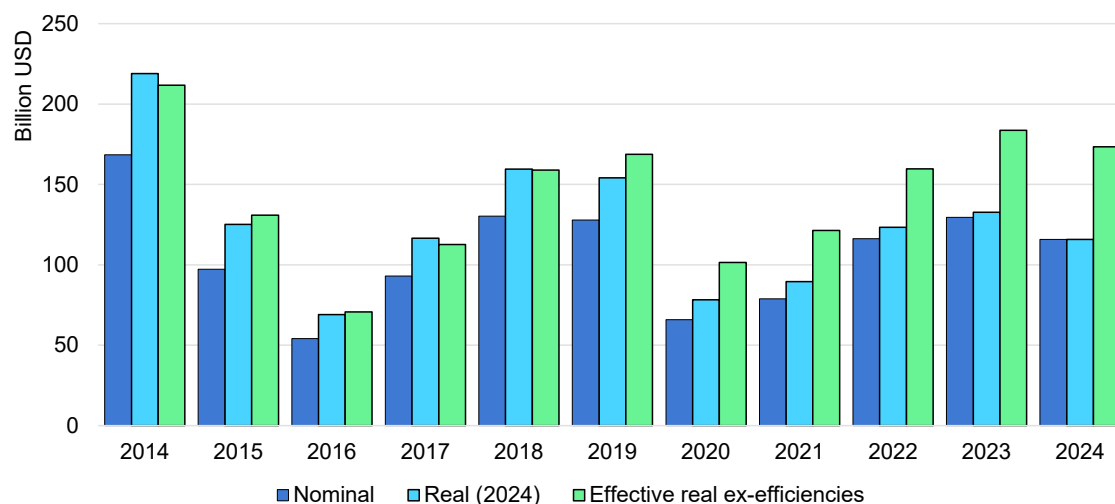
The increase in upstream investment in 2024 was largely concentrated in projects by national oil companies (NOCs) in the Middle East, China and the Americas. Middle Eastern and Asian NOCs accounted for over 60% of the global capex spend this decade, up 10% from a decade ago. Saudi Aramco increased its upstream capex by 19% y-o-y to USD 39 billion in 2024, primarily to maintain its 12 mb/d of crude oil production capacity and grow its natural gas, NGLs and condensate output. In the UAE, the Abu Dhabi National Oil Co (ADNOC) plans to raise crude supply capacity to 5 mb/d by 2027, spending USD 150 billion between 2024 and 2027 to achieve this goal.

Upstream investment among international oil companies (IOCs) held relatively steady in 2024, and is expected to remain flat in 2025. However, levels are still well below those seen a decade ago, in part due to a reduction in offshore facility spend and in part due to cost deflation, with 2024 cost indexes still only at 80% of 2014 levels.

Capital expenditures by US independents have recovered since 2020 but remain below pre-Covid levels on a real and nominal basis as LTO capex has not fully returned. Analysis of well productivity gains in the shale patch shows that since 2014 hard-won efficiencies have saved a cumulative USD 220 billion of capex, of which USD 60 billion accrued in 2024. In other words, had drilling and completion operations' capital efficiency stayed static since 2014, to deliver the same amount

of footage drilled and completed US shale spend would have been USD 175 billion in 2024, not USD 115 billion.

US upstream tight oil and gas capex and the impact of well efficiencies, 2014-2024



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Notes: Efficiencies taken as change in feet drilled per day and feet completed per day normalised to Q1 2014 then applied on a capital weighted basis by basin. Capital allocation for US LTO assumed at 30% for drilling, 60% for completions and 10% for facilities. Facility efficiency held constant.

Sources: IEA analysis based on data from the [US EIA](#), Rystad Energy's ShaleWellCube and UCube.

Projects, declines and resources

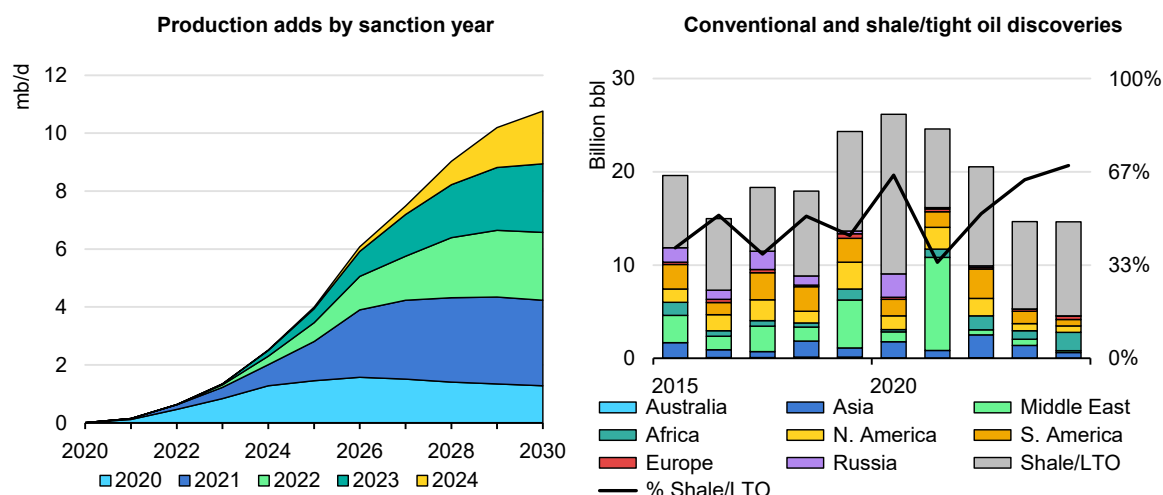
Peak production from conventional projects sanctioned during 2021-24 is expected to average 2.5 mb/d on annual basis, with the 2024 cohort of sanctioned projects adding 2.2 mb/d of annual supply by 2032. This slightly exceeds the 2015-19 previous five-year average peak output of 2.1 mb/d.

Less encouraging has been the result of global exploration efforts over the last decade. Average exploration capital from 2015-24 was 30% lower than the previous ten-year average, at USD 35 billion. Additionally, the correlation factor between exploration capex and oil prices changed from 0.9 over 2000-14 to effectively zero from 2015-24, corresponding with the rise in US LTO. Indeed, the percentage of discovered resources being shale/tight oil has increased from 40% of the global total in 2015 to close to 70% in 2024.

According to Rystad Energy, just under 5 billion barrels of conventional liquid resources were discovered in 2024 – replacing 19% of the conventional production last year. Furthermore, just four projects accounted for approximately half of the discovered volumes. Namibia saw two large discoveries, including Galp Energia's 900 million barrels of oil-equivalent (boe) Mopane and TotalEnergies' 500 million barrels (mb) add at its Venus oil field. ExxonMobil added 400 mb at its

Stabroek Block in Guyana. Rounding out the top five were two Eni discoveries – both 175 mb – at Saasken-Sayulita in Mexico and Calao in Côte d'Ivoire.

Conventional production additions by sanction year and discoveries by region



IEA. CC BY 4.0.

Source: IEA analysis based on Rystad Energy UCube data.

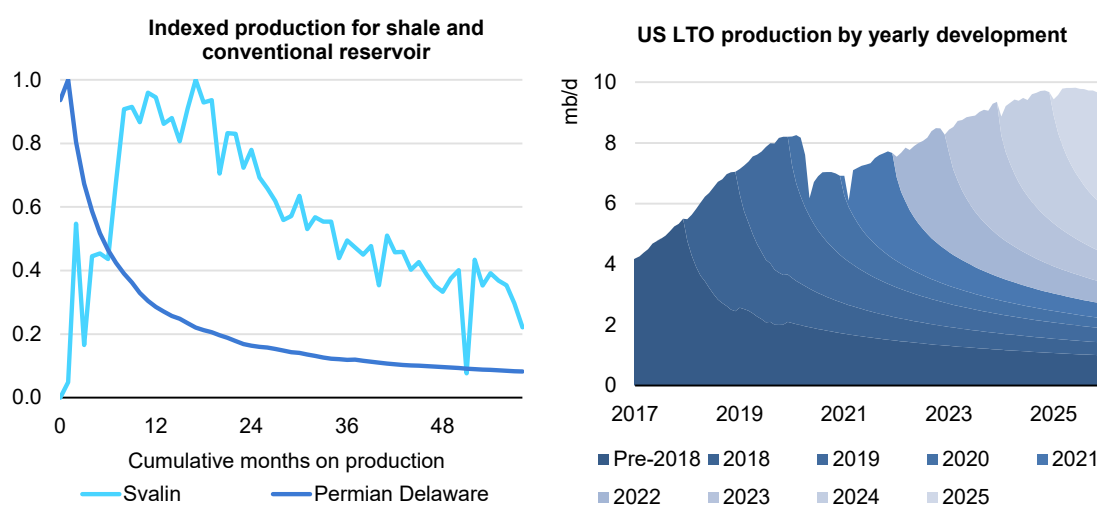
While conventional discovered resources volumes are decreasing, companies are reallocating capital and operating expense (opex) dollars to stem output losses at producing fields. Well capex and field direct opex levels are key determinants of decline rates. Analysis of Rystad Energy data shows that 2024 saw the highest ever level of field direct opex. These investments are critical for operators to maintain existing wells and facilities, as well as for boosting short-term supply by increasing incremental recovery from producing reservoirs through sidetracks, major well workovers and other well interventions. Nominal well capex and field direct opex in 2024 was close to 90% of 2014 highs and the third highest in the history of the industry, whereas facility capex still sat at 75% of its 2014 peak.

This rotation in capital allocation has been due to the shift toward short-cycle investment in shale wells and infrastructure-led offshore expansions, both of which inherently have higher percentages of spend directed towards wells. Additionally, there have been savings in facility spend thanks to a push towards standardisation and 'right-sizing' offshore facilities compared to the last decade.

Observed decline rates at mature fields, accounting for all capex and opex, average close to 6% globally, meaning that around 5 mb/d of supply needs to be replaced annually to hold production flat. The increase in opex since the 2020 nadir has helped hold replacement volumes steady even as more output comes from high-decline shale barrels.

Between January and December 2024, there was a 2.8 mb/d decline in production from existing US shale wells on a base of 8.2 mb/d, while the remainder of non-OPEC+ oil output declined by 2 mb/d on a 40 mb/d base. Due to the high initial decline rate of shale wells compared to conventional wells, continuous drilling and fracking is required to maintain production, and key to determining the path of global supply and investment. The Q1 2025 [Dallas Federal Reserve Energy Survey](#) recently noted that for operators in its region (primarily the Permian and Eagle Ford), existing wells need an average of USD 41/bbl to cover operating expenses and USD 65/bbl to profitably drill new wells.

Indicative field declines and US LTO by development year



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Notes: Permian Delaware median well type-curve. Svalin is a Norwegian Paleocene sandstone reservoir developed in a single stage with no additional development during the period shown. Production is indexed with peak output equal to 1.

Source: IEA analysis based on historical data from Rystad Energy ShaleWellCube and [Norskpetroleum](#).

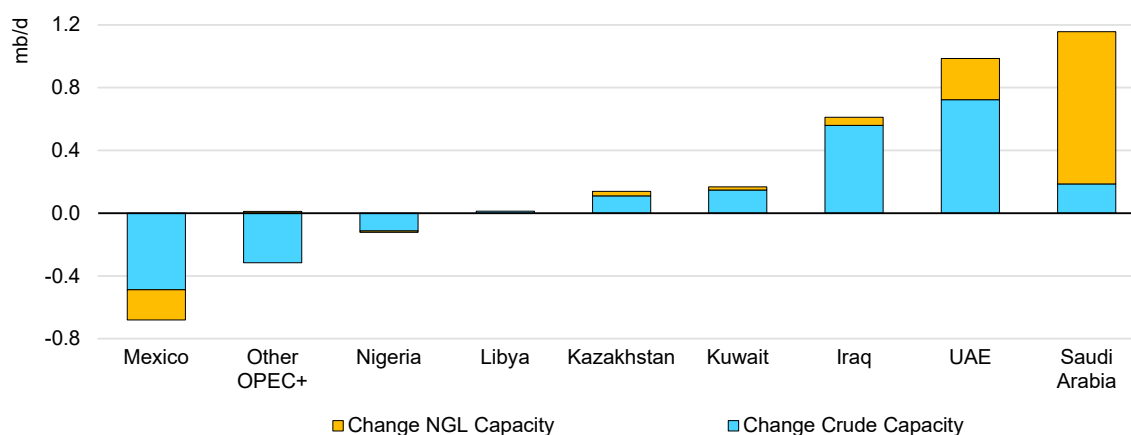
OPEC+ supply

Middle East boosts OPEC+ capacity gains

OPEC+ oil production capacity, including condensates and NGLs, is forecast to grow by a net 2 mb/d from 2024 to 2030, led by Saudi Arabia, the UAE and Iraq (see Table 6 in Annex). Growth in OPEC+ condensates and NGLs capacity – not subject to quotas – comprises over 60% of the net additions, primarily from the first development phases of Saudi Arabia's giant unconventional Jafurah field. The increase in NGLs and condensate output corresponds with the country's strategy to increasingly shift to natural gas production to reduce crude use in power generation, boost feedstocks for expanding petrochemical supply chains and build out NGLs export capacity.

OPEC+ net crude capacity rises by over 810 kb/d between 2024 and 2030. Kazakhstan leads the expansion early in the forecast, following the January 2025 start-up of its Tengiz expansion (260 kb/d). Together, the UAE and Iraq increase crude capacities by 1.3 mb/d from 2024 to 2030, while Mexico is set to dominate total oil capacity losses globally, declining by 630 kb/d to 1.3 mb/d.

Selected OPEC+ crude and NGLs production capacity changes, 2024 vs 2030



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Note: Other OPEC+ = Algeria, Azerbaijan, Bahrain, Brunei, Congo, Equatorial Guinea, Gabon, Malaysia, Oman, South Sudan and Sudan. Iran and Venezuelan capacities are held constant over the forecast. OPEC+ NGLs include condensates.

Total oil production from the OPEC+ alliance fell by 800 kb/d to 49.9 mb/d in 2024. Saudi Arabia led the declines, cutting output by 480 kb/d on average, while Kuwait trimmed 110 kb/d. Production disruptions and outages hit Russia (-260 kb/d), Mexico (-130 kb/d) and Libya (-90 kb/d). Iran and Venezuela, both exempt from supply quotas, boosted output by a combined 530 kb/d on the year, running at their highest levels since 2019. NGLs output from the group, including condensates, reached 8.2 mb/d in 2024. NGLs and condensate supplies from Middle East OPEC+ producers are expected to rise by 1.4 mb/d by the end of the decade, to reach 6.8 mb/d, with Saudi Arabia alone set to increase by 970 kb/d.

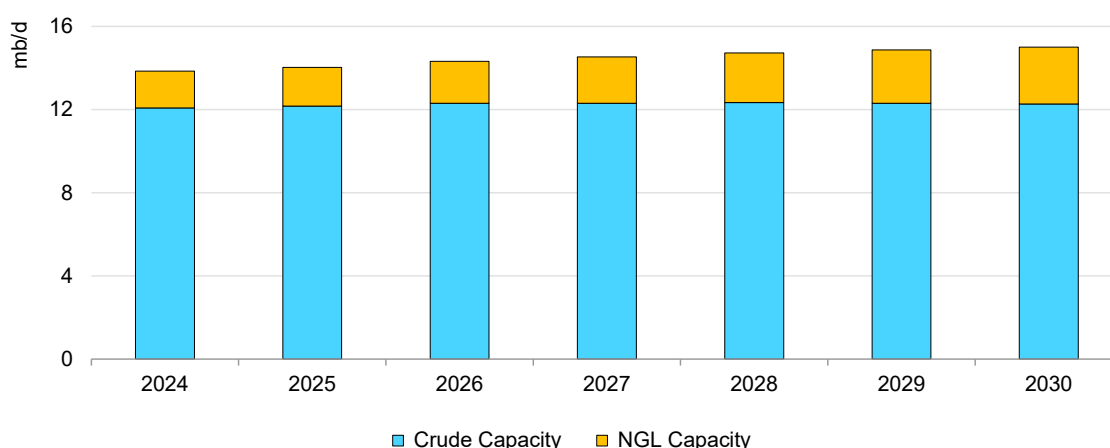
The group of eight OPEC+ countries – Saudi Arabia, Russia, Iraq, the UAE, Kuwait, Algeria, Oman, Kazakhstan – that agreed in November 2023 to cut production by 2.2 mb/d saw their market share decline from 39% in 2022 to 36% in 2024. While overall compliance has been mediocre (in 2024 Russia, the UAE, Iraq and Kazakhstan overproduced by an average of 1.2 mb/d), Saudi Arabia has shouldered the bulk of cuts, with its spare capacity rising to 3 mb/d.

This group of eight countries announced three substantial monthly target increases since April 2025, reaching a total of 1.4 mb/d for the May-July period. Given current levels of overproduction and capacity limits, we assess that only

Saudi Arabia has room to pump substantially higher volumes, bringing an additional 180 kb/d each month on the market from May to July. OPEC+ has noted that it may revise guidance on production levels based on changing market conditions, therefore we have maintained OPEC+ production levels effective from July 2025 and included increased output only for countries producing below their monthly production targets and with spare capacity.

Saudi Arabia is set to lead OPEC+ overall capacity growth through the end of the decade, gaining almost 1.2 mb/d. Saudi total oil production declined by 500 kb/d y-o-y to 10.9 mb/d. In early 2024, Saudi Aramco downgraded its 13 mb/d crude capacity goal to a 12 mb/d maintenance plan (excluding Neutral Zone production). At the same time, it shifted its focus to develop its natural gas resources as well as renewable energy supply, which will allow the country to reduce its reliance on oil burn for power generation. Aramco envisions over 4 bcf/d growth in gas capacity from 2024 to 2030. Coupled with the build-out of renewables, this will displace around 1 mb/d of oil used in power plants of which 500 kb/d is crude oil (see *Saudi plans lead push to slash oil-fired electricity generation*). Phases 1 and 2 of the massive Jafurah unconventional resource play account for half of the growth in natural gas capacity plus a 270 kb/d gain in ethane and 630 kb/d of NGLs and condensates by the end of the decade.

Saudi Arabia crude and NGLs production capacity, 2024-2030



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Notes: Crude capacity includes Saudi Arabia's share of Neutral Zone production. NGLs capacity includes condensates.

While Saudi Arabia's crude oil production capacity is largely steady over the forecast period, Aramco is investing heavily to offset field declines. Over 1.2 mb/d of crude maintenance projects are reported to come onstream between 2025 and 2026. The Dammam Phase 1 water injection project of 25 kb/d is slated for a 2025 start-up followed by a 50 kb/d incremental Phase 2 in 2027. Offshore crude

megaprojects, Berri and Marjan, are scheduled to bring 550 kb/d of combined production onstream, with NGLs processed at the Tanajib Gas Plant. Tanajib is planned to handle 2.6 bcf/d of natural gas offtake from the Marjan and Zuluf fields. Arabian Heavy crude grades will see a lift from 2026 onwards with the commissioning of the 600 kb/d Zuluf central processing facility. The additional heavy crude supply would offset losses of Mexican heavy grades over the forecast period.

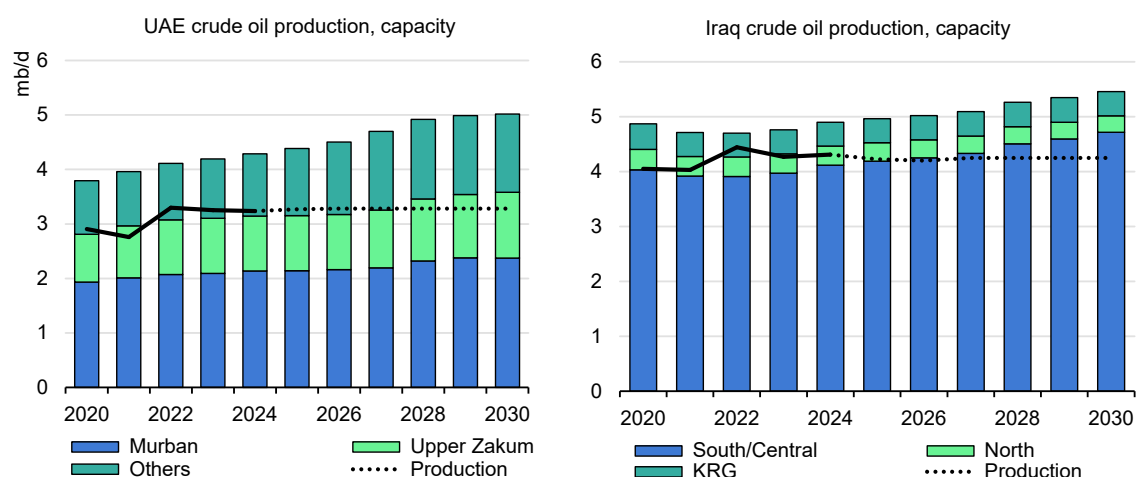
The **UAE** is set to lead global crude oil capacity additions, adding over 720 kb/d by 2030. The country will also increase NGLs capacity by 260 kb/d as it charts its path to become a natural gas net exporter by 2030. Oil production, including NGLs and condensates, was roughly flat at 4.2 mb/d in 2024. The Emirati's ambitious 5 mb/d crude capacity plan by 2027 received a boost last month after agreements were signed with operator ExxonMobil and partner INPEX/JODCO for the Upper Zakum expansion project, designed to raise production by 200 kb/d to 1.2 mb/d.

The Abu Dhabi National Oil Company's (ADNOC) investments have focused on offshore crude capacity growth in recent years. Efforts to boost Upper Zakum supply have lifted output over the last decade from about 500 kb/d to 1 mb/d. Last year, the 45 kb/d Belbазem offshore block started up. Contracts have also been awarded to develop incremental production, including the 20 kb/d project to help stabilise output at the 450 kb/d Lower Zakum field, with a further 50 kb/d expansion planned by 2027. At the Umm Shaif field, an uplift of 115 kb/d, split into three separate phases, is scheduled to raise capacity to 390 kb/d by 2027.

Recently, attention has turned back onshore with planned incremental capacity projects at Bu Hasa (+100 kb/d) and the Bab field (+90 kb/d). Like Saudi Arabia, the UAE is actively appraising its unconventional resources. In onshore Abu Dhabi, ADNOC Drilling is utilising its Turnwell joint venture, in partnership with SLB and Patterson-UTI, to drill over 140 wells by the end of 2025 and test the extent and productivity of the country's unconventional resources. In May 2025, EOG Resources, one of the largest US shale producers, signed exploration licenses for the massive Unconventional Onshore Block 3 in the Al Dhafra region. While no additional unconventional capacity growth volumes are held in our forecasts, success could support the UAE's crude and gas ambitions.

With the Emirates targeting gas self-sufficiency by 2030, ADNOC Gas is raising its natural gas processing capacity with an associated increase in NGLs. The Maximising Ethane Recovery and Monetisation (MERAM) project, expected to start-up in late 2025, will extract around 120 kb/d of ethane at Habshan that can be used as feedstock at Borouge's expanding fleet of steam crackers. Several upcoming projects, including the 1.5 bcf/d Rich Gas Development and 1.8 bcf/d Bab Gas Cap project, have had contracts recently awarded and their liquids capture are not yet included in capacity estimates.

UAE and Iraqi crude oil production and capacity, 2020-2030



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Note: Forecast production for 2025-2030 is based on OPEC+ supply targets effective July 2025 and estimated levels of compliance.

Iraq is forecast to increase its capacity by 560 kb/d to 5.4 mb/d by 2030, short of its 6 mb/d target. Crude expansion plans are hindered by a lack of water injection capacity, midstream challenges and limited export capacity from its southern fields. Most capacity growth is driven by foreign companies working in the south as Iraq seeks to streamline the Basra port and provide alternative crude export options. The recent re-entry of BP at the Kirkuk field, after its 2019 exit, and the potential re-opening of the Iraq-Türkiye Pipeline (ITP), provide options for longer-term gains. The country's total oil production gained 40 kb/d y-o-y to reach 4.5 mb/d in 2024.

A lack of water injection capacity for pressure support across many Iraqi fields constrains the country's ability to increase production. TotalEnergies is developing infrastructure and local capabilities at its 5 mb/d seawater treated injection project for the southern fields of Zubair and West Qurna. The company also intends to raise output at the Ratawi oil field from 85 kb/d to 210 kb/d.

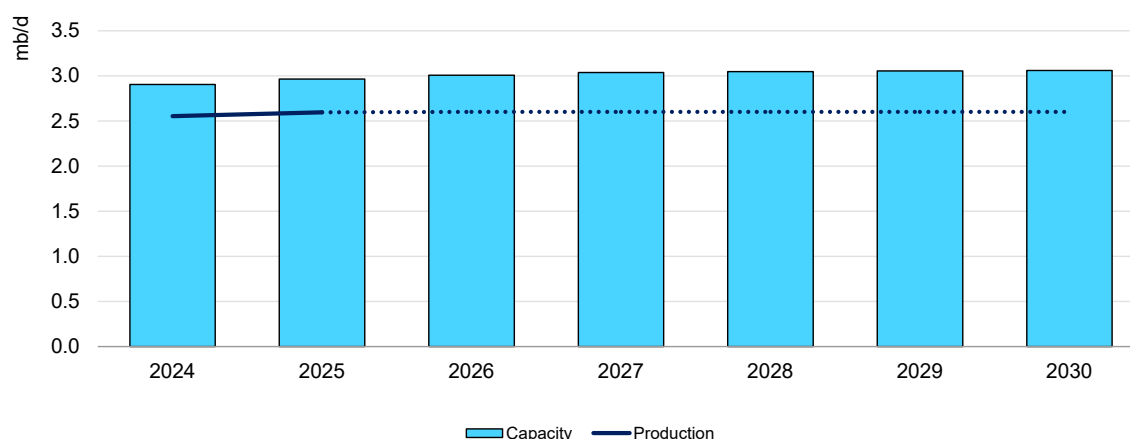
Iraq has made significant strides in the last year to debottleneck the Basra port. In April, Basra Oil Company signed a contract with Italy's Micoperi and the Turkish firm ESTA to build a third subsea export pipeline, design capacity 2.4 mb/d, and construct loadings options including a floating single point mooring platform. At the end of 2024, the Iraqi government approved the 2.25 mb/d capacity, 685 km Basra-Haditha pipeline project that will run parallel to the dilapidated 800 kb/d Strategic Pipeline. The new line will act as a conduit for oil from the Basra region to demand centres in central Iraq and, in the long-term, could connect southern fields to northern export routes.

Including Ratawi, Iraq has over 1.2 mb/d of potential capacity additions in different phases of maturity being worked across southern fields. For example, PetroChina is proceeding with expansion plans at West Qurna-1 to raise production from a reported 550 kb/d towards 800 kb/d near the end of the decade, but no formal project sanctioning has been announced. In late 2023, Lukoil and Baghdad agreed to extend the service contract for the West Qurna-2 field to 2045, with plans to gradually boost capacity from 480 kb/d to 800 kb/d. Eni continues work on the 250 kb/d expansion at the 450 kb/d Zubair field. Basra Oil Company is progressing work on Majnoon, currently producing around 130 kb/d, to reach 450 kb/d, but project delays have continued to plague the field's expansion.

The northern Kirkuk oil fields and the capacity that is controlled by the Kurdistan Regional Government (KRG) may see a resurgence in the coming years if shipments of 450 kb/d along the ITP to Ceyhan ultimately resume following long months of negotiation between Federal Iraq, the KRG, and foreign oil companies in the north and Türkiye. Flows on the line have been halted since the end of March 2023, but around 300 kb/d is reportedly moving into the local market or trucked across Iraqi borders. BP stopped work on the Kirkuk re-development in 2019 due to security concerns, but its recent re-entry may buoy output from the world-class resource into the next decade.

Kuwait is projected to deliver a 150 kb/d increase in capacity to nearly 3.1 mb/d by 2030, while endeavouring to deliver its 3.5 mb/d capacity target by 2035. Annual average crude oil production in 2024 slipped 110 kb/d to 2.55 mb/d including contributions from the Neutral Zone. Kuwaiti capacity has been on the decline, losing 220 kb/d since its peak in 2018.

Kuwait crude oil production and capacity, 2024-2030



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Note: Forecast production for 2025-2030 is based on OPEC+ supply targets effective July 2025 and estimated levels of compliance.

Kuwait Oil Company (KOC) has focused its attention to the south at its giant, 80-year-old Burgan oil field, but efforts to stem declines have proven challenging. Last year, KOC brought on the northern Jurassic gas development, adding about 50 kb/d of light crude oil capacity. Gains at the Ratqa heavy oil field, supported with technical work by Shell, are boosting capacity by another 20 kb/d this year. Recent exploration success offshore Kuwait may yet provide a more direct path to achieving the country's 2035 targets.

Capacity in the Neutral Zone, shared equally between Kuwait and Saudi Arabia, is expected to remain around 260 kb/d for each country through 2030. Recent strides to increase production could add some upside, however.

Oil output in **Oman**, including condensates and NGLs, was around 1.1 mb/d in 2024. Omani capacity is forecast to increase by 40 kb/d over the outlook period with continued investments in heavy oil fields and offshore developments including Yumna offshore. A recently signed USD 30 billion agreement with Occidental Petroleum, extending its development plans for the Mukhaizna field, is expected to help curb declines. Oman's focus on natural gas, including the anticipated start-up of Marsa LNG's 1 Mtpa train in 2028, will provide a small boost to the country's condensate and NGL supply.

Looming Caspian capacity declines for Kazakh and Azeri crude by 2030

In **Kazakhstan**, total oil supply in 2024 edged lower by 50 kb/d y-o-y to 1.9 mb/d, due to maintenance at its largest producing fields Tengiz, Kashagan and Karachaganak. The Future Growth Project (FGP) at Tengiz, led by Chevron, has expanded field capacity by 260 kb/d to 860 kb/d in January 2025, bringing overall crude capacity to 2.1 mb/d. Following the 2023 launch of the Tengiz Wellhead Pressure Management Project and the FGP start-up, Tengiz now accounts for over 40% of Kazakh oil output. Kazakh crude capacity is projected to fall to 1.9 mb/d by 2030 until larger field expansions, such as those at Kashagan, are given a firm go-ahead.

In October 2024, the Kazakhstan senate further strengthened ties with Qatar, approving the establishment of a long-term strategic partnership across key sectors including energy and agriculture. Agreements were sanctioned for QazaqGaz and UCC Holdings, a major Qatari construction and infrastructure company, to finish a 100 million standard cubic feet per day (MMscf/d) sour gas processing plant that supports the Kashagan Phase 2A development. The plant and its associated pipelines will help raise crude capacity by an estimated 25 kb/d to 440 kb/d, with a reported start-up in late 2026. Agreements for subsequent gas plants associated with the Kashagan development were signed but are not yet assumed as firm in the forecast.

Azerbaijan's total oil supply dipped 20 kb/d in 2024 to 600 kb/d. Output stays at roughly this level as BP continues to ramp-up the 100 kb/d capacity Azeri Central East (ACE) offshore platform over the next few years. ACE, along with Shah Deniz compression, is expected to help bolster the country's total oil capacity in the near-term and cushion declines before slumping to 520 kb/d by 2030.

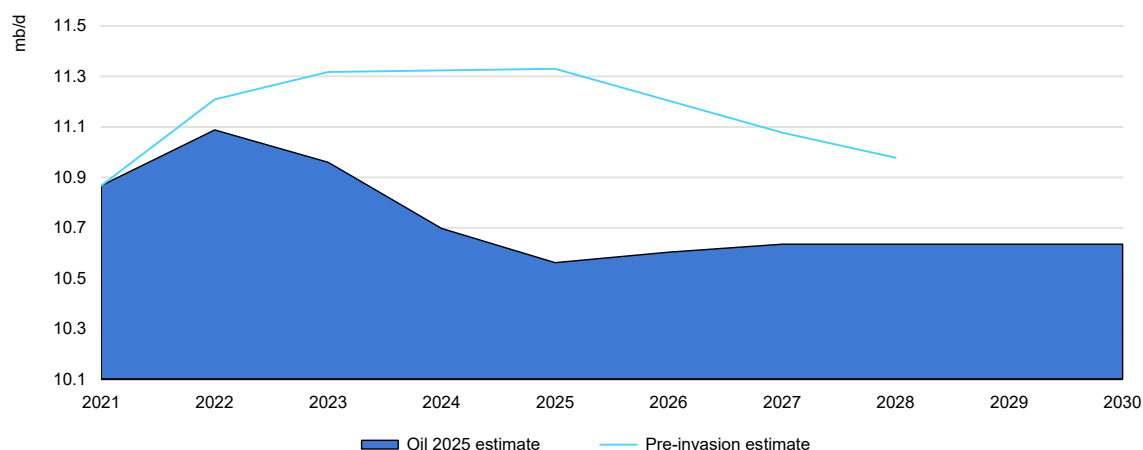
Sanctions stifle capacity gains but barrels still flow

The increased tensions between Iran and Israel and the impact of western sanctions on Russia, Iran and Venezuela remain a key uncertainty in our forecast. Access to technologies, financing and supply chain inputs, as well as changes as to who is buying sanctioned barrels, may be further impacted depending on the outcome of ongoing negotiations. For Russia and Iran, we have held capacity and production at current levels throughout the forecast period. In the case of Venezuela, current sanctions guidance from the US government has put a dent into the country's production capacity outlook.

Russian oil production dipped by a further 260 kb/d in 2024, to 10.7 mb/d, taking total production losses since the outset of the invasion of Ukraine to 600 kb/d. Nonetheless, the Russian oil sector has proven surprisingly resilient, with reports of western Siberian drilling ticking up. Yet sanctions appear to have muted near-term growth ambitions. We hold Russian capacity broadly flat at 10.9 mb/d to 2030, but further delays to Rosneft's giant, 600 kb/d Vostok Oil project may dampen Russian output in the medium term. Significant strides in building up domestic supply chains and key technologies, such as heavier drill rigs to target longer horizontal wells and development of new fracking completion fluids, have supported efforts to sustain production, but a dearth of equipment for hard-to-recover reserves in the Arctic and lack of shipbuilding technologies have pushed back the start-up of key projects. The war and the resulting economic impact have also held back investment due to all-time high interest rates, higher corporate taxation and tight Russian labour markets.

Moscow's Arctic reserves could provide a significant boost to capacity, but this has proven difficult amid tightening sanctions. In Q1 2025, Rosneft pushed back the start-up of its Vostok Oil project from 2024 to 2026. Development of Vostok is split roughly evenly between the currently producing Vankor and nearby Suzunskoye and Lodochnoye fields, with the other half from Payakha, Ichemminskoye and Baikalovskoye. While oil reserves for Vostok Oil were bumped up last year by nearly 10% to over 50 billion barrels, the ability to export oil from the fields is constrained. Progress continues on the 770 km pipeline, with just under half completed at the end of 2024, and on the receiving oil terminal at Sever Bay on the Arctic Kara Sea. Arctic LNG-2 has been scaled back and delayed from 2026 to 2028 as Novatek works on a liquefaction technology and sources LNG icebreaker tankers.

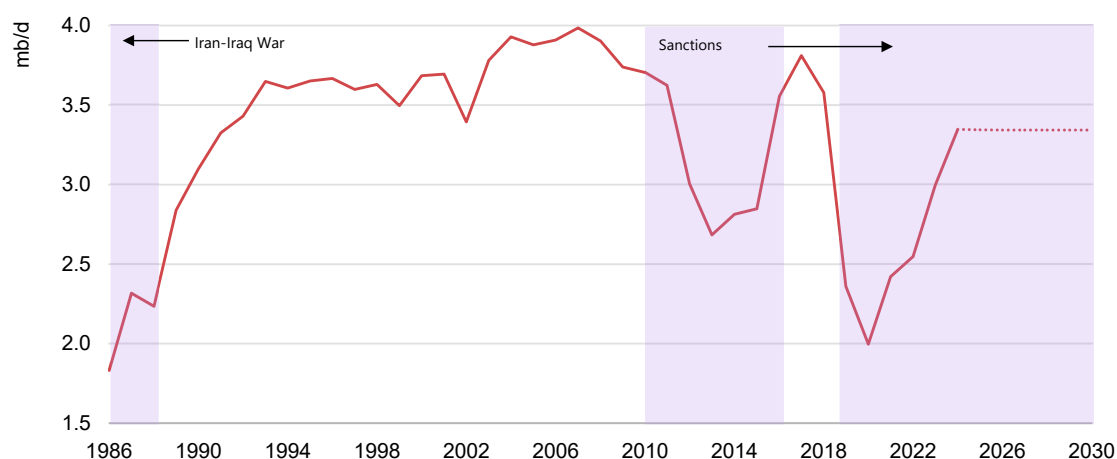
Russia estimated total oil supply, 2021-2030



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Iran's oil supply has, thus far, continued to flow at a relatively robust levels in the face of increasing US sanctions and tensions with Israel, making it the second largest source of supply growth after the United States for two years running. Total oil production last year reached its highest level since 2017, rising 420 kb/d y-o-y to 4.7 mb/d, of which 1.3 mb/d was condensates and NGLs. Crude exports averaged 1.6 mb/d in 2024. Nearly all Iranian oil exports land in China, with crude destined for use in independent “teapot” refineries. US sanctions have grown progressively more severe in recent months with nearly all aspects of Iranian supply chains currently targeted.

Iran crude oil production, 1986-2030



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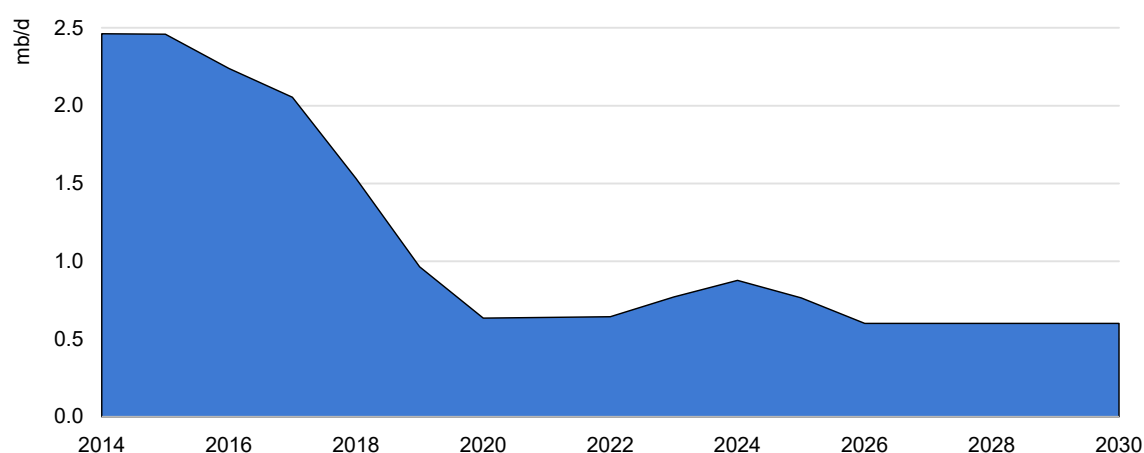
The ramp-up in Iranian output has left the country with slim spare capacity. Tehran relies heavily on domestic firms to try to stem field declines. Since the start of last year, Iran has awarded over USD 33 billion worth of contracts locally to raise production from its natural gas and oil fields. Additionally, the country continues to make strides to develop its capabilities to manufacture and operate technologies critical for improved oil recoveries, such as large field compressors.

There are several sources of Iranian capacity growth on the horizon with its core West Karun fields, together with Yaran and Yadavaran, capable of driving around 1 mb/d of growth. Contracts were awarded in March 2024 to expand onshore field capacity, aimed at raising production from 400 kb/d to 620 kb/d. Iran is also working on options to increase output at smaller fields in West Karun, including Jofeyr and Sepehr.

Oil continues to flow, on pace with 2024 levels, but recent hesitancy from Chinese buyers has reduced crude imports from Iran by 30% in May over last year's average. Ongoing negotiations with the United States on the future of Iran's nuclear programme cast a shadow of uncertainty on the path forward for sanctions. Swift enforcement of US sanctions in 2018 slashed Iranian crude exports from an average of 2.2 mb/d to less than 500 kb/d but enforcement of US sanctions this time around has proven more challenging.

Venezuela increased its total oil supply by nearly 110 kb/d last year, to 960 kb/d, its highest level since 2019. Venezuelan crude capacity is estimated at over 1 mb/d in 2025, with the largest capacity gains from western fields (+30 kb/d) and heavy oil producers (+70 kb/d), including joint ventures with Chevron, China National Petroleum Company (CNPC) and Rosneft.

Venezuela crude oil production, 2014-2030



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In Q1 2025, the United States announced it was reversing the US Department of Treasury's Office of Foreign Asset Control's (OFAC) 2022 Venezuela General License 41 (GL 41) effective 3 April, and other licenses that had allowed western companies including Chevron, Eni, Repsol and others to operate, trade and pay taxes in Venezuela. The United States doubled down on 2 April, announcing secondary sanctions on buyers of Venezuelan oil. Chevron's general license expired on 27 May. Western operators in the country continue to discuss their path forward with the US government.

PDVSA is highly dependent on diluent imports (particularly naphtha or light condensate) for blending Venezuelan heavy crude which constitutes over half of the country's crude capacity. While imports of diluent dipped after the latest US announcements, new supplies from Russia, China and Iran have been sourced since May. The establishment of new diluent supply chains not reliant on western sources allows Venezuela to maintain its upstream activities. However, the risk of secondary sanctions and US tariffs on buyers of Venezuelan oil has slowed oil exports by 220 kb/d since February. While the situation remains fluid, we have assumed crude production declines of 30 kb/d per month to 600 kb/d by year-end.

African OPEC+ members entice foreign interest

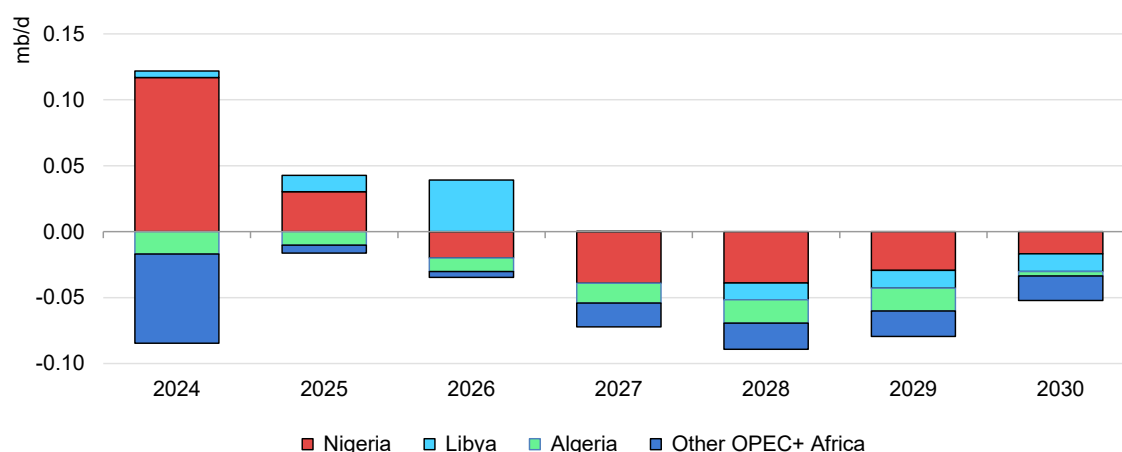
African OPEC+ members are expected to see crude capacity fall by 260 kb/d to 4.2 mb/d by 2030, largely due to difficulties in attracting investments needed to offset field declines and raise production. However, recent changes to contract terms in several countries may yet encourage foreign capital inflows towards the end of the decade. In Nigeria, a broad series of reforms across the petroleum sector were introduced in 2024 and 2025, including new tax incentives for deepwater exploration and production, and the first fiscal framework changes since 1991. Algeria revised contracts for its auction last year, reducing government take by about one-third. Libya, as part of its first exploration round in 18 years, introduced a new production sharing agreement that could see foreign company returns increase from 2.5% up to 35.8%.

Nigerian total oil production ticked up markedly towards the end of 2024 to 1.6 mb/d, at 1.4 mb/d of crude and 210 kb/d of condensates, in part due to concerted government action against theft and sabotage and supported by the start-up of the Utapate field (+40 kb/d). In our 2024-30 forecast, Nigerian crude oil capacity is projected to average near 1.6 mb/d in the next few years based on plans to sustain production at existing projects. Further out, sanctioned projects will help slow, but not reverse, steep declines, with crude capacity ending the decade just below 1.4 mb/d.

President Bola Ahmed Tinubu, elected on promises of reform to the country's energy sector in May 2023, implemented major policy, regulatory and fiscal

reforms aimed at revitalising investment in the country's oil and gas sector. The government's new policies mark a seismic shift to focus on the country's rich oil and gas offshore, with more favourable contract, fiscal and tax terms for investors. Changes in the framework of production sharing contracts (PSC) and joint operating agreements (JOA) are intended to remove bureaucratic bottlenecks, streamline licensing processes, provide improved fiscal terms for deepwater developments, and incentivise production of natural gas and its associated liquids. Additionally, in April President Tinubu dissolved the board of the Nigerian National Petroleum Company and appointed a new board and executive leadership team, with a mandate to enhance operational efficiency and commercial viability. The government has set highly ambitious targets for the country's total oil production, at 2 mb/d in 2027 and 3 mb/d by 2030.

OPEC+ Africa crude oil production capacity (y-o-y change), 2024-2030



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While Nigeria has seen the exit of most major foreign companies from its onshore portfolio, including ExxonMobil's sale of Mobil Producing Nigeria Unlimited to Seplat Energy and Shell's sale of its onshore assets to Renaissance, the government's sweeping reforms have sparked cautious optimism and increased investor interest. ExxonMobil announced plans in May to invest USD 1.5 billion to revitalise production at its flagging Usan deepwater oil field, with a final investment decision (FID) expected in late Q3 2025.

This follows the sanctioning in late 2024 by a Shell-led consortium of the Bonga North development with a capacity of about 110 kb/d, aimed at sustaining production at the Bonga floating production storage and offloading (FPSO) facility, with first oil expected towards the end of the decade. Shell is also reviewing the potential sanctioning of Bonga Southwest (150 kb/d crude and condensate capacity) with an FID expected in 2027. However, in late May TotalEnergies

entered into an agreement with Shell to sell its non-operating interest in the Bonga fields, which includes Bonga North and Bonga Southwest, as the companies work their respective strategies to optimise and upgrade their portfolios. Despite Nigeria's expansive new incentive and contract terms, international companies appear to be taking a watchful approach, waiting to see how the implementation and delivery of the reforms are executed.

Libyan total oil output for 2024 was down 90 kb/d y-o-y to 1.1 mb/d following a six-week shut-in of fields during the country's banking crisis in August and September of last year. Well workovers during the shut-in helped to push crude output up to 1.2 mb/d at year-end. With over 90% of Libya's revenues coming from the oil and gas sector, the fields and oil terminals are frequently targeted by political factions.

In March 2025, Libya announced for the first time since 2007 significantly improved production sharing agreement terms for foreign companies, along with a licensing round for 22 exploration areas which so far has attracted over 40 applicants. Contract awards are scheduled for 22-30 November 2025. However, any exploration and development pick up in Libya will fall outside of our forecast period.

Libyan capacity reaches 1.25 mb/d this year before declining below 1.2 mb/d by 2030. Smaller capacity gains, such as those at the recently restarted Mabruk field, will add about 25 kb/d, following the field's shut-in since 2015. Further near-term gains may come from Akakus (a joint venture of the Libya's NOC, Repsol, TotalEnergies, OMV and Equinor) as it works an ambitious plan to increase capacity at the Sharara field blocks NC-186 and NC-115, from 300 kb/d to 350 kb/d, by the end of 2025.

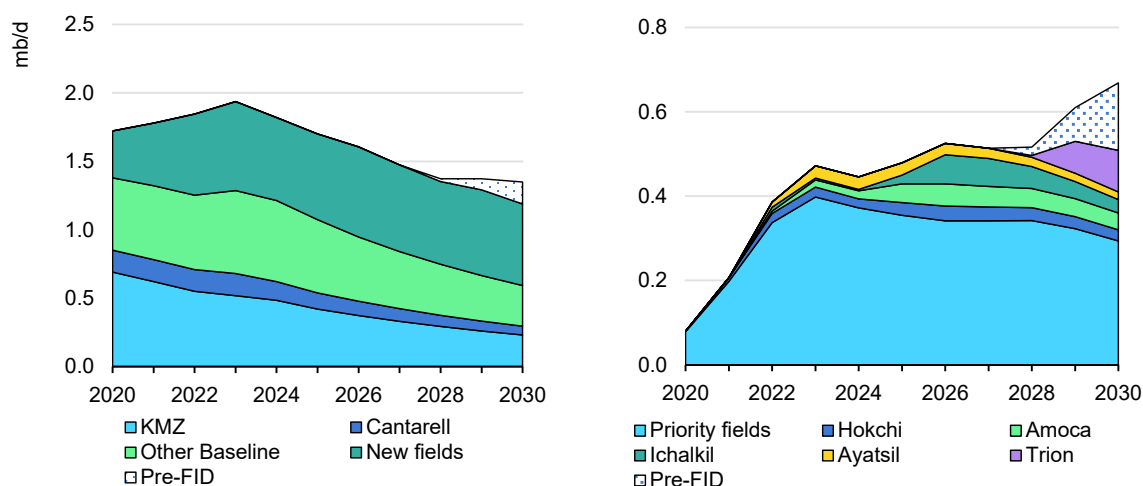
Mexico risks becoming a net importer as output declines

Mexico posts the largest drop in output, not just among the OPEC+ alliance, but out of all producers globally – falling 680 kb/d to 1.3 mb/d over the forecast period. Paired with this *Report's* demand forecast, this could lead the country to become a net importer of close to 500 kb/d by 2030. Its long-term oil production decline showed a brief respite from 2021-23 as the Quesqui condensate field ramped up in earnest. Pemex severely curtailed planned investments during the pandemic, with the previous administration requesting the state oil company focus on quick crude growth from onshore and shallow-water fields to the detriment of larger deepwater reservoirs. As of 2024, over half of Pemex's production came from just seven of its 240 fields.

Looking forward, challenges remain with Pemex carrying a high debt load and only one major project slated to see first oil by 2030. Output declined by close to 160 kb/d y-o-y in 1H 2025. Fiscal changes, large unpaid debts to its suppliers and

upstream budget cuts have seen oil rigs slashed from 50 in October 2024 to fewer than 20 in less than six months – although recently some payments reportedly have been made and five rigs have returned to work. Woodside's 100 kb/d Trion project is still targeting first oil in 2028 while Zama and the Ku-Maloob-Zaap (KMZ) expansion still await sanctioning. Yet the window to see production from these two developments before the end of our forecast is closing.

Mexico total production and growth from new fields, 2020-2030



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Non-OPEC+ supply

Non-OPEC+ Americas quintet to continue supply growth

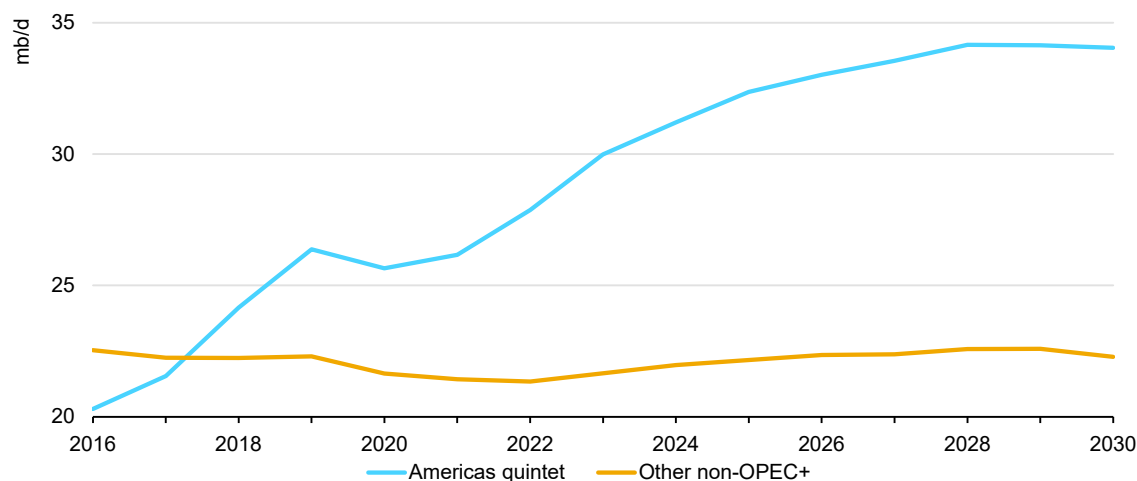
The Americas dominates total non-OPEC+ supply increases of 3.1 mb/d through the medium-term, even as the pace of growth slows markedly. Following gains of as much as 11 mb/d since 2016, the quintet – comprised of the United States, Canada, Brazil, Guyana and Argentina – will boost production a further 2.8 mb/d by 2030, with their combined output approaching one-third of global oil supply.

The United States remains the largest single source of non-OPEC+ gains, even as US LTO growth stalls. Canada comes in a close second as expanded pipeline egress helps facilitate bitumen expansions and shale drilling brings incremental NGL volumes.

Guyana, powered by the ExxonMobil-led Stabroek Block, continues to execute its ambitious growth targets, while neighbouring Suriname will see the TotalEnergies-led offshore Block 58 produce its first barrels by the end of the decade. Further south, Brazil continues to ramp up production, albeit at a slower pace than expected previously, with Petrobras and other large oil companies

continuing the FPSO factory development of pre-salt reservoirs. Vaca Muerta growth will be boosted by new takeaway capacity and as economic reforms make their way through the Argentinian economy.

Americas quintet has grown while other non-OPEC+ has stayed flat, 2016-2030



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Note: Chart includes total oil supply (crude, condensates, NGL, biofuels and non-conventional). Americas quintet includes the United States, Canada, Brazil, Guyana and Argentina. Global biofuel production included in other non-OPEC+.

Total non-OPEC+ supply (mb/d)

	2024	2025	2026	2027	2028	2029	2030	2024-30
OECD	29.9	30.7	31.0	31.2	31.2	31.1	30.9	1.0
OECD Americas	26.3	27.0	27.3	27.7	27.8	27.8	27.9	1.6
OECD Europe	3.2	3.3	3.3	3.1	3.1	2.9	2.7	-0.5
OECD Asia Oceania	0.4	0.4	0.4	0.4	0.4	0.4	0.4	-0.1
Non-OECD	17.5	17.9	18.2	18.5	19.1	19.2	18.8	1.4
Eurasia	0.3	0.3	0.3	0.3	0.3	0.3	0.3	-0.0
Europe	0.1	0.1	0.1	0.1	0.1	0.1	0.1	-0.0
China	4.3	4.4	4.4	4.4	4.3	4.3	4.2	-0.1
Other Asia	2.0	2.0	1.9	1.9	1.9	1.9	1.9	-0.1
Non-OECD Americas	6.4	6.8	7.2	7.3	7.7	7.8	7.6	1.2
Middle East	1.9	1.9	2.0	2.0	2.3	2.4	2.5	0.6
Africa	2.4	2.4	2.4	2.5	2.5	2.4	2.3	-0.1
Non-OPEC+ Oil	47.4	48.6	49.2	49.7	50.4	50.2	49.8	2.4
Processing Gains	2.4	2.4	2.5	2.5	2.5	2.5	2.5	0.1
Global Biofuels	3.4	3.5	3.7	3.8	3.9	4.0	4.1	0.7
Total Non-OPEC+ Supply	53.2	54.5	55.4	55.9	56.7	56.7	56.3	3.1
Annual Change	1.5	1.4	0.8	0.6	0.8	-0.0	-0.4	

Note: OECD Americas excludes Mexico, which is included in OPEC+ data.

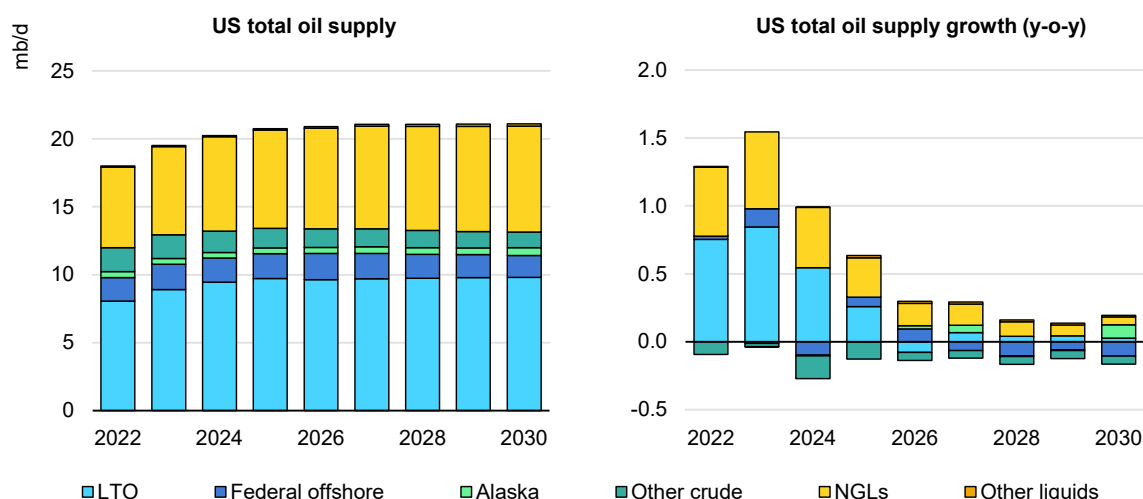
In the Middle East, Qatar is set to notch impressive increases in associated liquids from their North Field East and South LNG expansions. Global biofuels output will grow by 680 kb/d, with non-OECD countries accounting for 75% of the gains. Recent major discoveries in Namibia and Côte d'Ivoire are advancing through the exploration and appraisal lifecycle, while other small West African producers continue to see growth. Angolan output is flat over the forecast period, as new projects offset declines at mature assets. The Lake Albert development in Uganda continues to progress, with first oil scheduled for 2026.

Other non-OPEC+ regions, however, continue to slump. North Sea supplies are set to decline without a robust queue of new projects to be sanctioned. A decade-long downtrend in ex-China Asia Pacific oil supply continues, as companies prioritise natural gas developments. Other legacy South American producers have seen increased political risk hampering an already challenging development outlook.

US shale growth slows precipitously

US liquids output, excluding biofuels, expanded by 720 kb/d in 2024, and by a total 3.7 mb/d since the Covid-19 trough in 2020. Oil supply is expected to rise by a further 510 kb/d in 2025, reaching a fourth consecutive annual record high. While US oil production is set to increase every year through the forecast period, the pace of growth will slow markedly as crude plateaus. Greater shareholder distributions, producer consolidation and increasingly complex wells amid a possible 'lower for longer' price environment act as headwinds to the industry.

US growth expectations moderate over the remainder of the decade, 2022-2030



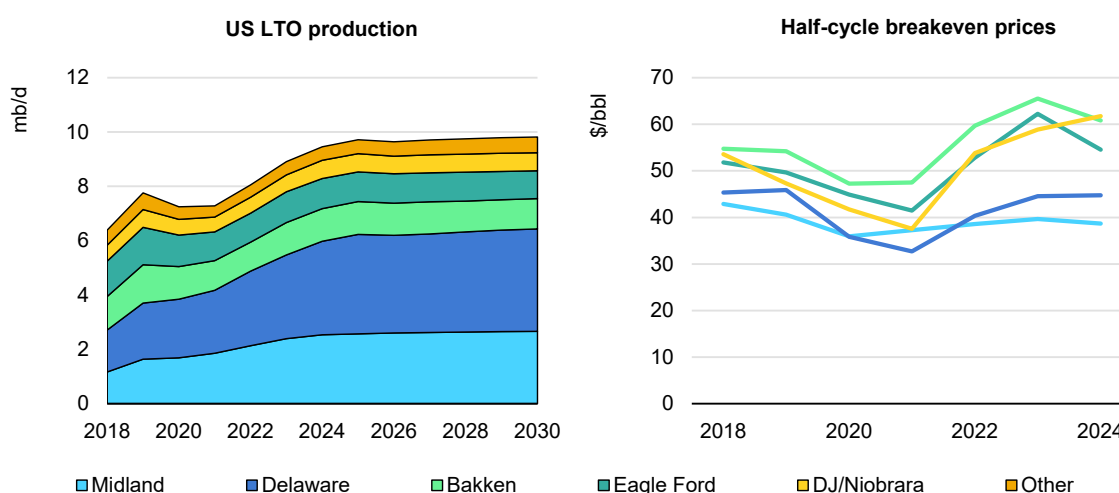
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Even then, the United States is the second-largest contributor to medium-term supply capacity growth (behind Saudi Arabia), adding 880 kb/d over the 2024-30 period, to reach 21.1 mb/d by the end of the forecast. Crude oil production is expected to decline by 60 kb/d to 13.1 mb/d. NGLs from processing plants are set to rise by 860 kb/d to 7.8 mb/d, led by both higher exports and domestic petrochemical facility utilisation as associated gas from the Permian and Appalachia Basins continues to drive growth. Non-conventional volumes make up the balance.

US LTO production increases by 360 kb/d from 2024 to 2030, reaching 9.8 mb/d, while conventional Lower 48 supply is expected to decline by 420 kb/d over the same time frame. Federal offshore output is forecast to rise by 160 kb/d between 2024 and 2026 to 1.9 mb/d, before retreating to 1.6 mb/d by 2030. Alaskan volumes are set to increase by 160 kb/d to 580 kb/d by the end of the decade.

While LTO remains the major engine of US oil supply growth, annual gains slow precipitously, with production essentially plateauing in the later years of the decade. Total shale production rises by a modest 0.5% annual average, with the Permian Basin essentially the sole contributor. The reassessment of US shale prospects compared to *Oil 2024* is due to lower prices that have prompted shale producers to scale back activity and industry consolidation. Indeed, data from Enverus show that over 70% of the sub-USD 50/bbl economic Permian Basin locations are operated by seven companies, with ExxonMobil and Diamondback holding over half the remaining locations in the Midland Basin.

Permian Basin drives US LTO growth with the lowest breakeven prices, 2018-2030



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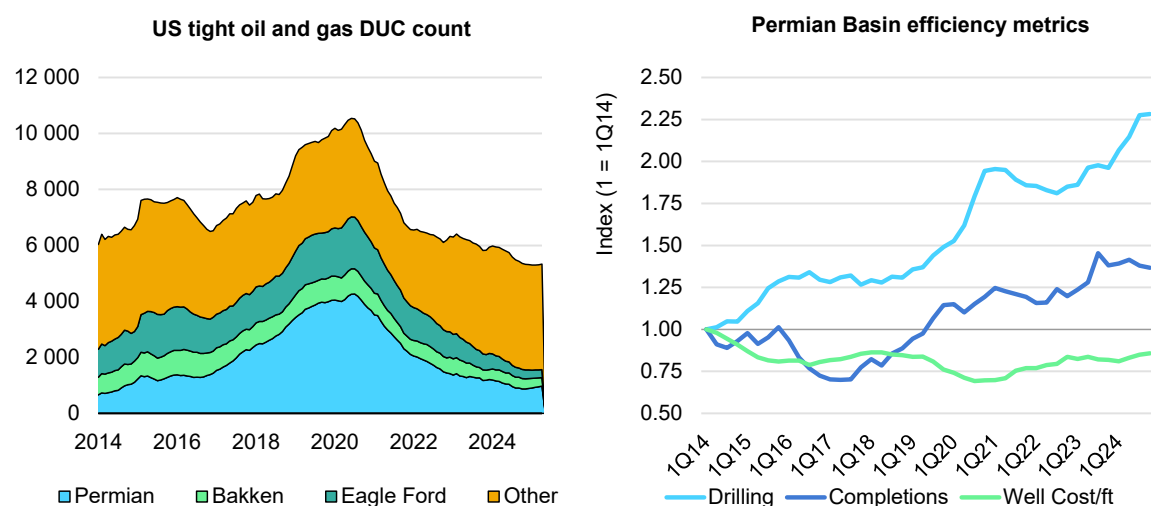
Note: Half-cycle breakeven prices are based on drilling and completion costs net of royalties, production tax, lease operating expenses, transportation costs and differentials, general and administrative charges and corporate tax.

Source: Half-cycle breakeven prices based on IEA analysis of data from Rystad Energy ShaleWellCube.

Additional headwinds stem from low levels of drilled but uncompleted wells (DUCs) within US shale basins. According to data from the Energy Information Administration's Short-Term Outlook (EIA STEO), DUC levels are at the lowest level since the government began tracking in 2013 and have fallen by 15% over the last two years to 5 300 wells, with the industry on pace to complete just shy of 9 000 wells this year. Key oil producing areas such as the Bakken, Eagle Ford and Permian have seen steep DUC drawdowns over the past three years, with the Permian Basin count currently sitting at only one-third of its pre-Covid average.

DUCs can normally be fracked and brought online in two months compared to the 9-12 months it takes to drill and bring on a new well. The continued just-in-time inventory model reduces operational buffers and complicates field development planning and optimisation. Combined with reduced activity, this presents headwinds to growth over the coming years.

US DUCs are at record lows while Permian Basin drillers keep getting better, 2014-2030



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Notes: DUC counts include oil and gas wells drilled in oil and gas formations. Drilling efficiency is the average total measured depth (TMD) for horizontal oil wells in the Permian Basin divided by average drilling days. Completion efficiency is TMD divided by active frac days. Well costs are on a total well basis normalised to TMD.

Source: IEA analysis based on data from the [EIA STEO](#) and Rystad Energy's ShaleWellCube.

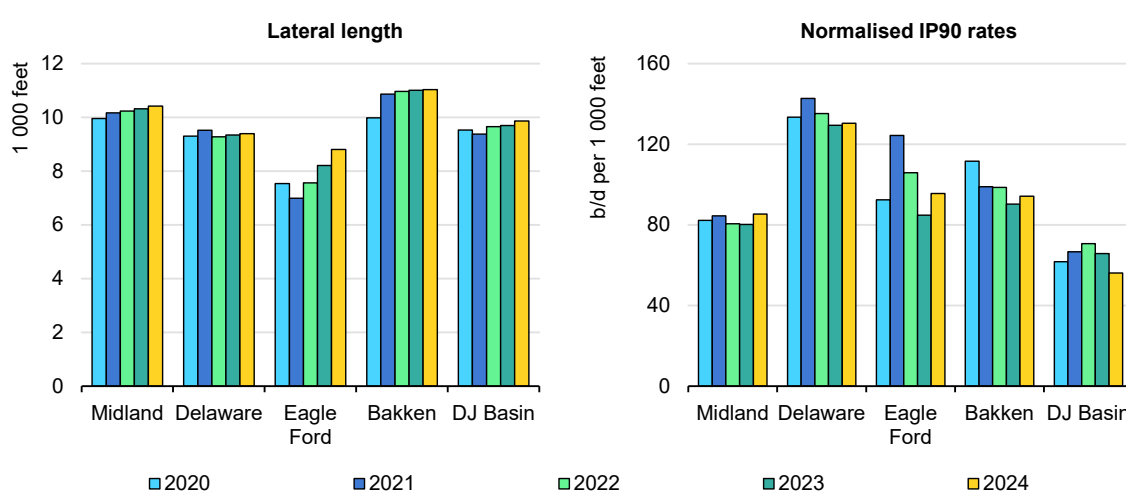
The high-grading of frack spreads, longer laterals and innovations such as simul-fracs, tri-fracs and horseshoe wells have allowed operators to continue to realise production gains while operating fewer drilling rigs. Since 2014 drillers in the Permian Basin are almost two and a half times more efficient and since 2019 they are over one and a half times more efficient. As a result, the 285 Permian Basin oil rigs that were active in Q4 2024 drilled an equivalent amount to 480 rigs five years ago and 650 rigs ten years ago.

When normalising for lateral length, well costs have begun to creep higher and initial production rates on a 90-day basis have, for the most part, flattened out. We

still assume that productivity gains will taper off, although some metrics may show an aggregate improvement as less efficient rigs and frack spreads are sidelined and rig-specific deflationary trends continue.

While decelerating, NGL growth will not slow as sharply as LTO as gas-oil ratios (GOR) have been increasing across key shale basins. US NGL production grows by 860 kb/d from 2024 through 2030. Data from Enterprise Products, an integrated midstream pipeline company and one of the largest NGL handlers in the world, show that every barrel of oil produced in the Permian Basin now has close to 30% more NGLs and natural gas associated with it when compared to 2022.

Productivity trends have tapered in key LTO basins, 2020-2024



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Note: Normalised IP90 rates are the median flow rates taken over 90 days then divided by the median lateral length.

Source: IEA analysis based on Rystad Energy ShaleWellCube data.

This is in part due to tight oil reservoir dynamics and the change in relative product mix over time, new drilling locations – including aerial expansions into gassier portions of the basin – as well as development strategies. There is more co-development drilling, where multiple stacked geologic zones – or benches – are developed concurrently leading to production from previously sub-economic zones that may be gassier from the start.

Shale oil price sensitivities

The oil prices used for modelling purposes in this *Report* are based on the NYMEX WTI forward curve in Q2 2025, rising from USD 60/bbl in 2026 to USD 62/bbl in 2030. For 2025, prices are pro-rated between realised WTI prices and the forward curve. These prices are then discounted to real terms. Additionally, some adjustments have been made to account for shale operators that are deemed less

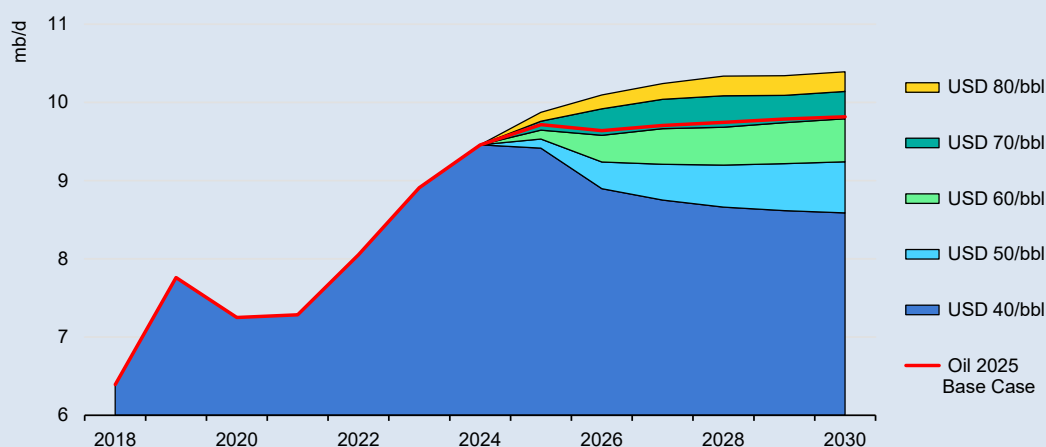
price responsive due to corporate planning assumptions, hedging programmes or other factors.

We also take into account changes seen in productivity, cost and DUC levels. Manual adjustments are used sparingly to tie to actual data and account for items such as repeated winter weather related shut ins.

Our modelling suggests that there are approximately 4-5 drilling rigs and 2-3 frack spreads at risk per dollar of sustained price movement between USD 45/bbl and USD 70/bbl. At higher price levels, recent behaviour suggests that additional cash flow from operations does not get reinvested at the same rate.

Base case LTO growth over the medium-term is expected to be 360 kb/d and, while sustained higher prices would drive increased activity levels and extra barrels, the production ceiling is lower than previous analysis showed. Should prices retreat further and stay below the threshold for drilling new wells, producers are expected to announce more activity cuts. It is estimated that a USD 80/bbl flat real WTI price would result in an additional 580 kb/d of growth by 2030 while USD 40/bbl flat real sustained WTI would see 1.2 mb/d of losses compared to the base case by 2030.

US LTO production at different flat real WTI price levels, 2018-2030



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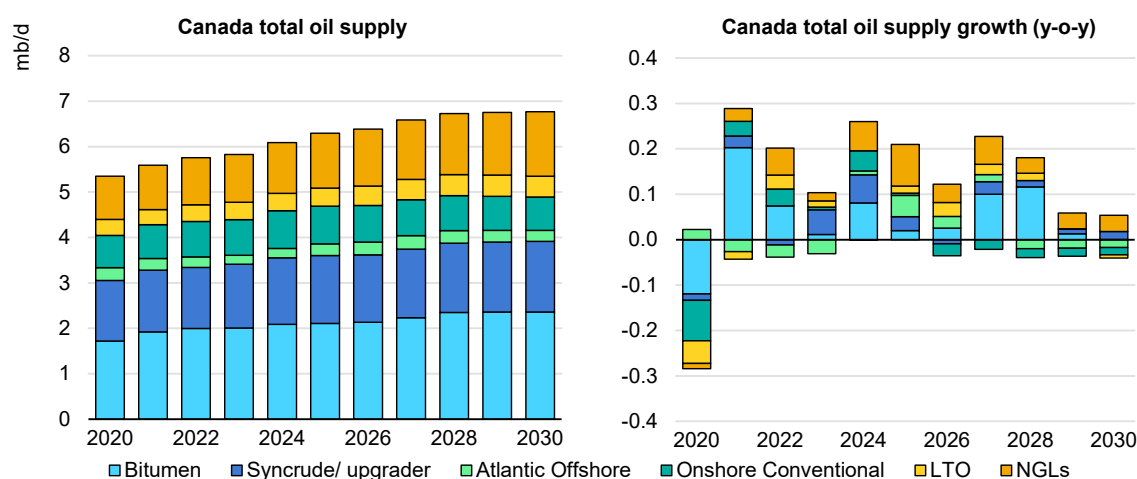
After rising by 160 kb/d to 1.9 mb/d in 2026, US federal offshore production is expected to fall to 1.6 mb/d by 2030, 170 kb/d below current levels. Following the start-up of 280 kb/d of new capacity in 2025, only 260 kb/d of additional capacity is sanctioned to come online from 2026 to 2030 – including Beacon Energy's Shenandoah Phase 2, BP's Kaskida and Shell's Vito infill and Sparta projects. An additional 400 kb/d of pre-FID developments could come online by the end of the decade if sanctioning decisions are made over the next two years.

Two Alaskan projects with 230 kb/d of capacity are also slated to see first oil during the latter half of the decade. The 80 kb/d Santos-operated Pikka Phase 1 development, and the 150 kb/d ConocoPhillips Willow project are expected to start in 2026 and 2029, respectively. Alaskan output will increase by 35% to 580 kb/d as these fields start up.

Discounts and debottlenecking of Canadian crude

Canadian production is set to continue its upward trajectory over the forecast period, buoyed by increases in bitumen and NGL output. The Western Canada Sedimentary Basin (WCSB) will account for most of the growth through the end of the decade, bringing total liquids to 6.8 mb/d. Out of the 680 kb/d total increase, 270 kb/d will be bitumen and 300 kb/d NGL.

Canadian oil supply and annual changes, 2020-2030



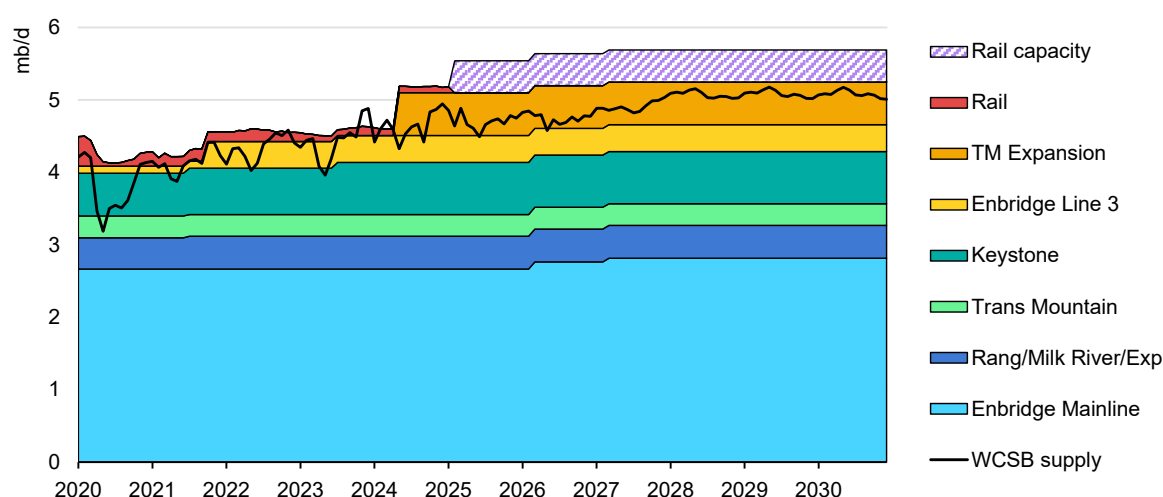
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Bitumen supply rises to 2.4 mb/d in 2030, driven by optimisation and debottlenecking of operations at oil sands projects, while new capital projects and expansions are limited in size and scope. NGL growth will come from associated liquids as LTO and shale gas developments continue across the Montney, Duvernay and Cardium plays. Total NGL output, including pentane plus, will top 1.4 mb/d by 2030 while LTO increases by 80 kb/d to 460 kb/d.

Canadian offshore output will finish the decade 30 kb/d higher at 240 kb/d after peaking in 2027 at 300 kb/d as Cenovus brings on its 50 kb/d West White Rose project. Exploration interest is muted with Equinor, the only company to have recently drilled a successful exploration well in the area, putting any field development on hold.

Takeaway capacity for the WCSB was expanded by 590 kb/d when the Trans Mountain Expansion (TMX) pipeline entered into service in Q2 2024, offering some relief to producers by increasing egress options and tightening local price differentials relative to WTI. Based on data from *Argus Media*, Western Canadian Select (WCS) Hardisty crude traded at a USD 12.60/bbl discount on average for the twelve months since TMX came into service compared to negative USD 17.20/bbl for the year prior.

Western Canada Sedimentary Basin liquids takeaway capacity, 2020-2030



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Notes: Western Canada Sedimentary Basin (WCSB) supply includes a 35% diluent ratio for non-upgraded bitumen and is net of 550 kb/d of non-export refining capacity. Nameplate pipeline capacity assumed, does not account for changes in available capacity due to maintenance or other items. Enbridge Mainline optimisation phase 1 included.

Source: IEA analysis based on data from the [Canadian Energy Regulator](#), [Enverus](#) and company statements.

The Latin America FPSO factory shows signs of slowing

Total non-OPEC+ Latin American oil supply will grow by 1.2 mb/d to 7.6 mb/d in 2030, after peaking at 7.8 mb/d in 2029. Growth will continue to be driven by developments in Brazil's offshore pre-salt, Guyana's Stabroek Block and Argentina's Neuquén Basin. These resources, along with Block 58 in Suriname, will more than offset declines from mature fields in the region. Additional barrels could come online if further projects are sanctioned in Guyana and Brazil's pre-salt, while just over the horizon are frontier areas such as Brazil's Equatorial Margin or Argentina's North Argentine Basin.

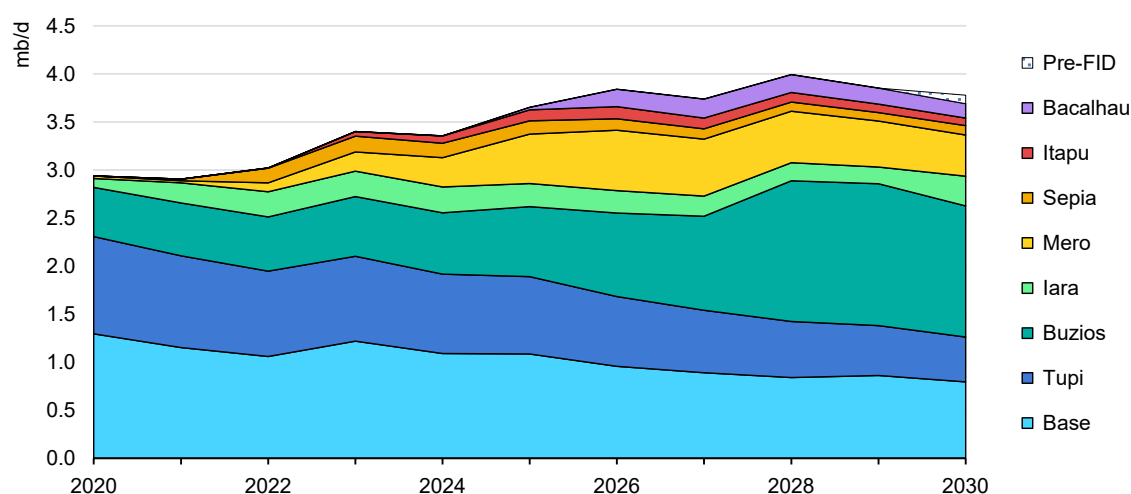
Brazilian output is on track to increase by 640 kb/d from 2024 to 2028 when it reaches 4.1 mb/d before retreating to 3.8 mb/d in 2030 in the absence of additional project start-ups. Petrobras operated fields are expected to contribute most of the gains, while TotalEnergies, Shell, Equinor, China National Offshore Oil Company (CNOOC) and CNPC also expand their footprint in Brazil's prolific offshore. The Santos Basin, home to 70% of the country's current crude production, will continue

to lead the expansion. Yet persistent project delays, operational issues and a tightening of maintenance spend could put Brazil's projected growth at risk.

Last year saw operational issues and labour action by a regulatory body impact 300 kb/d of output, while Petrobras' updated strategic plan pushed back six FPSO start dates. Along with a reduction in maintenance spend in response to lower oil prices, 2030 Brazilian oil production has been reduced by 500 kb/d compared to *Oil 2024*.

That is not to discount the monumental FPSO factory development plan that is mid-flight within Brazil. The Mero and Búzios fields will deploy a total of 15 FPSOs by 2028, including nine already in service. Petrobras plans to bring online eight FPSOs between now and 2030, including five additional Búzios installations. Once completed, the 11 Búzios FPSOs will have capacity of close to 2 mb/d. There are only two pre-FID projects that may see first oil by 2030 – Equinor's 100 kb/d Bacalhau Phase 2 and Petrobras's 100 kb/d BRC/CRT revitalisation.

Delays and declines weigh on future Brazilian growth, 2020-2030



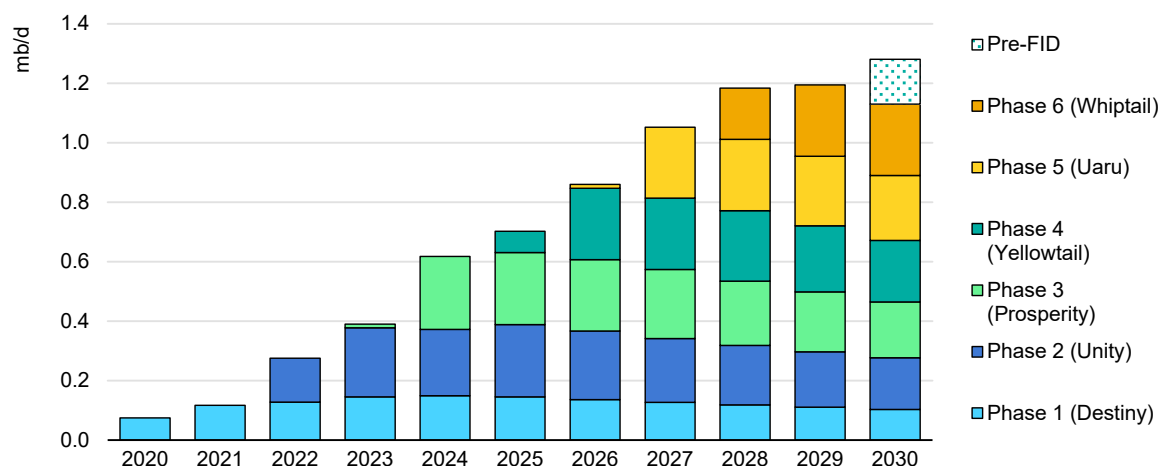
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The ExxonMobil-led consortium made further discoveries in **Guyana's** prolific Stabroek Block in 2024. Current estimates of recoverable oil equivalent resources are close to 12 billion barrels and a seventh development phase is scheduled to be sanctioned later this year. An eighth phase is slated to be approved in 2027. Based on the current sanctioned project pipeline, and in the absence of accelerated schedules, production should reach 1.2 mb/d in 2029 – double the 600 kb/d of supply in 2024.

Suriname's first offshore barrels will come from TotalEnergies and APA Corporation's recently sanctioned Block 58 development. The parcel sits adjacent to Guyana's Stabroek Block and the 220 kb/d FPSO is expected to produce first

oil in 2028, bringing the country's output from 20 kb/d of onshore production in 2025 to 220 kb/d at the end of the decade. Three other blocks in the country look promising, but even if exploration results yield commercial resources, those volumes would likely only materialise in the second half of the next decade.

Guyana growth driven by phased development of the Stabroek Block, 2020-2030



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Argentina's supply is forecast to grow by 430 kb/d to 1.3 mb/d through 2030 as its main shale patch the Vaca Muerta in the Neuquén Basin roars ahead. LTO is projected to rise by 500 kb/d, to just shy of 900 kb/d, while conventional supplies decline to 230 kb/d. New takeaway capacity, robust drilling and fracking activity as well as economic reforms underpin the LTO growth story in Argentina.

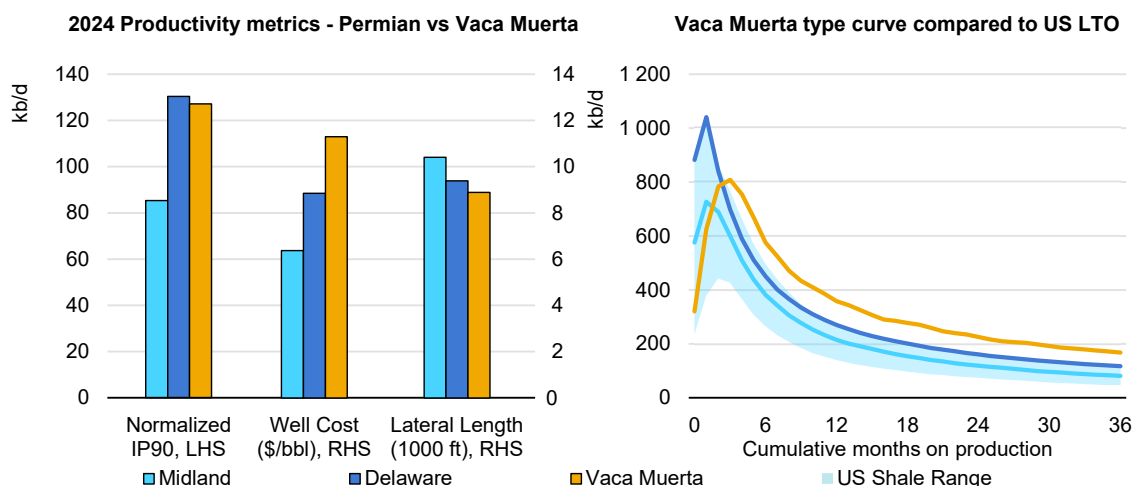
On 14 April 2025, President Javier Milei removed most capital and currency controls, alleviating international companies' concerns that profits would be trapped in Argentina. This continues the path of sectoral reforms along with the Régimen de Incentivo para Grandes Inversiones (RIGI) law passed [last autumn](#) which will provide 30 years of legal and regulatory stability in addition to tax breaks and export exemptions for projects greater than USD 1 billion.

Additional takeaway capacity is under construction, with the first phase of the 440 km Vaca Muerta Sur pipeline expected to enter service by the end of 2026. The 700 kb/d pipeline will bring LTO barrels for export to Punta Colorada in the Rio Negro state. The first phase will have capacity for 180 kb/d. The following year sees the project expand to 550 kb/d, with full capacity targeted for 2028.

The reforms and pipeline build-out help to unlock Vaca Muerta production. The play has high quality unconventional reservoirs that also benefit from a decade of derisking and operational learnings from North America. Indeed, the normalised initial 90-day average oil production (IP90) competes with the Delaware sub-basin

within the Permian and outperforms the Midland sub-basin. While lateral lengths are still shorter than in the Permian, there remains upside in reducing well costs.

Argentina's LTO well productivity competes with the Permian, costs have room to fall



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Notes: In the lefthand chart values are for the median well. In the right-hand chart, average type-curves are for the 2022-2024 drilling programme. Includes oil only and key US basins (Permian, Eagle Ford, Bakken and DJ-Niobrara). Actual and forecast data included.

Source: IEA analysis of data from Rystad ShaleWellCube.

By contrast, supply in the rest of Latin America is expected to decline due to a lack of investment and new projects. **Peru**, currently producing 120 kb/d of crude and NGLs, announced plans to boost upstream investment over the next five years. This has already begun to mitigate losses, but the roadmap to increased volumes remains unclear.

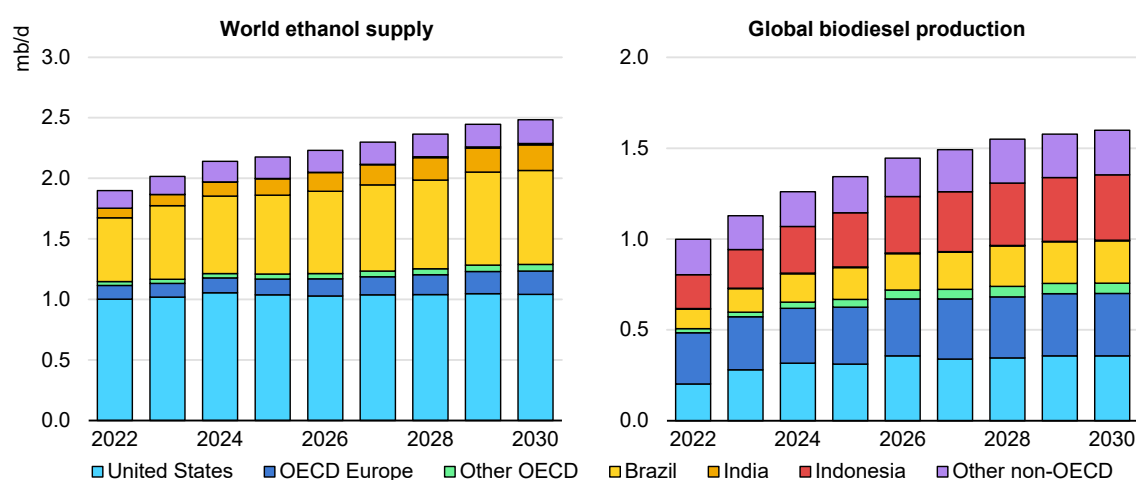
Ecuador and Colombia have seen reduced oil and gas investment levels in recent years as their governments have prioritised clean energy developments. In **Ecuador**, the passing of a 2023 referendum to close the 60 kb/d Ishpingo-Tambococha-Tiputini (ITT) field has been scaled back to ban new activity but allow existing wells to produce. More recently, optimism around renewed private investment dimmed as Sinopec missed the deadline to acquire the right to operate the 80 kb/d Sacha field. This *Report* forecasts output to fall 110 kb/d to 370 kb/d by 2030.

After the 2022 elections, the **Colombian** government halted new oil and gas exploration licenses and completely banned hydraulic fracturing (fracking) in tight oil and gas reservoirs. While mature oilfields have seen improved maintenance operations that have lessened underlying decline rates, output is still projected to fall from 790 kb/d in 2024 to 630 kb/d at the end of the decade.

Biofuels growth driven by developing and emerging countries

Global biofuels growth of 680 kb/d, split equally between ethanol and biodiesel is expected from 2024-30. Brazil and India lead ethanol production growth, adding 140 kb/d and 100 kb/d, respectively, and account for 70% of the total increase. Biodiesel gains are more geographically dispersed with Indonesia and Brazil forecast to deliver the largest contributions of 100 kb/d and 80 kb/d, respectively, through the end of the decade. The United States will add an additional 40 kb/d and Canada 20 kb/d.

Global biofuel supply split by ethanol and biodiesel, 2022-2030



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Notes: Biodiesel production includes renewable diesel and other distillate fuels such as biojet and renewable heating oil.

Brazilian growth, representing one-quarter of global increase, comes on the heels of its “Fuel of the Future” programme, which aims to increase domestic biofuel supply and develop a regulatory framework for sustainable aviation fuels (SAF), biomethane and biorefining. A decision to raise gasoline blending requirements to 30% from 27.5% could come as early as 1 July 2025 at the next National Energy Policy Council (CNPE) meeting, while the government plans to gradually increase biodiesel blending annually, aiming to reach 20% by 2030.

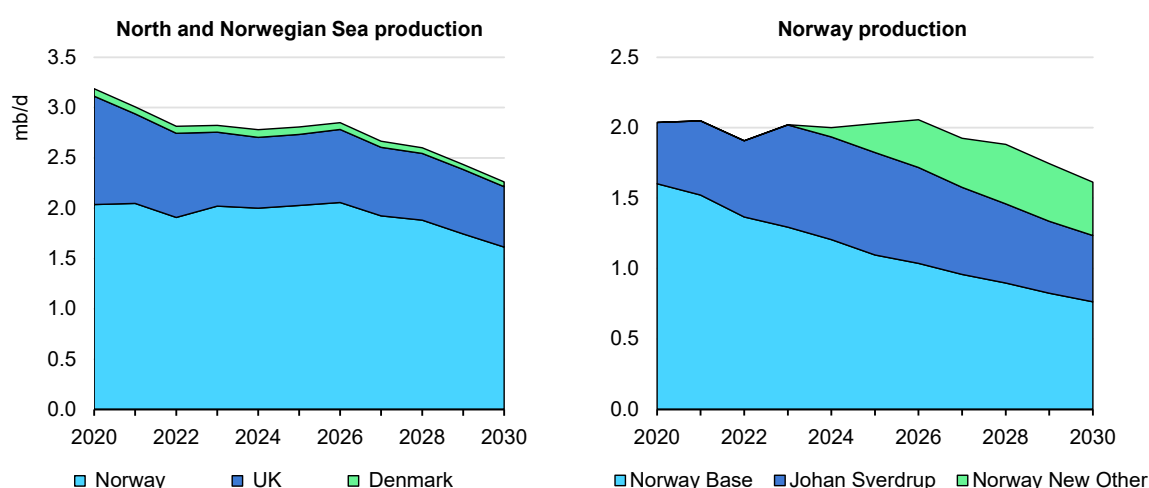
North Sea oil in regulatory and political focus

Net-zero ambitions as well as social pressure on governments, major financial institutions and insurers are shaping the production profile of the North Sea. Regional assets are already undergoing a structural shift as private equity and joint venture corporate spin-offs change the operating environment. Fortunes diverge between the United Kingdom and Norway, as the former continues to

operate in harvest mode, while the latter has seen more project approvals and field developments in recent years.

The **United Kingdom's** five-year production decline is expected to pause in 2025 at 710 kb/d before continuing its downward trend through our forecast period. Neptune's 2023 Seagull project, Shell's recent Penguins redevelopment along with BP's Clair Ridge project arrested declines last year and are expected to support supply this year as well. However, these developments, along with Equinor's 2028 Rosebank start-up – pushed back a year to account for the court ruling that more time is needed to assess its impact on the climate – are not sufficient to offset years of weak investment. Despite a few other smaller developments, output is set to reach a 50-year low of 600 kb/d by 2030.

Norwegian investments delay North Sea declines, 2020-2030



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Norway's crude oil production is expected to average 2.1 mb/d this year and next, underpinned by the large Johan Sverdrup project and the recently started-up Johan Castberg field. While a 2024 court ruling invalidated three field licences, producers are moving forward with confidence that the legal challenges will not impede production. Norway has substantial remaining resources, robust infrastructure and low-carbon intensity for oil production. Yet, without the sanctioning of new projects, output will fall to 1.6 mb/d by the end of the decade.

Chinese barrels support a sagging Asia Pacific region

Non-OPEC+ Asia Pacific oil supply has fallen by 650 kb/d over the last decade and is poised to drop by another 340 kb/d to 6.4 mb/d by 2030. While the region has struggled with ageing oil fields, changing ownership structures and